

# **Custom Compiler™ Packaging Designs User Guide**

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# About This User Guide

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This user guide explains how to use the Custom Compiler™ packaging features to create and route packaging designs.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

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## New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

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## Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Synopsys® Custom Compiler™

Synopsys 3DIC Compiler

Synopsys Fusion Compiler™

Synopsys IC Compiler™ II

Synopsys IC Compiler™

Synopsys IC Validator™

## Conventions

The following conventions are used in Synopsys documentation.

| Convention            | Description                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Courier               | Indicates syntax, such as <code>write_file</code> .                                                                                                                                                                                      |
| <i>Courier italic</i> | Indicates a user-defined value in syntax, such as <code>write_file design_list</code>                                                                                                                                                    |
| <b>Courier bold</b>   | Indicates user input—text you type verbatim—in examples, such as<br><code>prompt&gt; write_file top</code>                                                                                                                               |
| Purple                | <ul style="list-style-type: none"> <li>• Within an example, indicates information of special interest.</li> <li>• Within a command-syntax section, indicates a default, such as <code>include_enclosing = true   false</code></li> </ul> |
| [ ]                   | Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>                                                                                                                                                      |
| ...                   | Indicates that arguments can be repeated as many times as needed, such as<br><code>pin1 pin2 ... pinN</code> .                                                                                                                           |
|                       | Indicates a choice among alternatives, such as<br><code>low   medium   high</code>                                                                                                                                                       |
| \                     | Indicates a continuation of a command line.                                                                                                                                                                                              |
| /                     | Indicates levels of directory structure.                                                                                                                                                                                                 |
| <b>Bold</b>           | Indicates a graphical user interface (GUI) element that has an action associated with it.                                                                                                                                                |
| <b>Edit &gt; Copy</b> | Indicates a path to a menu command, such as opening the <b>Edit</b> menu and choosing <b>Copy</b> .                                                                                                                                      |
| Ctrl+C                | Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.                                                                                                                                                      |

## Customer Support

Customer support is available through SolvNetPlus.

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## Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

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## Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

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## Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

# 1

## IC Packaging Designs in the Custom Compiler Environment

---

*This chapter provides an overview of the Custom Compiler features and workflow that enable integrated circuit (IC) packaging design development. This chapter also provides an overview of the menu commands that you would use for packaging designs within the Custom Compiler environment.*

This chapter describes the following topics:

- [About IC Packaging Designs](#)
- [Regions in a Packaging Design](#)
- [Software User Interface for Packaging Designs](#)

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### About IC Packaging Designs

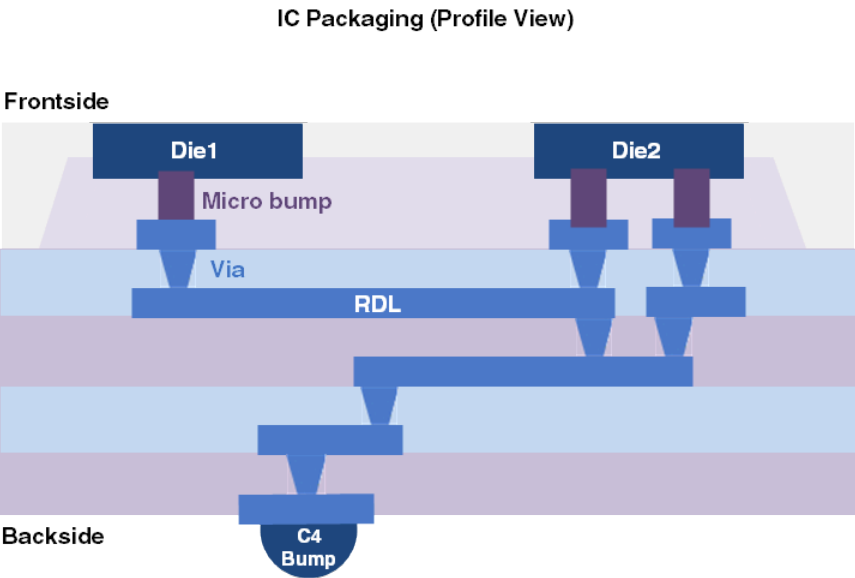
Integrated circuit (IC) packaging is the process in which physical enclosures are wrapped around integrated circuits to protect them from damage and corrosion. Exposed electrical contacts (microbumps) permit signal distribution and connections to other components.

Advanced IC packaging is the technology that packages multiple dies together to:

- Reduce the package size, thickness, and power consumption.
- Increase signal speed and heat dissipation.
- Accommodate a large number of external contacts.

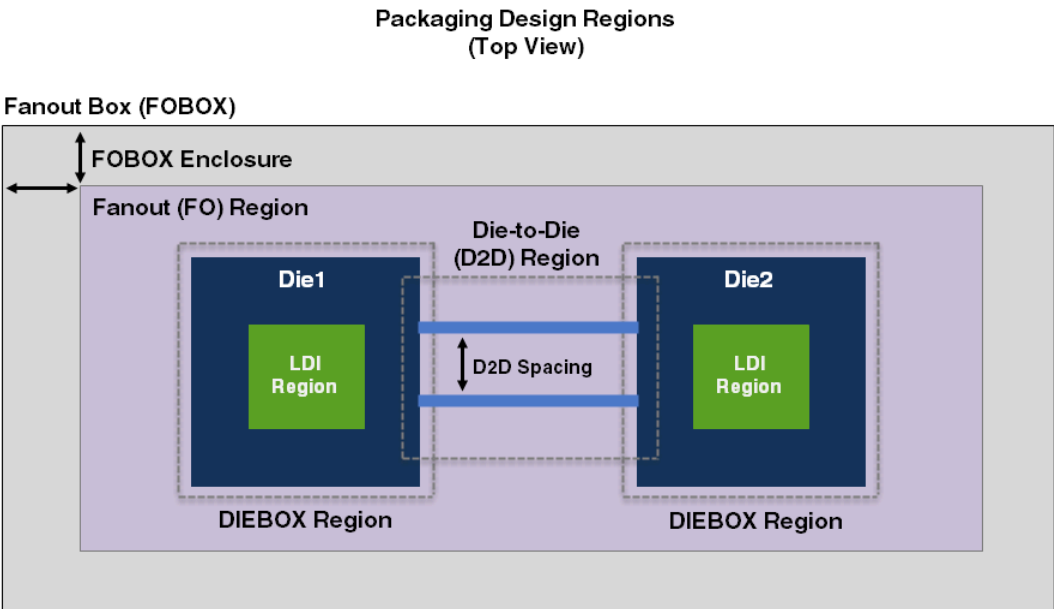
With minimal form factor and optimal performance, the technology is ideal for meeting small mobile computing device requirements.

Wafer-level packaging is an advanced IC packaging technology that integrates dies and C4 bumps with high-density redistribution layers (RDLs). Frontside dies can connect to other frontside dies through RDLs and through-silicon vias (TSVs). Frontside dies can also connect to backside C4 bumps using RDLs and TSVs. Typically, routing through RDLs occur at 45°, although they can be placed at any angle as required.

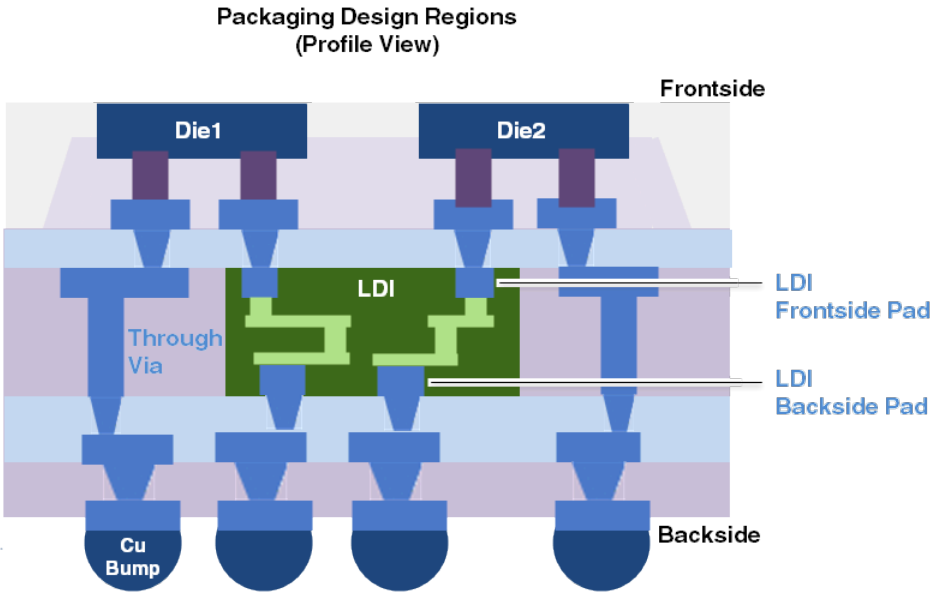


## Regions in a Packaging Design

Wafer-level packaging designs consist of different regions. The regions have different routing requirements.



| Region                       | Description                                                  |
|------------------------------|--------------------------------------------------------------|
| Fanout box (FOBOX)           | Packaging design boundary.                                   |
| Fanout (FO)                  | FOBOX minus the FOBOX enclosure.                             |
| Die box (DIEBOX)             | Region that contains a die.                                  |
| Die-to-die (D2D)             | Region that contains die-connecting routes.                  |
| Local die interconnect (LDI) | Region for routes that require optimized signal performance. |



## Software User Interface for Packaging Designs

This topic provides an overview of the Custom Compiler user interface for packaging designs:

- [Co-design User Interface](#)
- [Setup and Routing User Interface](#)
- [Routing Constraint User Interface](#)

## Co-design User Interface

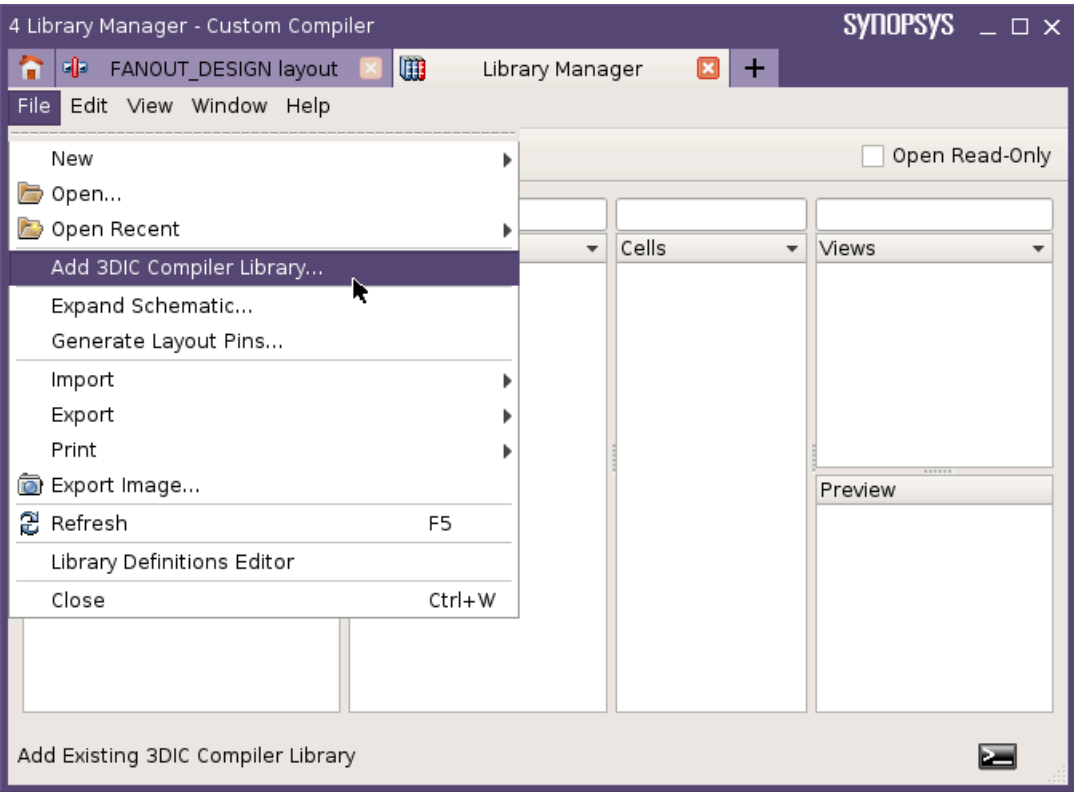
During the co-design flow, you use the Library Manager to add 3DIC Compiler design libraries into the Custom Compiler environment. Within the Custom Compiler environment, you can open the 3DIC Compiler tool to review the modifications you made.

This topic describes the following:

- [Library Manager](#)
- [Layout Editor](#)

## Library Manager

To open 3DIC Compiler libraries in the Custom Compiler, you must first add the existing 3DIC Compiler libraries in the Custom Compiler environment. You use the Custom Compiler Library Manager to add 3DIC Libraries.

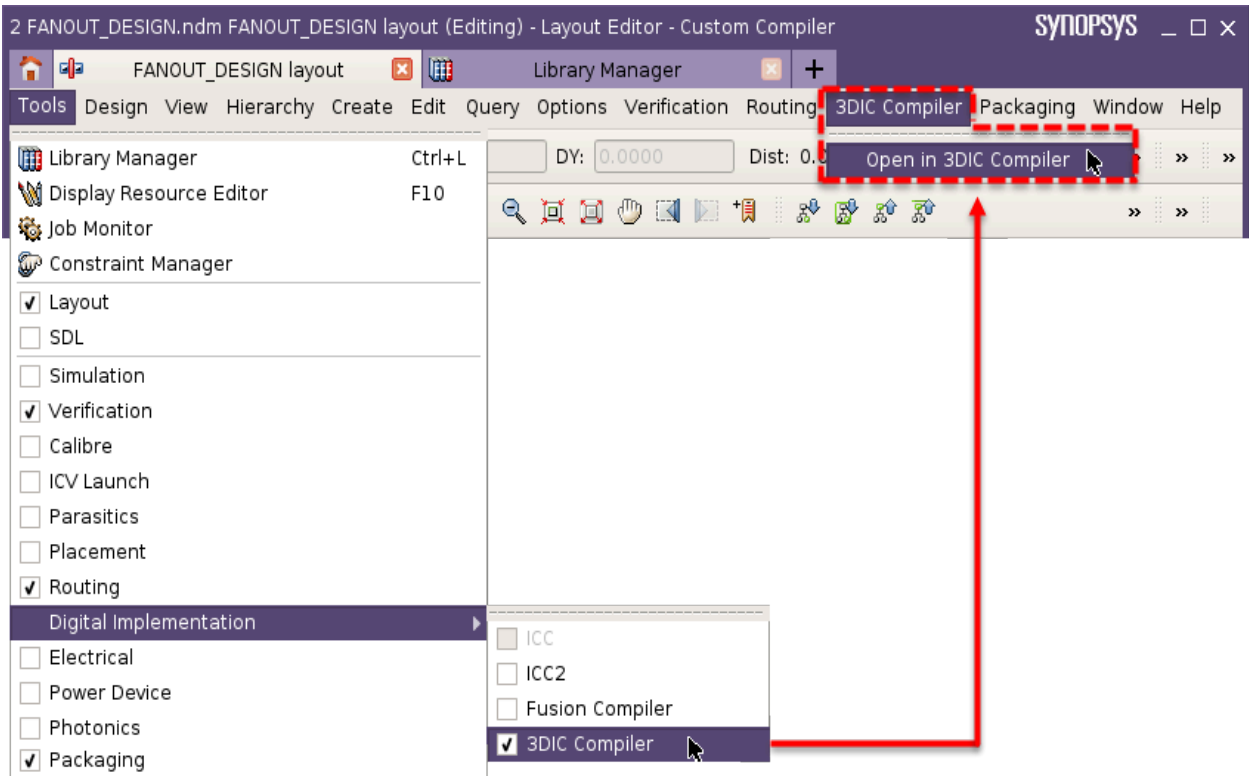


| Command                          | See...                                                                                                                                                                          |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File > Add 3DIC Compiler Library | Opens existing 3DIC Compiler libraries in the Custom Compiler environment. See <a href="#">Adding 3DIC Compiler Libraries</a> .<br>Tcl command: <code>dm::add3DICLibrary</code> |

After you add the 3DIC Compiler libraries in the Custom Compiler environment, you can use the Library Manager to view information about the designs. You can also open and manage libraries, cells, and views for the design process. For detailed information, see the Library Manager Tool.

Layout Editor

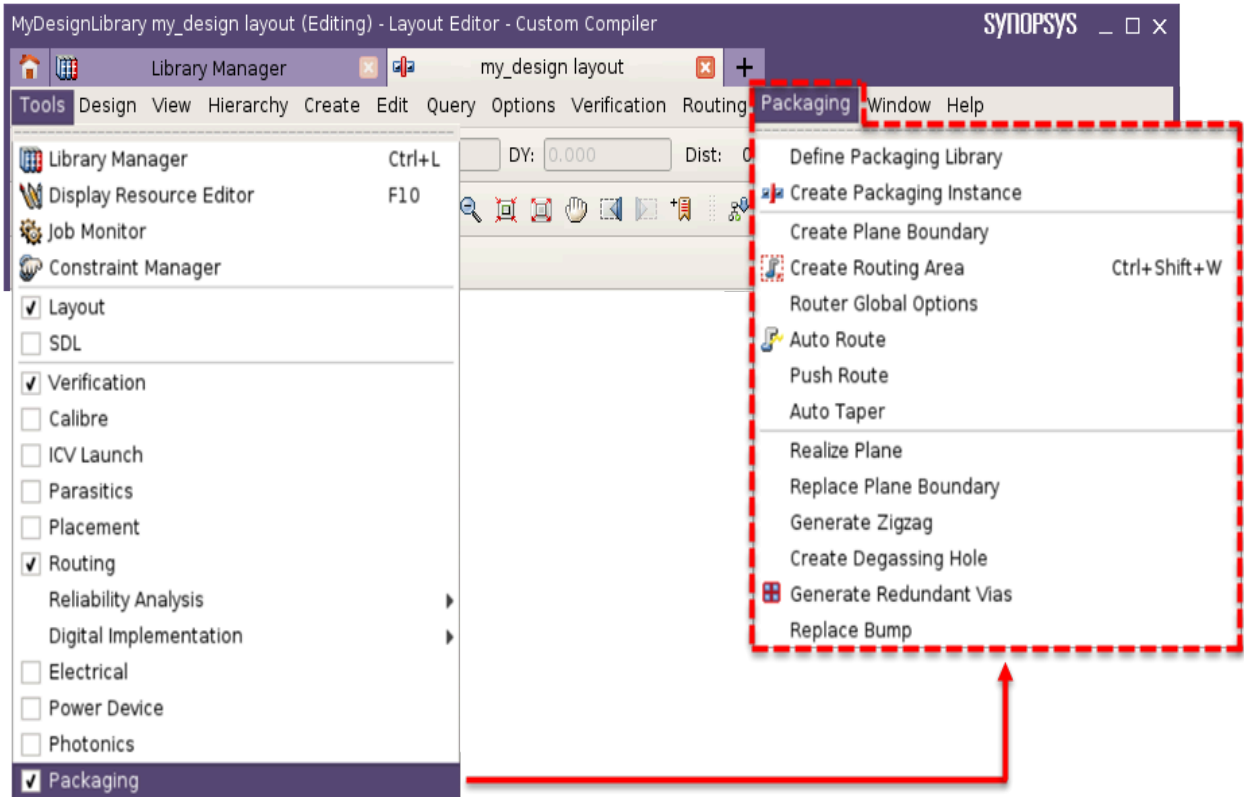
The Layout Editor **3DIC Compiler > Open in 3DIC Compiler** command lets you open and view the design using the 3DIC Compiler tool while you are still in the Custom Compiler environment.



| Command                            | See...                                                                                                                                                                                     |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DIC Compiler > Open 3DIC Compiler | Opens the 3DIC Compiler design in the 3DIC Compiler tool while you are in the Custom Compiler environment. See <a href="#">Reviewing Custom Compiler Edits in the 3DIC Compiler Tool</a> . |

## Setup and Routing User Interface

For packaging design setup, creation, and editing tasks, use the Layout Editor **Packaging** commands:



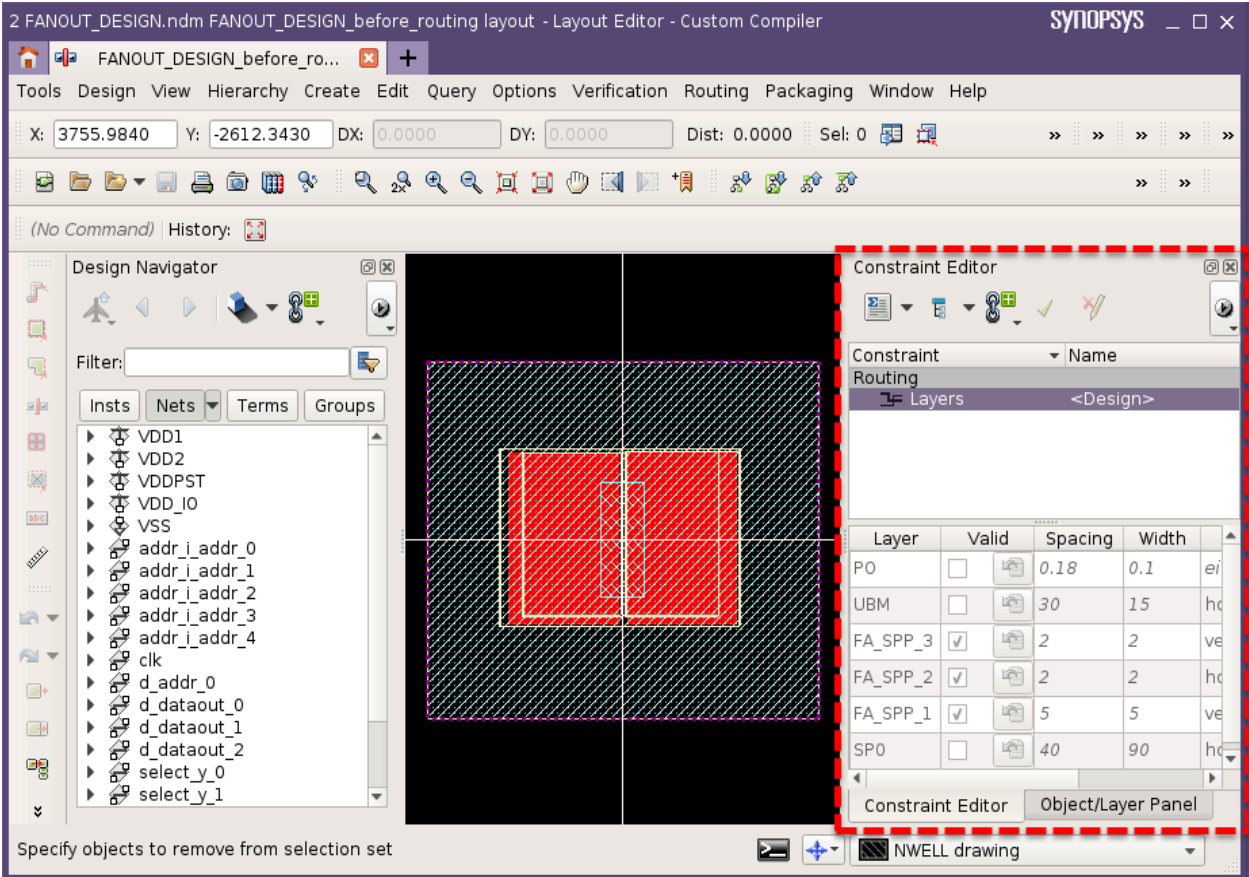
| Packaging Command         | See...                                                               |
|---------------------------|----------------------------------------------------------------------|
| Define Packaging Library  | <a href="#">Setting Up for Packaging Designs</a>                     |
| Create Packaging Instance | <a href="#">Creating Packaging Instances</a>                         |
| Create Plane Boundary     | <a href="#">Creating PG Plane Boundaries Manually</a>                |
| Create Routing Area       | <a href="#">Defining the Routing Area in the Packaging Design</a>    |
| Router Global Options     | <a href="#">Setting Global Routing Options for Packaging Designs</a> |
| Auto Route                | <a href="#">About the Packaging Auto Router</a>                      |
| Push Route                | <a href="#">Fixing Spacing Violations</a>                            |

| Packaging Command       | See...                                                    |
|-------------------------|-----------------------------------------------------------|
| Auto Taper              | <a href="#">Fixing Tapering Violations</a>                |
| Realize Plane           | <a href="#">Realizing PG Plane Boundaries</a>             |
| Replace Plane Boundary  | <a href="#">Replacing PG Plane Boundaries</a>             |
| Generate Zigzag         | <a href="#">Generating Zigzags</a>                        |
| Create Degassing Hole   | <a href="#">Creating Dummy Metals and Degassing Holes</a> |
| Generate Redundant Vias | <a href="#">Creating and Reporting Redundant Vias</a>     |
| Replace Bump            | <a href="#">Replacing Microbumps</a>                      |

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## Routing Constraint User Interface

For packaging design routing, use the Constraint Editor to set Routing Layer constraints. See [Setting Routing Constraints for Packaging Designs](#).



# 2

## Getting Started

---

*This chapter describes the high-level workflow for packaging designs, and explains how to enable the 3DIC Compiler and Custom Compiler co-design assistant.*

*This chapter describes the following:*

- [Packaging Design Workflow](#)
- [Starting Custom Compiler Sessions and Displaying the Library Manager](#)

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### Packaging Design Workflow

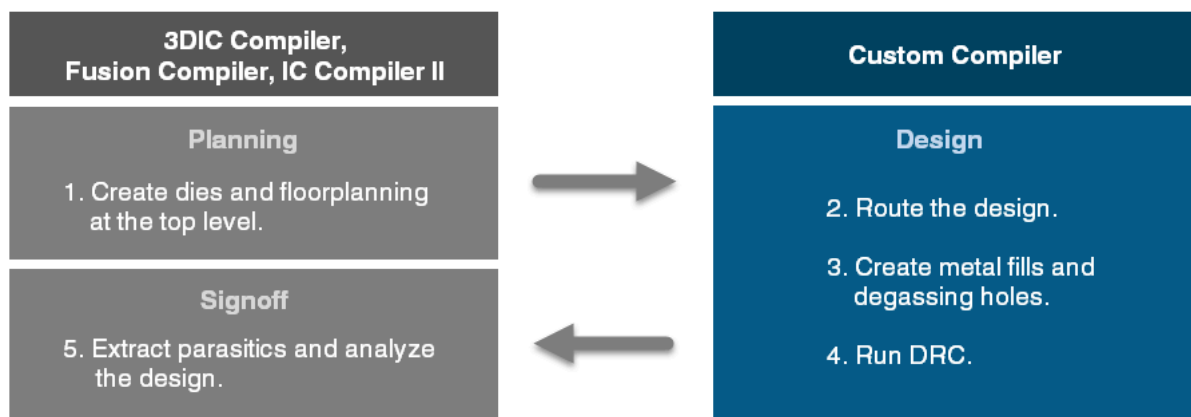
This topic describes the following:

- [Packaging Co-design Workflow](#)
- [Custom Compiler Packaging Design Workflow](#)

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### Packaging Co-design Workflow

A packaging design is developed in multiple phases using multiple tools:

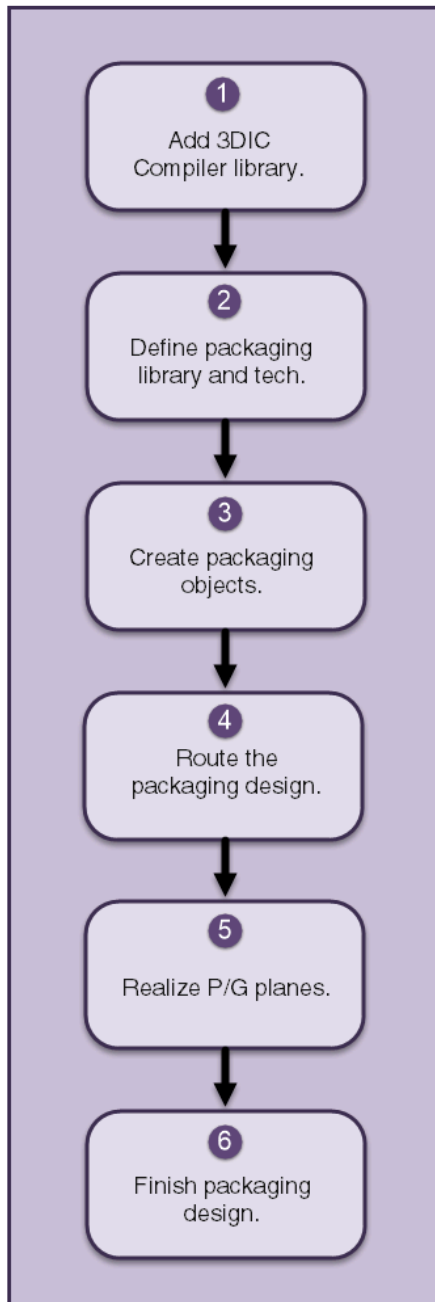


| Step |                             | Description                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Digital tool                | Create dies and floorplanning at the top level. Tasks include die placement, microbump placement, and netlist editing.<br>See: <ul style="list-style-type: none"><li>• <a href="#">3DIC Compiler user documentation</a></li><li>• <a href="#">Fusion Compiler user documentation</a></li><li>• <a href="#">IC Compiler II user documentation</a></li></ul> |
| 2-4  | Custom Compiler environment | Route individual dies, place power and ground planes, add metal fills and degassing holes, and run IC Validator design rule checks (DRC).<br>See <a href="#">Packaging Design Workflow</a> .                                                                                                                                                               |
| 5    | Digital tool                | Extract parasitics and analyze the design.<br>See one of the following: <ul style="list-style-type: none"><li>• <a href="#">3DIC Compiler user documentation</a></li><li>• <a href="#">Fusion Compiler user documentation</a></li><li>• <a href="#">IC Compiler II user documentation</a></li></ul>                                                        |

---

## Custom Compiler Packaging Design Workflow

The following diagram shows the high-level task sequence in the Custom Compiler environment. Click a task to view the instructions for that step.



---

## Starting Custom Compiler Sessions and Displaying the Library Manager

To route the packaging design in the Custom Compiler environment, you must start a Custom Compiler session, and display the Library Manager. The Library Manager allows you to view and select the design you want to open.

1. Start a Custom Compiler session. See [Starting Sessions](#).
2. Display the Library Manager. See [Displaying the Library Manager](#).

# 3

## Preparing Packaging Designs for Editing

---

*To open 3DIC Compiler designs in the Custom Compiler environment, you must first add the existing 3DIC Compiler libraries in the Custom Compiler environment. If you want to edit the design using the 3DIC Compiler tool while in the Custom Compiler environment, you must enable the co-design assistant.*

This chapter describes the following:

- [Adding 3DIC Compiler Libraries](#)
- [Viewing 3DIC Compiler Library Attributes](#)
- [Editing Reference Library Associations](#)
- [Opening Packaging Designs](#)
- [Enabling the Co-design Assistant](#)

---

### Adding 3DIC Compiler Libraries

You must add 3DIC Compiler libraries in the Custom Compiler environment so that you can open and work with 3DIC Compiler designs.

Adding 3DIC Compiler libraries in the Custom Compiler environment allows you to:

- Open 3DIC Compiler libraries in the Custom Compiler environment.
- Store modifications you make within the Custom Compiler environment are stored in 3DIC Compiler libraries.

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### Before You Start

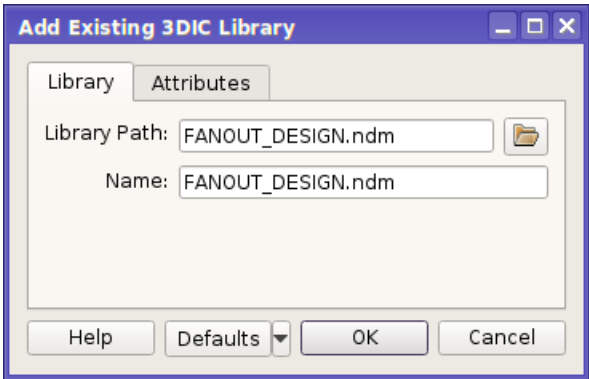
Make sure:

- You enable the Custom Compiler and 3DIC Compiler co-design assistant. See [Enabling the Co-design Assistant](#).
- The 3DIC Compiler library you are adding must already exist in the 3DIC Compiler tool.

Procedure

To add 3DIC Compiler libraries to the Custom Compiler environment:

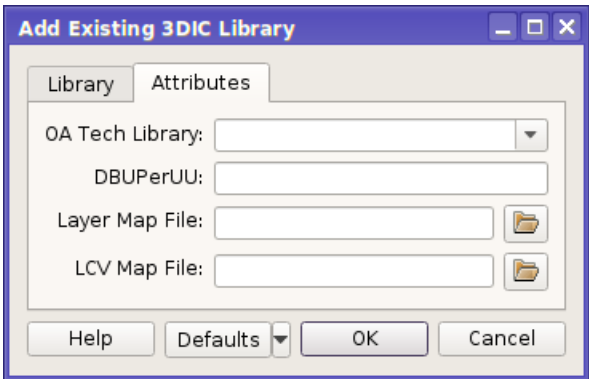
- 1. Start a Custom Compiler session and display the Library Manager. See Starting Sessions and Displaying the Library Manager.
- 2. In the Library Manager, choose **File > Add 3DIC Compiler Library**. The **Add Existing 3DIC Compiler Library** dialog box opens.



- 3. Provide the following information:

| Setting      | Description                                                                   |
|--------------|-------------------------------------------------------------------------------|
| Library Path | The directory or file that contains the 3DIC Compiler library.                |
| Name         | The new name of the 3DIC Compiler library in the Custom Compiler environment. |

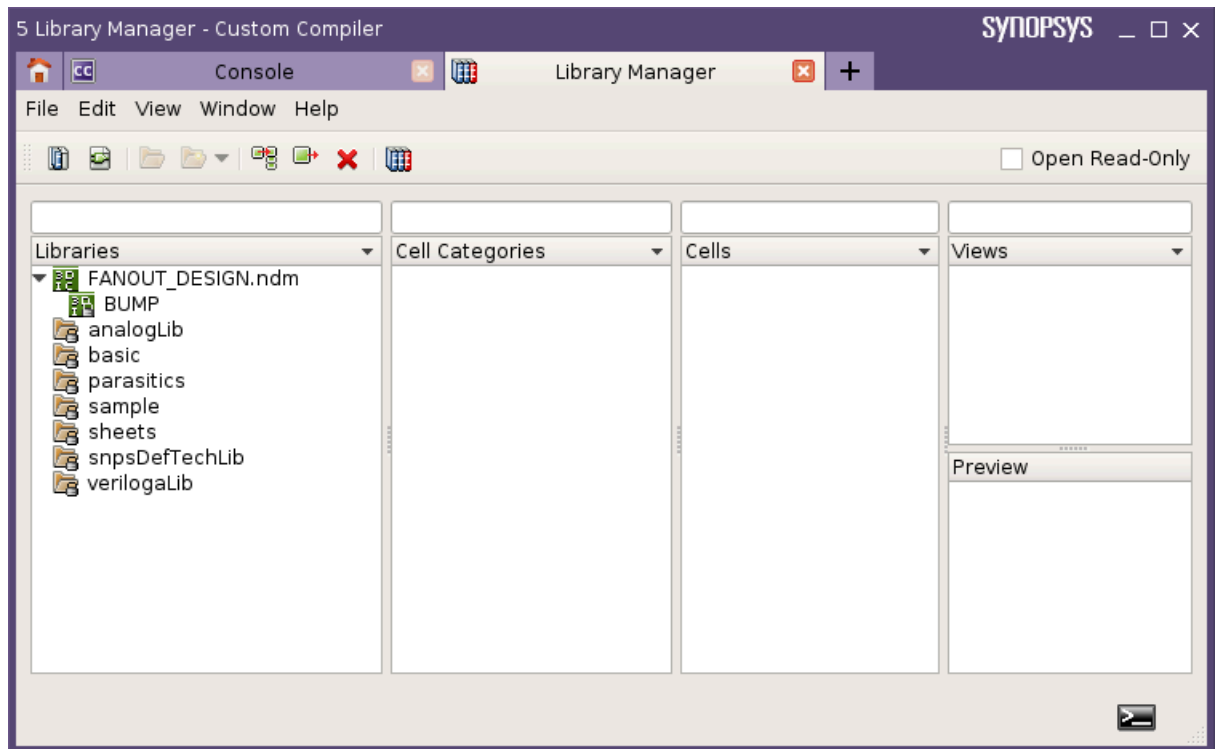
- 4. Click the **Attributes** tab, and provide the following information:



| Setting         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OA Tech Library | OA (Custom Compiler) technology library.<br>To ensure seamless co-designing, specify the OA technology library to use when in the Custom Compiler environment instead of the 3DIC Compiler library.                                                                                                                                                                                                                                                                                                                                                                                                        |
| DBUPerUU        | Design scale factor.<br>Mixed-signal designs use technology data that do not have the same DBUPerUU as in the 3DIC Compiler tool. Therefore, you must adjust the 3DIC Compiler design units to ensure smooth top-level integration within the Custom Compiler environment.<br>Synopsys recommends that you use the same DBUPerUU for the PDK and 3DIC Compiler technology libraries. To scale the design units, set DBUPerUU to the target PDK resolution.<br>For information about setting scale factors for designs in the 3DIC Compiler tool, see the 3DIC Compiler <a href="#">create_lib</a> command. |
| Layer Map File  | Layer mapping file.<br>Mixed-signal designs use OA technology, but the OA (Custom Compiler) technology layer names are not equivalent to the 3DIC Compiler technology layer names. Therefore, you must map 3DIC Compiler layer names to OA layer names so that you can edit and view digital blocks in the context of mixed-signal designs.<br>You must first prepare the layer mapping file. Then, set this option to reference the file.                                                                                                                                                                 |
| LCV Map File    | Library and cell name mapping file.<br>When working with digital components in the Custom Compiler environment, the digital library and cell names must be mapped to Custom Compiler (OA) library and cell names to ensure co-design compatibility and seamless top-level integration.<br>You must first prepare the library and cell name mapping file. Then, set this option to reference the file.                                                                                                                                                                                                      |

- When you are finished, click **OK**. The new library name appears in the Library Manager. The  icon in front of the library name indicates that it is a 3DIC Compiler library.
- To view the associated reference libraries, choose **View > Show 3DIC Compiler Reference Libraries**.

The list of 3DIC Compiler libraries change to a tree structure. To view the referenced libraries, click the 3DIC Compiler library name.

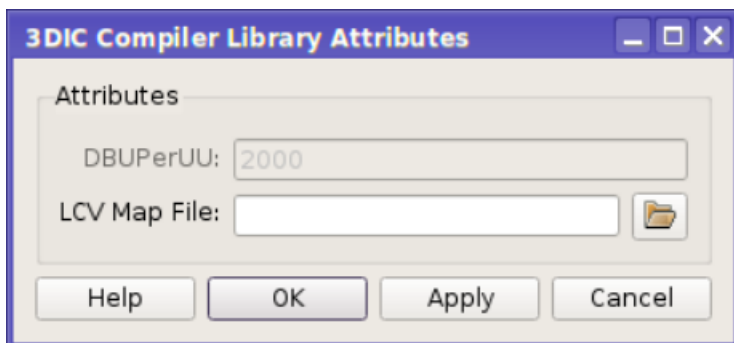


## Viewing 3DIC Compiler Library Attributes

After you add a 3DIC Compiler library, you can view the library attributes.

To view the 3DIC Compiler library attributes in the Custom Compiler environment:

1. In the Library Manager, right-click the 3DIC Compiler library name and choose **3DIC Compiler Library Attributes**. The **3DIC Compiler Library Attributes** dialog box opens.



2. Review the **DBUPerUU** information and the **LCV Map File**. The DBRPerUU information cannot be changed.
3. (Optional) Specify a different **LCV Map File**.
4. When you are finished, click **OK** to apply the changes and close the dialog box.

---

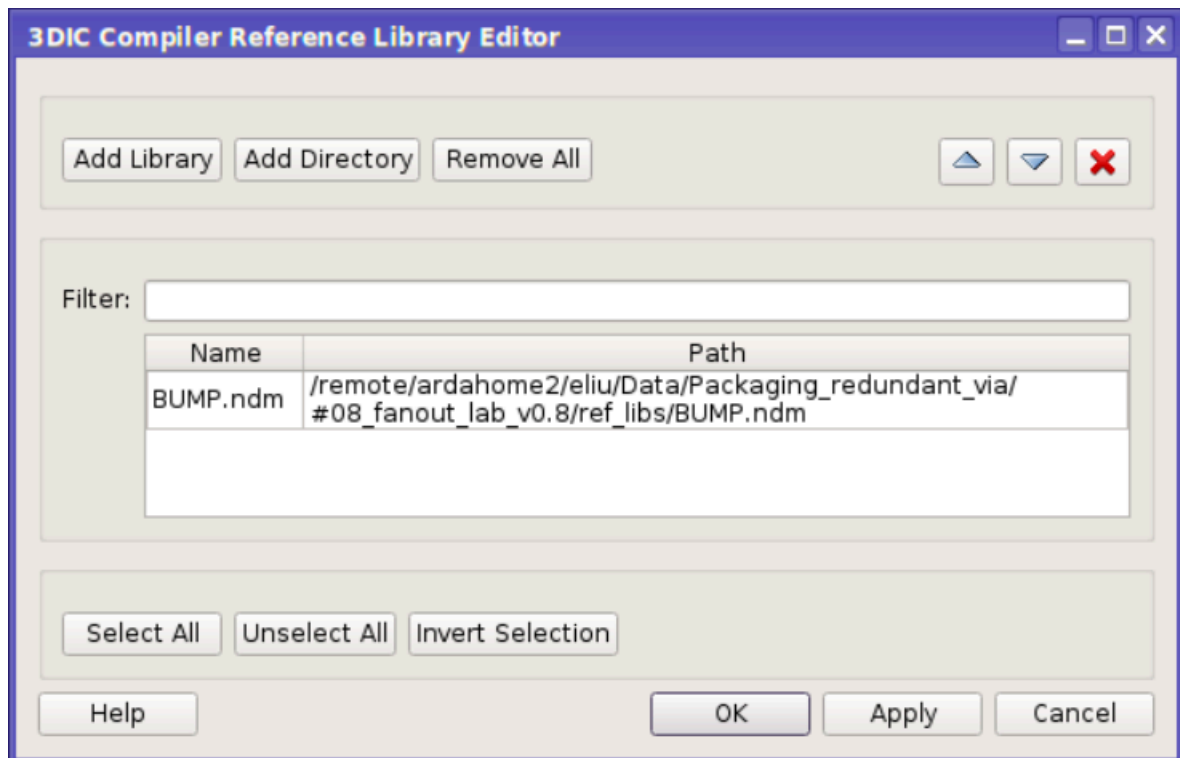
## Editing Reference Library Associations

After adding a 3DIC Compiler design in the Custom Compiler environment, you can edit its reference library associations as needed.

To edit the 3DIC Compiler reference library associations:

1. In the Library Manager, select the 3DIC Compiler library.
2. Choose **Edit > Edit 3DIC Compiler Reference Libraries**.

The **3DIC Compiler Reference Library Editor** dialog box opens.



3. To add one or more reference libraries:
  - Click **Add Library** to select and add a single reference library.
  - Click **Add Directory** to add all reference libraries in the directory you select.
4. To change the position of a reference library in the list, select the reference library and click  or .
5. To select one or more reference libraries, click one of the following:
  - **Select All**: Selects all reference libraries in the list.
  - **Unselect All**: Cancels the selection of reference libraries.
  - **Invert Selection**: Toggles the selection so that the libraries that were selected are no longer selected, and libraries that were not selected are now selected.
6. To remove a reference library from the list, select the reference library and click .
7. To remove all reference libraries from the list, click **Remove All**.
8. When you are finished, click **OK**.

---

## Opening Packaging Designs

After you add the 3DIC Compiler library to the Custom Compiler environment, you can open the design and start preparing it for routing. You open designs from the Library Manager.

### Note:

The source digital design is locked in the 3DIC Compiler tool to protect the data while you work with the design within the Custom Compiler environment.

This topic describes the following:

- [Opening Filtered Designs](#)
- [Opening an Entire Design](#)

---

## Opening Filtered Designs

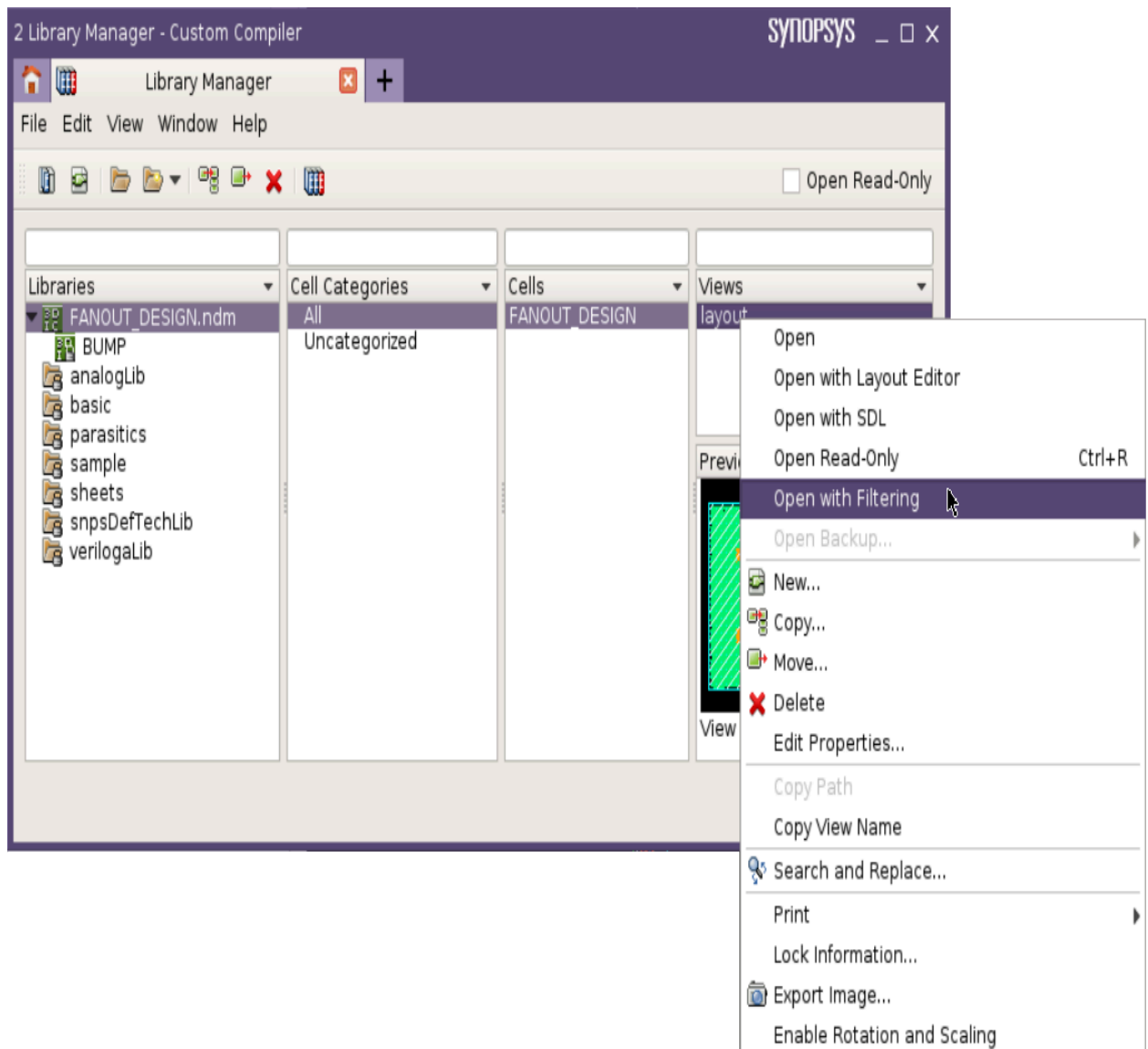
Loading and editing large 3DIC Compiler designs in the Custom Compiler environment can be cumbersome and impact tool performance. Instead, you can define a region of interest and load only that region in the Custom Compiler environment.

You define the region of interest with a set of filtering criteria. You can set up the filters while in the 3DIC Compiler tool or when you open the design in the Custom Compiler environment.

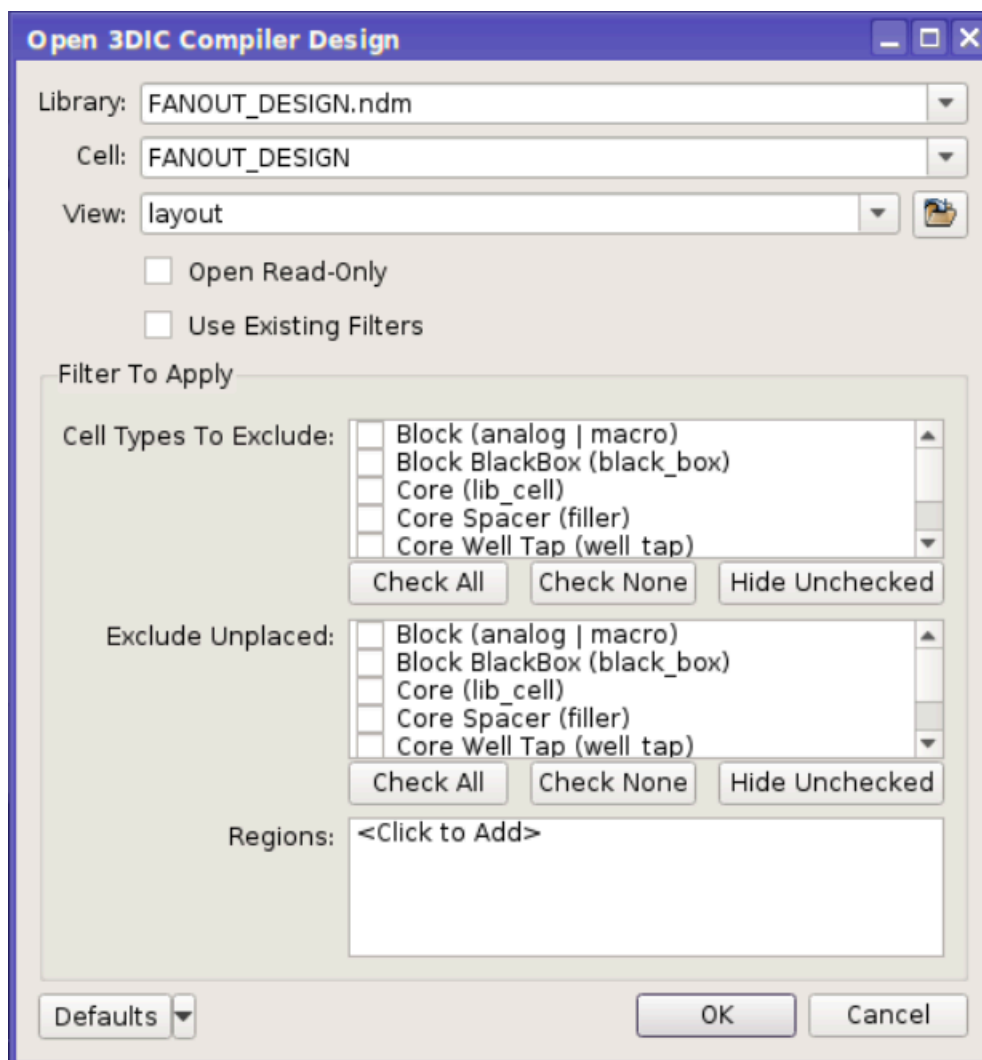
This section describes how to set up and apply filters while opening the design in the Custom Compiler environment.

To open a filtered design:

1. In the Custom Compiler [Library Manager](#), choose the library name in the **Libraries** pane. The associated Cell Categories and Cells appear.
2. In the **Cells** pane, choose the cell. The **Views** pane lists the available design views.
3. Right-click the **layout** view and choose **Open with Filtering**.



4. In the **Open 3DIC Compiler Design** dialog box that opens, specify the design **Library**, **Cell**, and **View** names.



5. Select **Open Read-Only** if you want to open the design in the read-only mode.
6. To use the filters you set up in the 3DIC Compiler tool, select **Use Existing Filters**.
7. To specify new filters and define the region of interest, set the following:

| Option                                    | Description                                       |
|-------------------------------------------|---------------------------------------------------|
| Filter to Apply:<br>Cell Types to Exclude | Filters the type of cell.                         |
| Filter to Apply:<br>Excludes Unplaced     | Excludes the selected unplaced design components. |

| Option                      | Description                                                                          |
|-----------------------------|--------------------------------------------------------------------------------------|
| Filter to Apply:<br>Regions | Specifies the coordinates (x, y) that define the vertices of the rectangular region. |

8. Click **OK**.

The Layout Editor opens and displays the filtered 3DIC Compiler design within the viewport of the canvas.

A mesh overlay lets you know that the regions beneath are not loaded, and you will not be permitted to edit the mesh-covered regions. The mesh overlay is treated as a blockage, and remains visible even if you disable the visibility of **Blockage** in the Object/Layer Panel.

**See Also**

- [Opening an Entire Design](#)

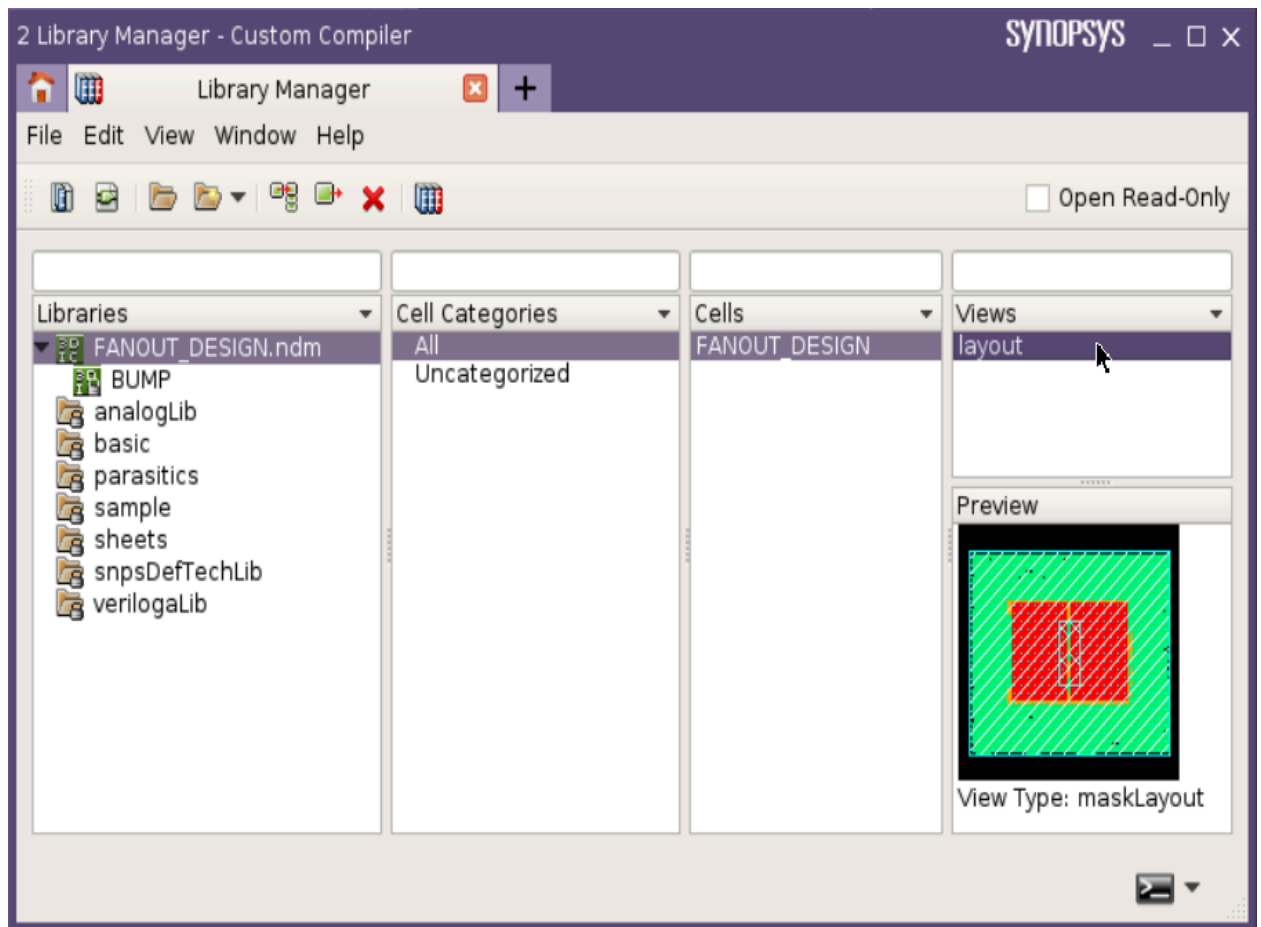
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## Opening an Entire Design

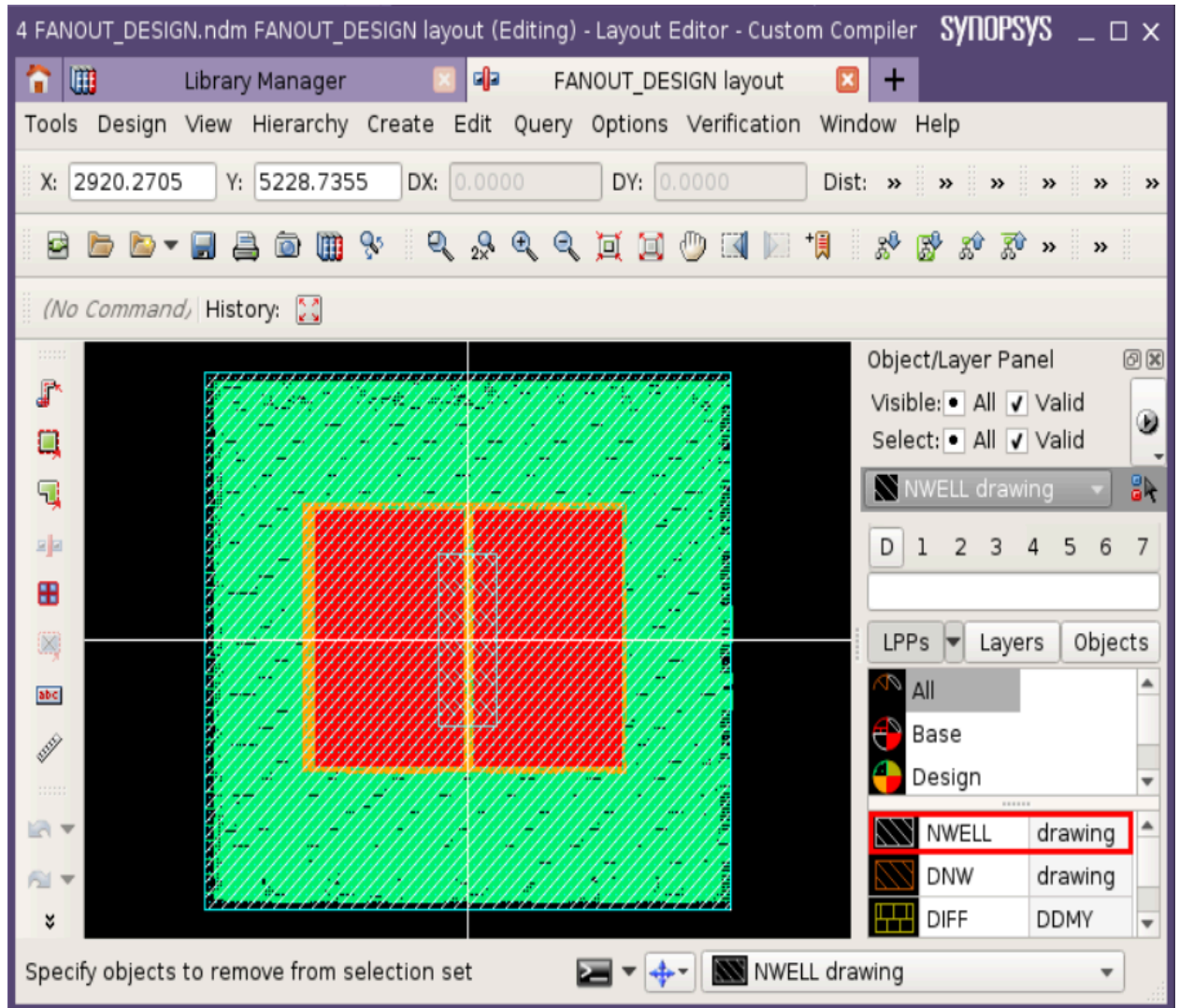
After adding a 3DIC Compiler library, you can open the associated 3DIC Compiler designs in the Custom Compiler environment. You can open the entire design, or you can open load and open only a region of the design.

To open an entire design:

1. In the Custom Compiler [Library Manager](#), choose the library name in the **Libraries** pane. The associated Cell Categories and Cells appear.
2. In the **Cells** pane, select the cell. The **Views** pane lists the available design views.
3. Double-click the **layout** view.



The Layout Editor opens and displays the 3DIC Compiler design within the viewport of the canvas. Notice that the names of the library, cell, and view appear at the top of the tab.



### See Also

- [Opening Filtered Designs](#)

## Enabling the Co-design Assistant

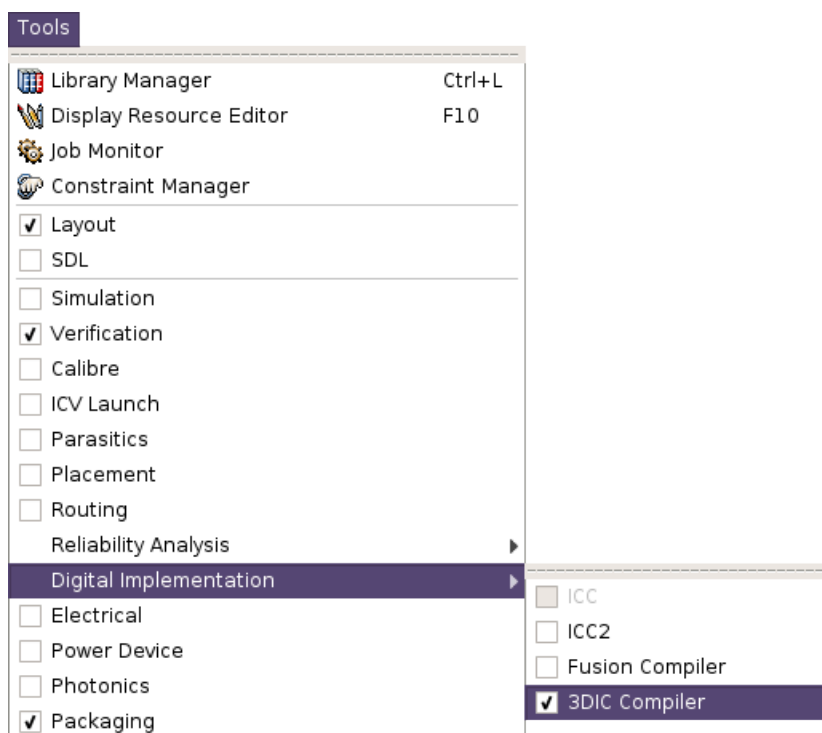
After opening a 3DIC Compiler design in the Custom Compiler environment, you enable the co-design assistant to view and edit the design using the 3DIC Compiler tool while in the Custom Compiler environment.

To enable the co-design assistant:

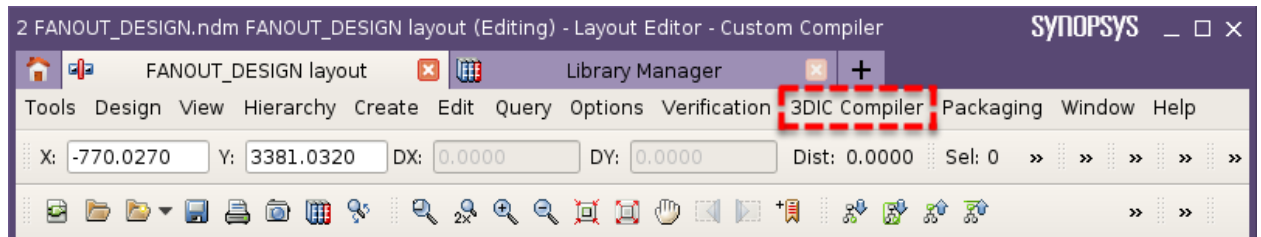
1. In the Custom Compiler Console, set the following preference:

```
db::setPrefValue xtDigitalImplementationTool -value 3dic
```

2. In the Layout Editor, choose **Tools > Digital Implementation > 3DIC Compiler**.



The **3DIC Compiler** menu appears on the design editor menu bar.



See [Software User Interface for Packaging Designs](#) for descriptions of the co-design menu commands and dialog boxes.

# 4

## Setting Up for Packaging Designs

---

*This chapter describes how to define packaging libraries and technology files.*

This chapter describes the following:

- [About Packaging Libraries and Technology Files](#)
- [Supported Shapes in Packaging Design Libraries](#)
- [Defining Packaging Libraries and Technology Files](#)

---

### About Packaging Libraries and Technology Files

Packaging designs depend on the following:

- **Packaging libraries:** Custom Compiler PyCell libraries that provide data and parameters to permit the creation and editing of special shapes specific to packaging designs, such as circle, teardrop, and oblong shapes. See [Supported Shapes in Packaging Design Libraries](#).
- **Packaging technology files:** Files that are provided by the foundry and contain global rules and constraints for packaging designs. Packaging technology rules, such as layer order, have priority over OA technology rules. Contact the foundry for information about the technology file.

To create a packaging design, you must first specify the location of the packaging library and technology file, and add the packaging library data to the `lib.defs` file. You use the Layout Editor **Packaging > Define Packaging Library** command for this task. See [Defining Packaging Libraries and Technology Files](#).

---

### Supported Shapes in Packaging Design Libraries

Packaging libraries consist of Custom Compiler PyCell libraries that define special shapes specific to packaging designs. Parameters in the PyCell libraries control the outcome of the shapes, such as layer, size, segment, and so on.

This topic describes the shapes and corresponding parameters that can be defined in the packaging libraries:

- [General Shape Parameters](#)
- [Circle and Teardrop Shapes](#)
- [Oblong Shapes](#)

---

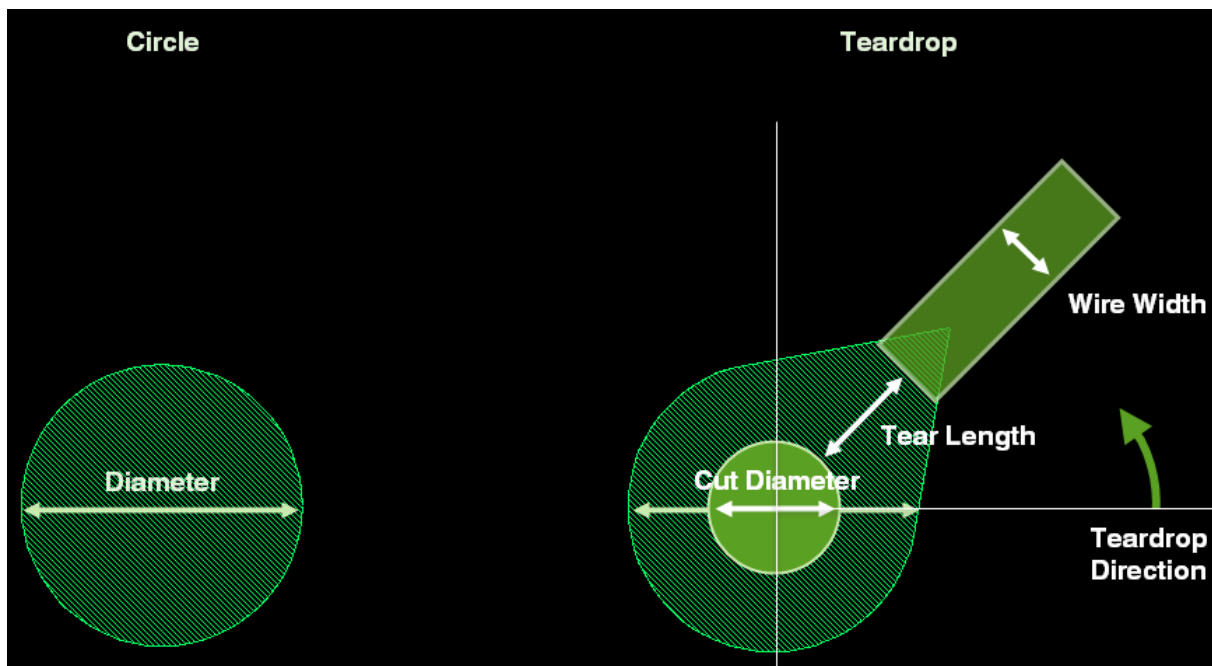
## General Shape Parameters

Packaging design shapes are defined by the following general parameters:

| Parameter    | Description                                                       |
|--------------|-------------------------------------------------------------------|
| numOfLayer   | Number of layers in the PCell.<br>Valid values: 1, 2, 3           |
| shapeType_%d | Shape type of the specific layer.<br>Valid values: circle, oblong |
| layerName_%d | Name of the specific layer-purpose pair (LPP).                    |
| piece_%d     | Number of points on the arc.<br>Valid value: >= 64                |

## Circle and Teardrop Shapes

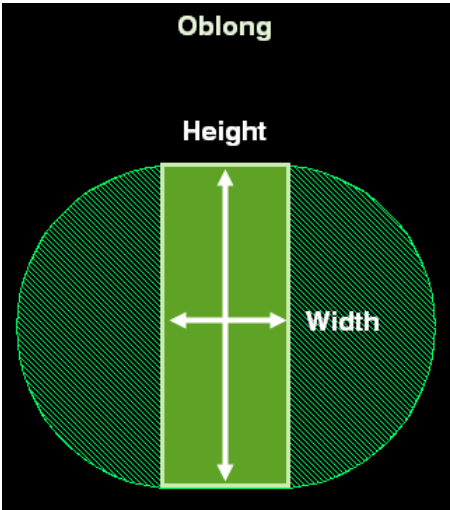
Teardrop shapes are derived from circle shapes. Circle and teardrop shapes are defined by the following parameters:



| Parameter                  | Description                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| diameter_%d                | Diameter of the circle.<br>Valid values: $\geq 10$                                                                                                                                                            |
| teardrop_%d                | Option to create a teardrop shape.<br>Valid values: true, false                                                                                                                                               |
| cutDiameter_%d             | Cut diameter. This parameter is available only if <code>teardrop_d%</code> is true.                                                                                                                           |
| wireWidth_%d               | Wire width. This parameter is available only if <code>teardrop_d%</code> is true.                                                                                                                             |
| teardrop_%d_tearLength     | Tear length. This parameter is available only if <code>teardrop_d%</code> is true.                                                                                                                            |
| teardrop_%d_directionAngle | Direction of the tear drop. The direction is counterclockwise and starts from the positive x-axis. This parameter is available only if <code>teardrop_d%</code> is true.<br>Valid values: 0 to 360, inclusive |

## Oblong Shapes

Oblong shapes are defined by the following parameters:

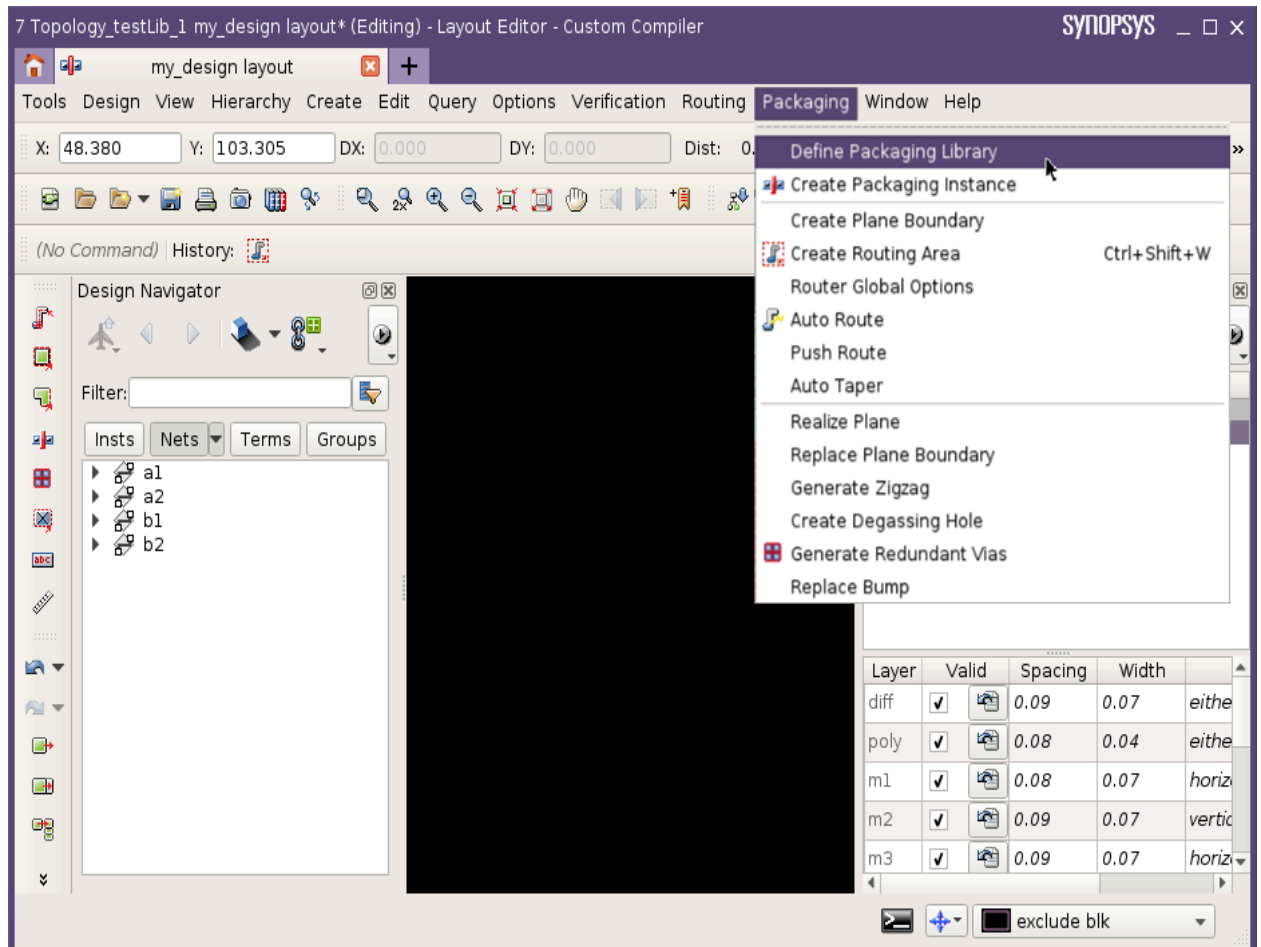


| Parameter        | Description                                                                             |
|------------------|-----------------------------------------------------------------------------------------|
| oblong_%d_width  | Width of the rectangle that fits within the oblong shape.<br>Valid values: >= 10        |
| oblong_%d_height | Height of the rectangle that fits within the oblong shape.<br>Valid values: true, false |

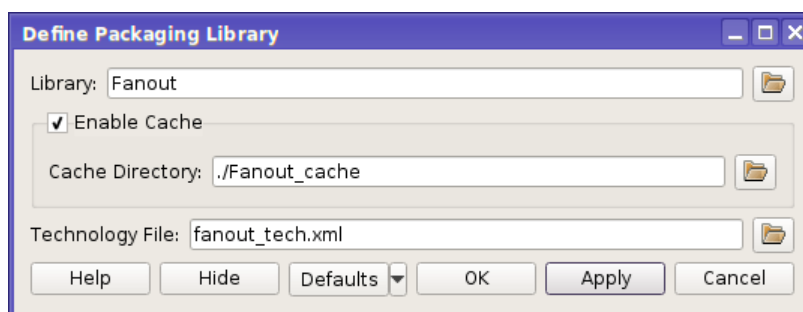
## Defining Packaging Libraries and Technology Files

To define the packaging library and the packaging technology file:

1. In the Layout Editor, choose **Packaging > Define Packaging Library**.



The **Define Packaging Library** dialog box appears.



2. Set the options in the **Define Packaging Library** dialog box.

| Field           | Description                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Library         | Path to the packaging library.<br>Click the file browser button to browse and select the path to the packaging library.                                                                                          |
| Enable Cache    | Enables PyCell caching for the packaging library.                                                                                                                                                                |
| Cache Directory | Path to the PyCell cache directory.<br><b>Note:</b><br>When you specify the path to the packaging library, the path to the cache directory is automatically set to <code>./packaging_library_name_cache</code> . |
| Technology File | Path to the packaging technology file.                                                                                                                                                                           |

3. Click **Apply** to add the specified packaging library data to the `lib.defs` file.

# 5

## Creating Packaging Objects

---

*This chapter describes how to use the Layout Editor Packaging commands to create objects in packaging designs.*

Before you start to route packaging designs, you might need to place component instances in the design. In addition, you must replace abstract pins with special patterns for packaging designs and create PG plane boundaries for power and ground supplies.

This chapter describes the following:

- [Creating Packaging Instances](#)
- [Replacing Microbumps](#)
- [Creating PG Plane Boundaries](#)

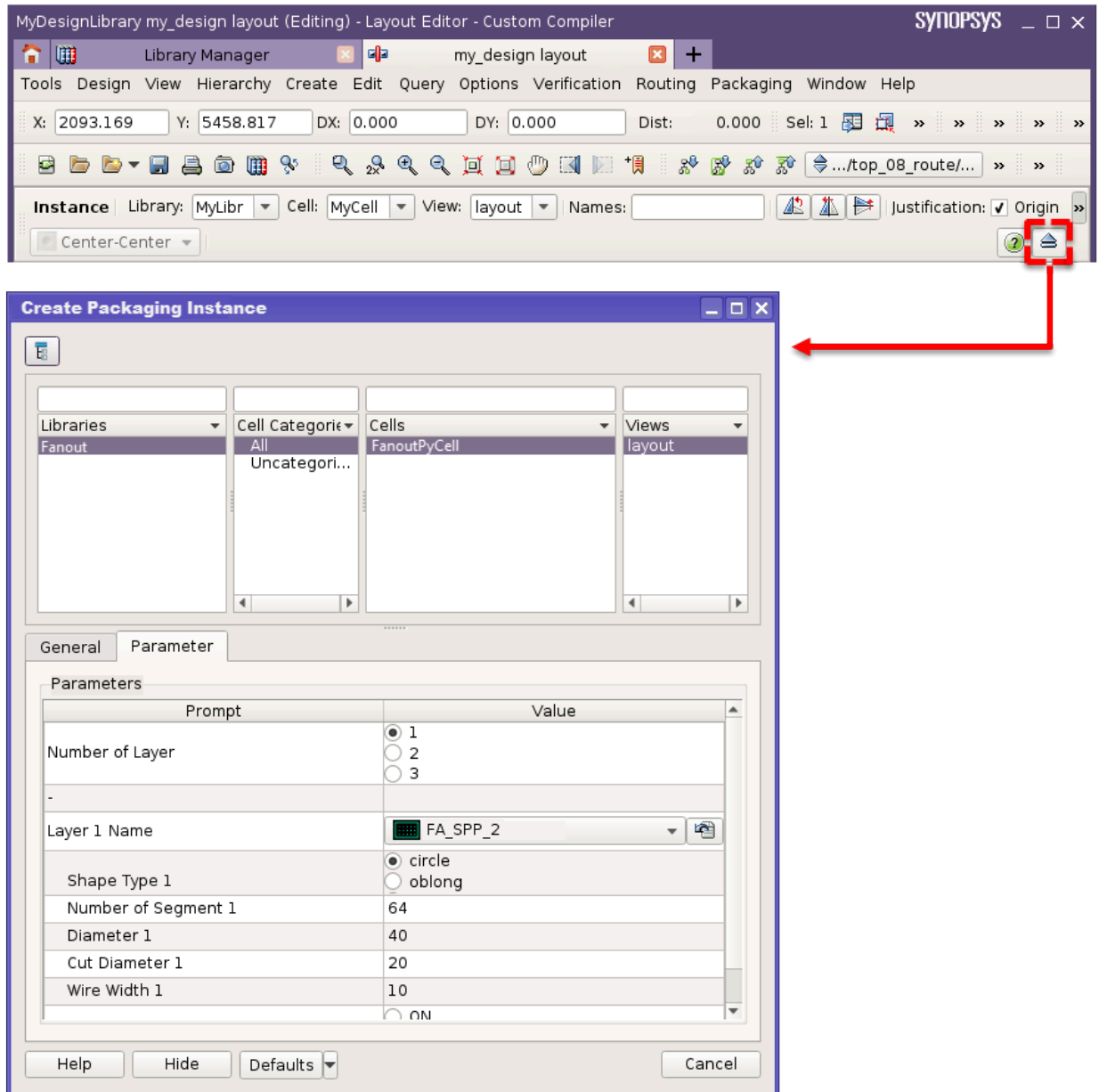
---

### Creating Packaging Instances

You use the Layout Editor **Packaging > Create Packaging Instance** command to create instances in packaging designs. The command, options, and procedure are the same as those in the **Create > Instance** command.

To create instances in packaging designs:

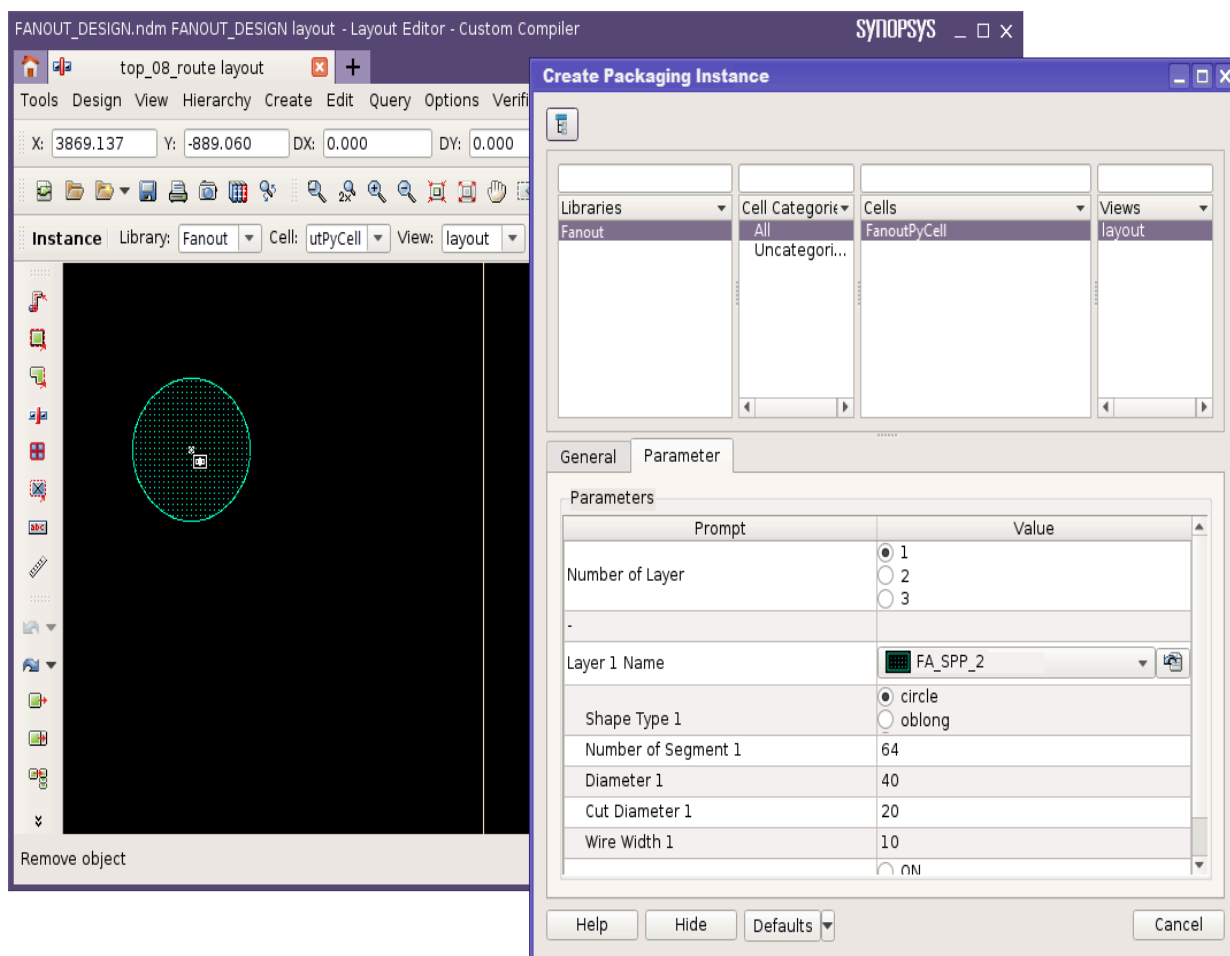
1. In the Layout Editor, choose **Packaging > Create Packaging Instance**. The **Create Packaging Instance** toolbar appears.
2. (Optional) On the **Instance** toolbar, click **Eject**  or press F3 to display the **Create Packaging Instance** dialog box. The dialog box contains additional options.



3. Set the options on the toolbar and in the dialog box.

Make sure the library and cell you specify are defined in the packaging library. For the remaining options, see the Instance Toolbar and Create Instance Dialog Box.

4. To place the instance, see Creating Instances.



## Replacing Microbumps

This topic describes the following:

- [About Microbump Replacement](#)
- [Preparing the Microbump Replacement Tcl Script](#)
- [Replacing the Abstract Pins](#)

### About Microbump Replacement

Packaging designs are developed using digital tools, such as the IC Compiler II tool, and the Custom Compiler tool. The IC Compiler II tool routes abstract pins in the design.

However, special patterns are required at pin locations for die-to-die routing in packaging designs. Therefore, you must replace the abstract pins with the special patterns at the pin locations before routing the design.

A [Tcl script](#) performs the pin (microbump) replacement procedure. You use the **Packaging > Replace Bump** command to specify the Tcl script file.

During the microbump replacement operation, the Tcl script:

- Searches for instances with the same name in the current design.
- If the instance name is not found, and if the name is in the `name[%d]` format, searches for any instance with `name%d`.
- If a matching instance is found, replaces the instance name with `packaging_library_name/InFOpyCell/layout`.

---

## Preparing the Microbump Replacement Tcl Script

The microbump replacement Tcl script must be an ASCII file that contains the following information:

```
LibraryName  
  
BlockName  
  
Instance_1_Name RoutingLayer RoutingDirection  
...  
Instance_n_Name RoutingLayer RoutingDirection
```

---

| Field           | Description                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------|
| LibraryName     | Name of the packaging design. This field is ignored during the microbump replacement process. |
| BlockName       | Name of the block. This field is ignored during the microbump replacement process.            |
| Instance_n_Name | Name of the instance.                                                                         |
| RoutingLayer    | Name of the routing layer.                                                                    |

| Field            | Description                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RoutingDirection | (Case-sensitive) Direction of the route, which is also used to determine the <a href="#">teardrop angle</a> . Valid values are: <ul style="list-style-type: none"><li>• DIR_E</li><li>• DIR_NE</li><li>• DIR_N</li><li>• DIR_NW</li><li>• DIR_W</li><li>• DIR_SW</li><li>• DIR_S</li><li>• DIR_SE</li></ul> |

## Replacing the Abstract Pins

To replace the abstract pins:

1. In the Layout Editor, choose **Packaging > Replace Bump**.

The **Replace Bump** dialog box opens.



2. Specify the **Routing Information File**, the [microbump replacement Tcl script file](#).
3. Click **OK** to start the replacement process.

## Creating PG Plane Boundaries

In packaging designs, power and ground (PG) supplies must be routed on boundary-based planes (layers). Plane boundaries are created on PG-plane layers to provide guidance during routing, which connects microbumps and C4 bumps to the net.

You can create PG plane boundaries using one of the following methods:

| Method                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Graphical user interface     | Using the <b>Create Plane Boundary</b> dialog box, you can specify the net, plane boundary priority, width threshold, and spacing constraints. You then draw the boundary on the layout canvas. Use this method if you want to create a small number of PG plane boundaries and want more precise control. See <a href="#">Creating PG Plane Boundaries Manually</a> .                                                                                  |
| Command-line interface       | Using the <code>SYNOPTSYS_PACKAGING_SUPPLY_PLANE_NET_LAYERS</code> environment variable, you only need to specify the net, layer, and priority information for multiple PG plane boundaries. The command generates the PG plane boundaries and determines the dimensions and locations. Use this method if you want to create multiple PG plane boundaries quickly. See <a href="#">Creating PG Plane Boundaries Using the Command-Line Interface</a> . |
| Automatically during routing | Turning on the <b>Auto Create Plane Boundary</b> option in the <b>Auto Route</b> dialog box lets the Packaging Design Router automatically do the following during the routing operation: <ol style="list-style-type: none"> <li>1. Create PG plane boundaries before routing starts.</li> <li>2. Assign routes to the plane boundaries during the routing operation.</li> </ol> See <a href="#">Setting Packaging Auto Router Options</a> .            |

## Creating PG Plane Boundaries Manually

Power and ground (PG) supplies must be routed on PG planes (layers) in packaging designs. This topic describes how to create individual PG plane boundaries manually using the graphical user interface. To use the command-line interface method, see [Creating PG Plane Boundaries Using the Command-Line Interface](#).

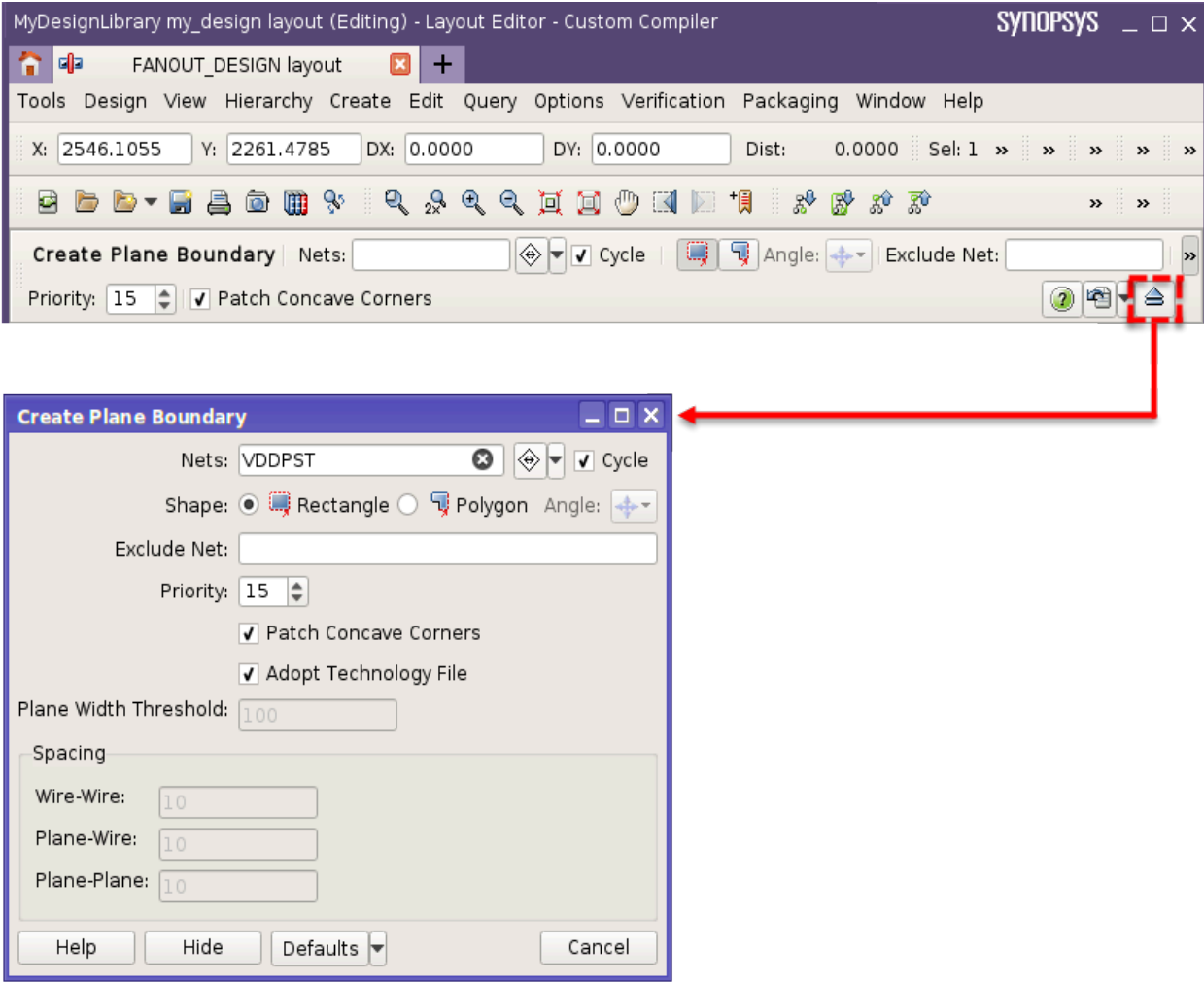
To create a PG plane boundary:

1. In the Layout Editor, choose **Packaging > Create Plane Boundary**.

The **Create Plane Boundary** toolbar appears. To view additional options, click **Eject**



to open the dialog box.



2. In the Design Navigator, select the power and ground nets to be included in the PG boundary plane.
3. Set the options in the **Create Plane Boundary** dialog box.

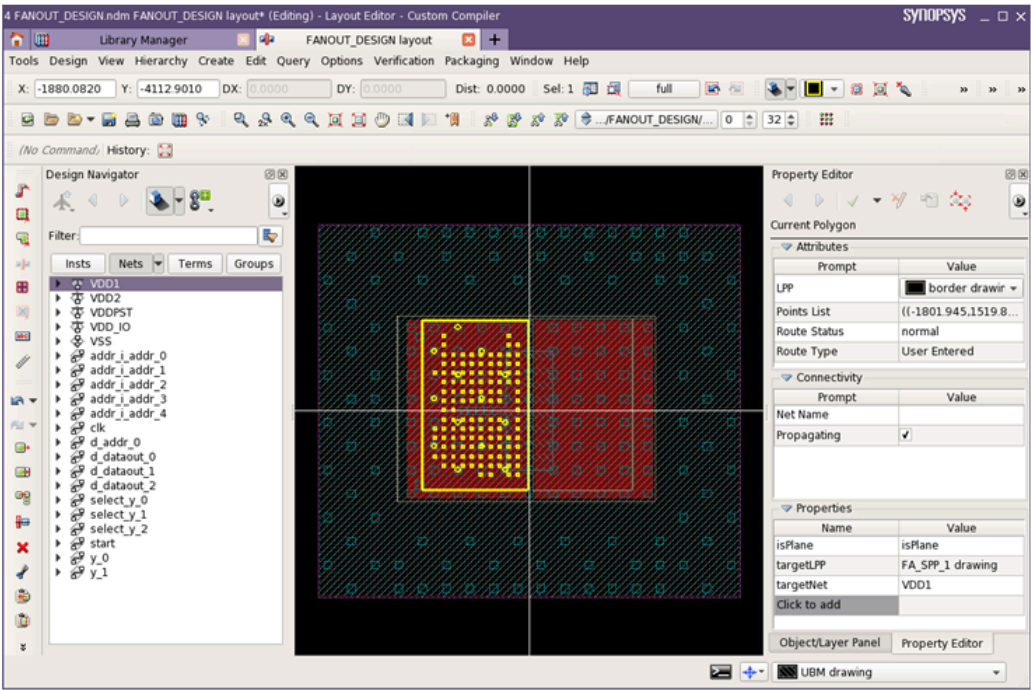
| Field | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nets  | <p>PG nets to be included in the PG plane boundary.</p> <p>If the nets you selected in the Design Navigator do no appear in the <b>Nets</b> box, click <b>Synchronize</b> .</p> <p>Select <b>Cycle</b> if you want to remove the first net name after the object is committed.</p> <p>Tcl preference: <code>leNetName</code>, <code>leCycleName</code></p> |

| Field       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shape       | Boundary-drawing mode: <b>Rectangle</b> or <b>Polygon</b> .<br>Tcl preference: <code>leCreatePlaneShapeType</code>                                                                                                                                                                                                                                                                                                                                                                                           |
| Angle       | ( <b>Shape: Polygon</b> only) Specifies the angle mode of the polygon.<br>Tcl preference: <code>leCreateAngleMode</code>                                                                                                                                                                                                                                                                                                                                                                                     |
| Exclude Net | Names of PG nets to be excluded automatically, because those nets have same-layer shapes and are included in another PG boundary plane. The exclusion ensures same-layer shapes on different planes comply with spacing rules, thus also enforcing plane-to-plane spacing rules.<br><div><div>VDD1 Plane Boundary, Exclude VDD2</div><div>VDD1 Plane Boundary, Include VDD2</div></div><br>Tcl preference: <code>lePackagingExcludeNet</code>                                                                |
| Priority    | Priority of the PG plane boundary. Larger numbers have higher priority over lower numbers.<br>When a supply net has multiple PG plane boundaries, the <b>Priority</b> setting determines which plane boundary to use for bump escapes. If multiple PG plane boundaries have the same priority, the layer that is closest to backside of the bump layer is used.<br>You can modify the priority using the <code>packagingPGPlanePriority</code> property in the Property Editor. See <a href="#">Step 6</a> . |

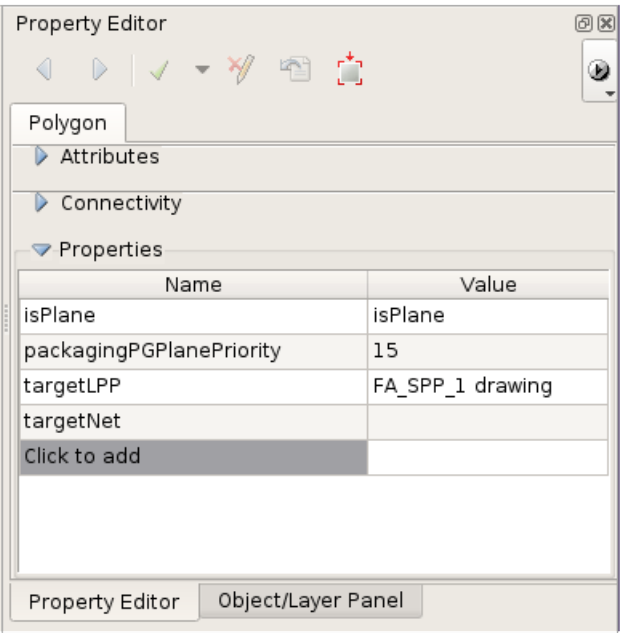
| Field                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patch Concave Corners | <p>Adds patches to rectangular PG plane boundaries, thus eliminating concave angles produced by the plane boundary creation.</p> <div><div>Before Patch</div><div>After Patch</div></div> <p>Tcl preference: <code>lePackagingPatchConcaveAcute</code></p>                                                                                                                                                                                                   |
| Adopt Technology File | <p>Uses only the packaging technology file and ignores all rules in the OA technology file.</p> <p>Tcl preference: <code>lePackagingAdoptTechnologyFile</code></p>                                                                                                                                                                                                                                                                                           |
| Width Max             | <p>Maximum width threshold. Any shape that has a width wider than this value is treated as a boundary plane. Otherwise, the shape is treated as a wire.</p> <p>Tcl preference: <code>lePackagingWidthMax</code></p>                                                                                                                                                                                                                                          |
| Spacing               | <p>Spacing specifications:</p> <ul style="list-style-type: none"><li>◦ <b>Wire-Wire:</b> Distance between two wires.</li><li>◦ <b>Plane-Wire:</b> Distance between the PG plane boundary and wire shapes.</li><li>◦ <b>Plane-Plane:</b> Distance between adjacent PG plane boundaries.</li></ul> <p>Tcl preference: <code>lePackagingWireWireSpacing</code>,<br/><code>lePackagingPlaneWireSpacing</code>,<br/><code>lePackagingPlanePlaneSpacing</code></p> |

4. In the Object/Layer Panel, set the active layer-purpose pair (LPP). The active LPP will become the target LPP, the layer on which the PG plane boundary will be **realized (placed)**.
5. On the layout canvas, draw the rectangle or polygon.

When you are finished, the PG plane boundary is created on the **border drawing** LPP, with proper spacing rules applied.



6. To view the PG plane boundary properties:
- a. On the layout canvas, select the PG plane boundary.
  - b. Display the Property Editor and scroll down to the **Properties** section.



The PG plane boundary has the following properties:

- **isPlane**: Identifies the shape as a PG plane boundary.
- **targetLPP**: LPP used for realizing the PG plane boundary.
- **targetNet**: Net used for realizing the PG plane boundary.
- **packagingPGPlanePriority**: Priority of the PG plane boundary, used for bump escapes during packaging design routing.

See [Realizing PG Plane Boundaries](#).

7. To edit the PG plane boundary, use the Layout Editor editing commands. See [Editing Objects - Basic Operations](#).

To route the packing design, see [Routing Packaging Designs](#). To report non-connecting vias, see [Reporting PG Plane Boundaries That Have Non-Connecting Vias](#).

---

## Creating PG Plane Boundaries Using the Command-Line Interface

Power and ground (PG) supplies must be routed on PG planes (layers) in packaging designs. This topic describes command-based PG plane boundary creation. To create individual PG plane boundaries manually using the graphical user interface, see [Creating PG Plane Boundaries Manually](#).

To create PG boundary planes using a Tcl command:

- In the Custom Compiler Console, type the following:

```
> set ::env(SYNOPSYS_PACKAGING_SUPPLY_PLANE_NET_LAYERS) "{net layer1  
layer2 ...}{net layer1 layer2...} ..."
```

Priority is based on the order in which the layers appear.

In the following example, for the VSS net, layer1 has higher priority than layer3.

```
> set ::env(SYNOPSYS_PACKAGING_PLANE_BOUNDARY_NET_LAYERS) "{VSS layer1  
layer3}{VDD layer2}"
```

To route the packing design, see [Routing Packaging Designs](#). To report non-connecting vias, see [Reporting PG Plane Boundaries That Have Non-Connecting Vias](#).

# 6

## Routing Packaging Designs

---

*The Custom Compiler Packaging Auto Router automates die-to-die and fanout signal routing in packaging designs. The router also generates a summary of routing results in the Console.*

This chapter contains the following topics:

- [About the Packaging Auto Router](#)
- [Workflow for Routing Packaging Designs](#)
- [Before You Start](#)
- [Setting Routing Constraints for Packaging Designs](#)
- [Accessing the Packaging Auto Router Commands](#)
- [Defining the Routing Area in the Packaging Design](#)
- [Setting Global Routing Options for Packaging Designs](#)
- [Setting Packaging Auto Router Options](#)
- [Starting the Packaging Auto Router](#)
- [Viewing the Packaging Routing Report](#)
- [Reporting PG Plane Boundaries That Have Non-Connecting Vias](#)
- [Fixing Routing Violations](#)
- [Troubleshooting Package Design Routing Errors](#)
- [Packaging Auto Route Toolbar and Dialog Box Reference](#)

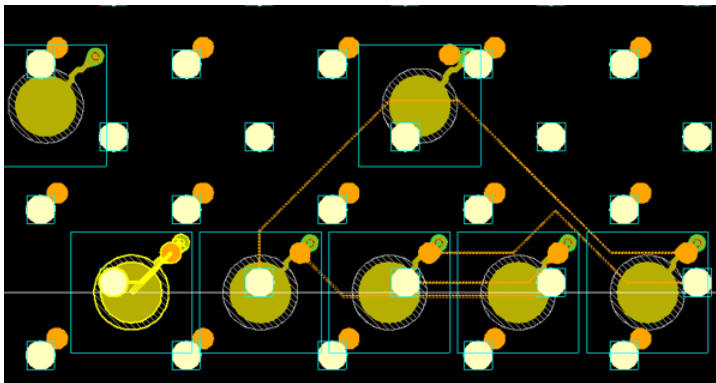
## About the Packaging Auto Router

You use the Custom Compiler Packaging Auto Router to route packaging designs. You can:

- Specify **regions** that you want to route in the packaging design.
- Select layer-purpose pairs (LPPs) to use in the regions.
- Add shielding to protect signal routes in die-to-die regions.
- Short route layers for short connections between neighboring microbumps.
- Route power and ground (PG) supplies.

The Packaging Auto Router:

- Automates die-to-die and fanout signal routing in packaging designs.
- Adds any-angle pin connections, 45° wiring, and zigzag patterns.
- Routes custom shapes, such as teardrops and oblong shapes.
- Adds teardrop-to-teardrop connections.
- Routes using multiple layers and changes layers automatically.
- Generates summary of routing results in the Console.



During routing, the Packaging Auto Router uses the layer information in the packaging technology file, including the layer order, to route the design. The Packaging Auto Router only adds routes on layers defined in the packaging technology. All other layers are ignored.

---

## Workflow for Routing Packaging Designs

The basic workflow for creating routes in packaging designs is as follows:

| Step |                                                | See...                                                                                                                                                                                                                                                                                                                   |
|------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Prepare for packaging design routing.          | <a href="#">Before You Start</a>                                                                                                                                                                                                                                                                                         |
| 2    | Set routing constraints for packaging designs. | <a href="#">Setting Routing Constraints for Packaging Designs</a>                                                                                                                                                                                                                                                        |
| 3    | Access the Packaging Auto Router commands.     | <a href="#">Accessing the Packaging Auto Router Commands</a>                                                                                                                                                                                                                                                             |
| 4    | (Optional) Define the routing area.            | <a href="#">Defining the Routing Area in the Packaging Design</a>                                                                                                                                                                                                                                                        |
| 5    | Set global routing options.                    | <a href="#">Setting Global Routing Options for Packaging Designs</a>                                                                                                                                                                                                                                                     |
| 6    | Set Packaging Auto Router options.             | <a href="#">Setting Packaging Auto Router Options</a>                                                                                                                                                                                                                                                                    |
| 7    | Start the Packaging Auto Router.               | <a href="#">Starting the Packaging Auto Router</a>                                                                                                                                                                                                                                                                       |
| 8    | Review the routing results and fix violations. | <ul style="list-style-type: none"><li>• <a href="#">Viewing the Packaging Routing Report</a></li><li>• <a href="#">Reporting PG Plane Boundaries That Have Non-Connecting Vias</a></li><li>• <a href="#">Fixing Routing Violations</a></li><li>• <a href="#">Troubleshooting Package Design Routing Errors</a></li></ul> |

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### Before You Start

Before you start the routing tasks, make sure you:

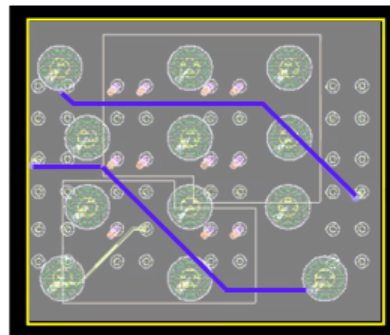
- Define a packaging library for the design. See [Setting Up for Packaging Designs](#).
- Create a plane boundary for the design. See [Creating PG Plane Boundaries](#).
- (Optional) Turn on the following command to minimize the way routes run across PG plane boundaries.

SYNOPSIS\_PACKAGING\_SUPPLY\_PLANE\_AWARE\_ROUTING\_EFFORT

**Plane Aware Routing = 0**



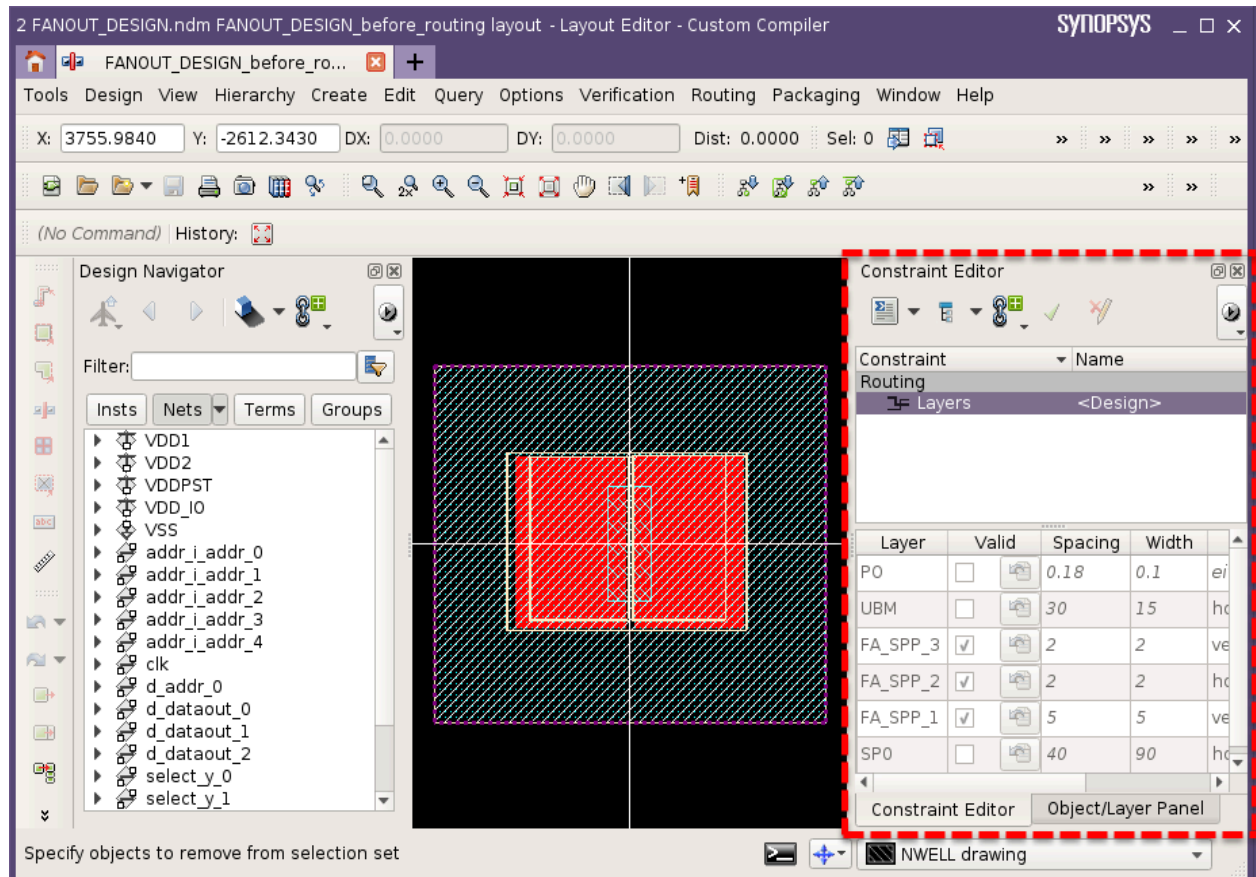
**Plane Aware Routing = 1**



---

## Setting Routing Constraints for Packaging Designs

Before you start routing packaging designs, you must set Routing Layers constraints to specify the valid routing layers. The Routing Layers constraint can be applied to an individual net or a group of nets.

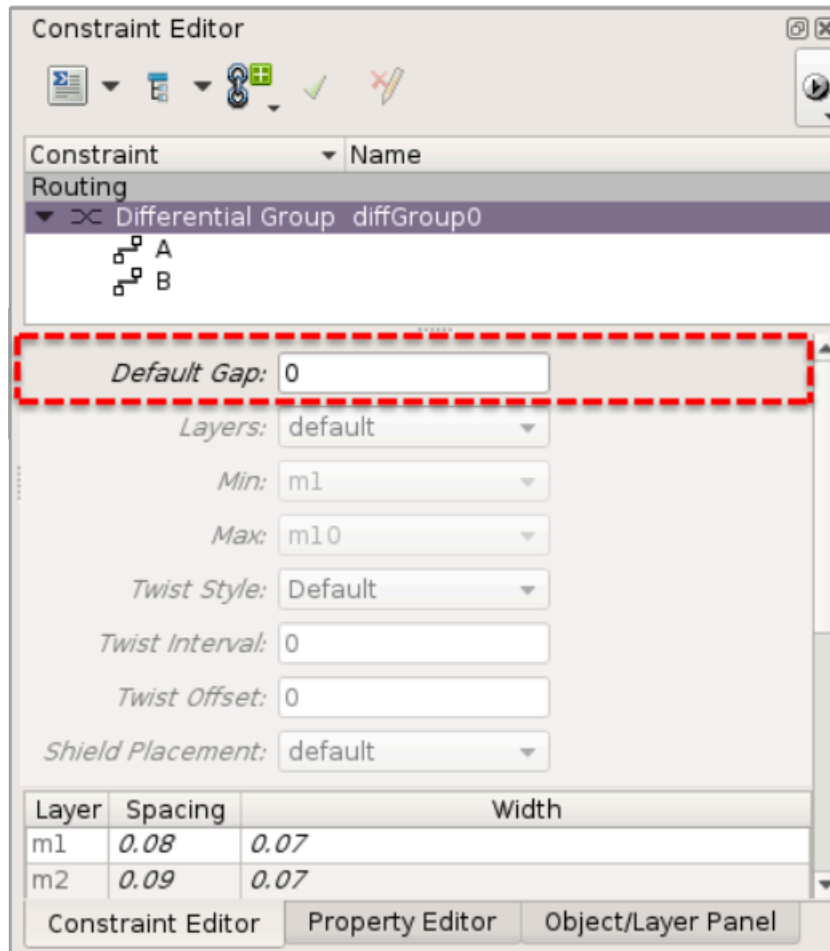


To set routing constraints:

1. Display the Constraint Editor.
2. Set one or more routing constraints.

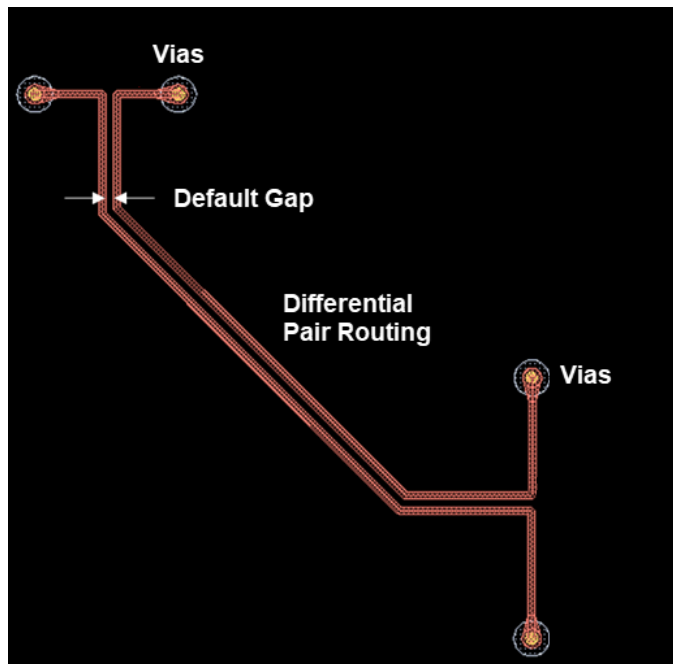
See the following topics for details:

- Creating Routing Constraints for general instructions.
- Layers Constraint for descriptions of the Routing Layers constraint parameters.
- Differential Group Constraint for descriptions of the Differential Pair constraint parameters. Note that you can only specify two nets and use only the **Default Gap** parameter for packaging designs. The minimum value you can specify for **Default Gap** is minSpacing.



The differential-pair routing begins and ends at a location that is midway between the vias of the two signal nets specified in the constraint. The routes follow a river-

style pattern, and the distance between the two paths is the Default Gap specified in the constraint.



---

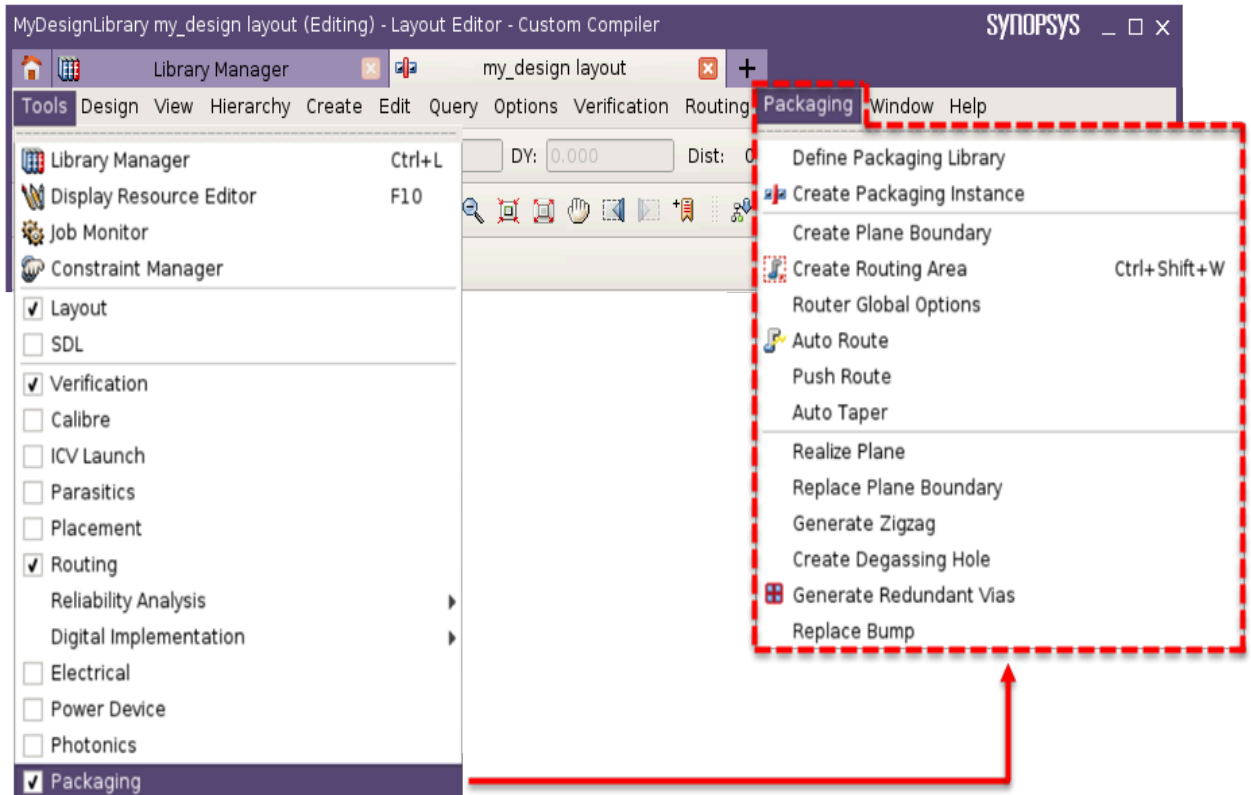
## Accessing the Packaging Auto Router Commands

The Packaging Auto Router commands allow you to define a region within the design for routing, set routing options, start the Auto Router, and fix violations.

To access the Packaging Auto Router commands:

- In the Layout Editor, choose **Tools > Packaging**.

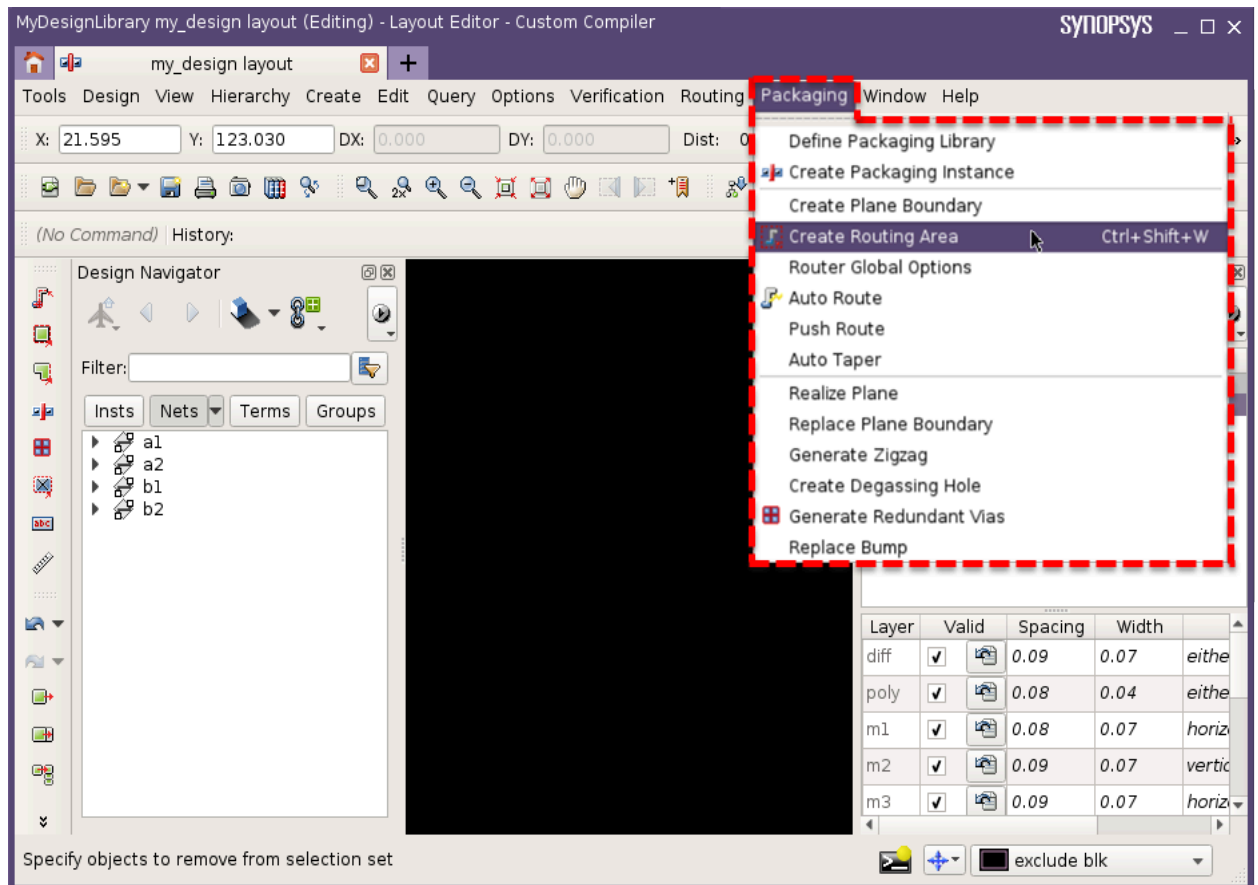
The **Packaging** menu appears on the menu bar. The routing commands are in the second section of the **Packaging** menu.



## Defining the Routing Area in the Packaging Design

To define a routing area in the design:

- In the Layout Editor, choose **Packaging > Create Routing Area**. For detailed instructions, see Defining a Routing Area.

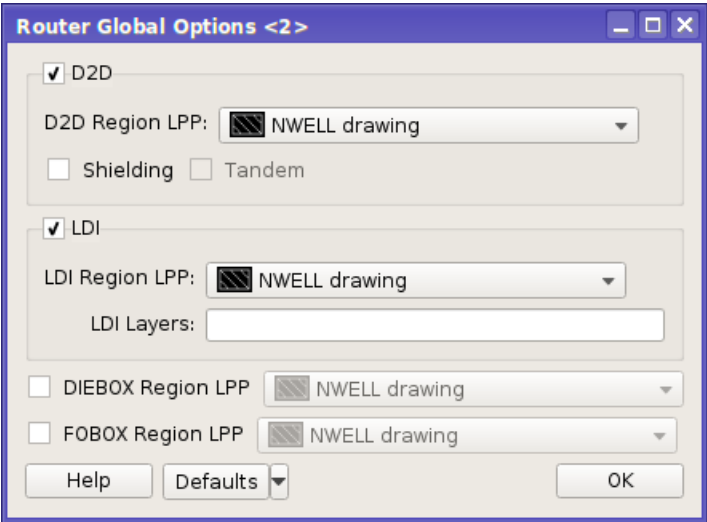


## Setting Global Routing Options for Packaging Designs

The Packaging Auto Router global options let you specify whether the design contains D2D, LDI, DIEBOX, and FOBOX regions, and the layers to use for routing in those regions.

To set the global routing options:

1. In the Layout Editor, choose **Packaging > Router Global Options**. The **Router Global Options** dialog box appears.

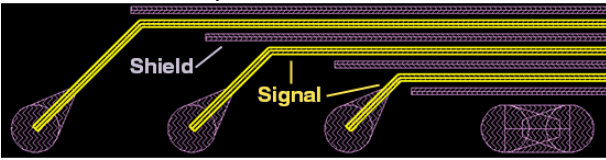


2. If the design contains [die-to-die regions](#), select **D2D** (`rtPackagingUseD2dRegionLpp`), and set the following:

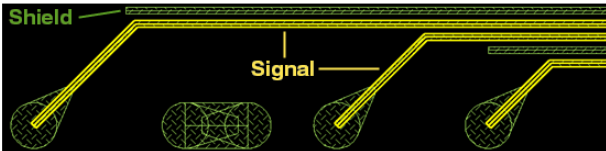
| Option         | Description                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| D2D Region LPP | Layer-purpose pair (LPP) that is used to define the die-to-die (D2D) connection region.<br>Tcl preference: <code>rtPackagingD2dRegionLpp</code> |

Shielding

Space permitting, adds floating shields between adjacent signal routes within the D2D region.



Otherwise, shield shapes are added where possible.



Tcl preference: `rtPackagingD2dUseShielding`

| Option | Description                                                                                                                                                                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tandem | <p>Spacing permitting, adds shields on layers above and below the signal route within the D2D region. The option is available only when you select <b>Shielding</b>.</p>  <p>Tcl preference: <code>rtPackagingD2dUseTandem</code></p> |

3. If the design contains [LDI regions](#), select **LDI** (`rtPackagingUseLdiRegionLpp`) and set the following:

| Option         | Description                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDI Region LPP | <p>Layer-purpose pair (LPP) that is used to define the local die interconnect (LDI) region.</p> <p>Tcl preference: <code>rtPackagingLdiRegionLpp</code></p>        |
| LDI Layers     | <p>Layers within the LDI region. Include the layers for the LDI frontside pads and LDI backside pads.</p> <p>Tcl preference: <code>rtPackagingLdiLayers</code></p> |

4. If the design contains [die box regions](#), select and set the following:

| Option            | Description                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DIEBOX Region LPP | <p>Layer-purpose pair (LPP) that is used to define the die box (DIEBOX) region.</p> <p>Tcl preferences: <code>rtPackagingUseDieboxRegionLpp</code>,<br/><code>rtPackagingDieboxRegionLpp</code></p> |

5. If the design contains [fanout box regions](#), select and set the following:

| Option           | Description                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FOBOX Region LPP | <p>Layer-purpose pair (LPP) that is used for the <a href="#">fanout box (FOBOX) region</a>.</p> <p>Tcl preferences: <code>rtPackagingUseFoboxRegionLpp</code>,<br/><code>rtPackagingFoboxRegionLpp</code></p> |

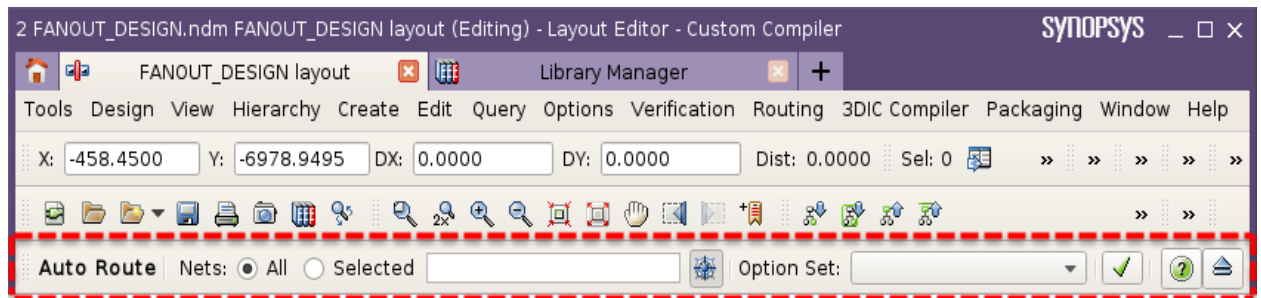
6. When you are finished, click **OK** to apply the settings.

## Setting Packaging Auto Router Options

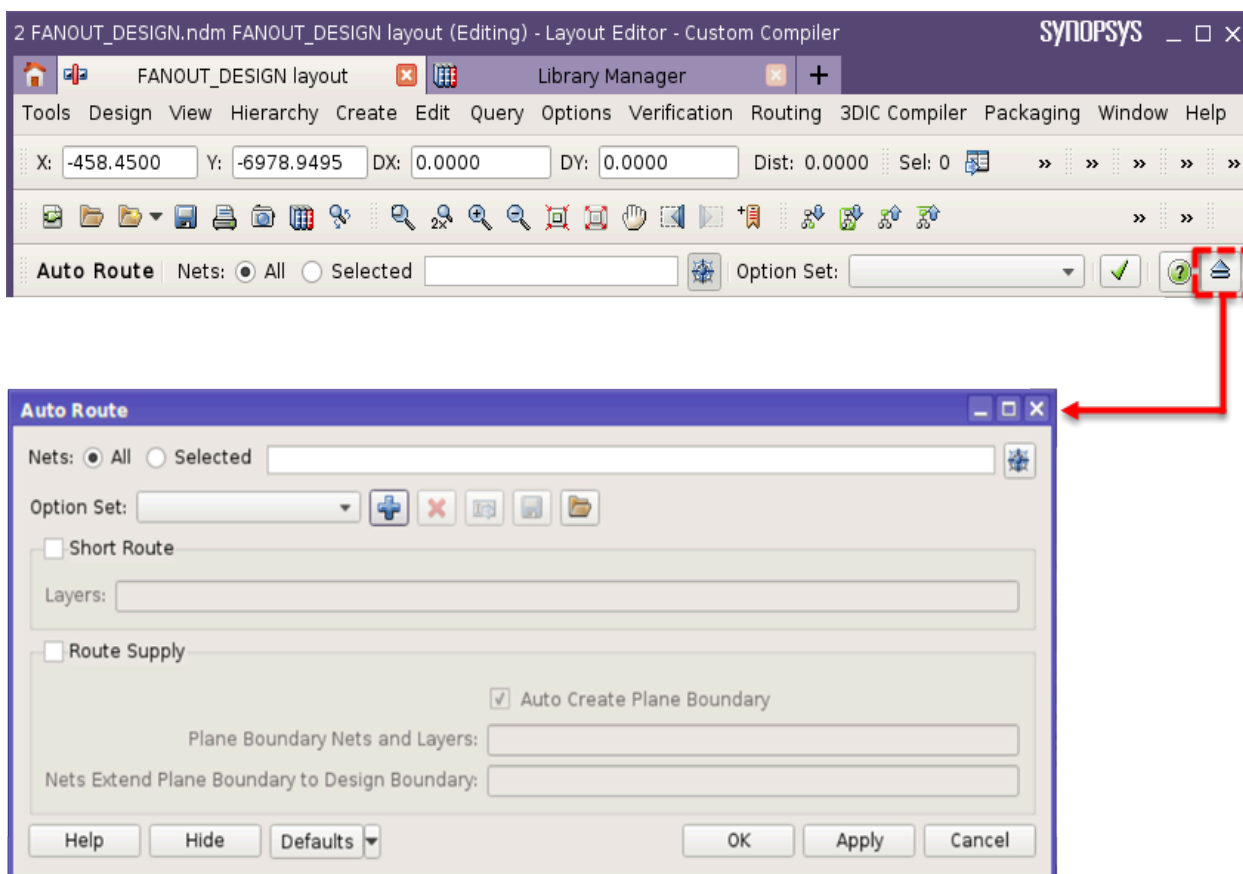
The **Packaging Auto Route** options lets you specify the nets to route, layers to use for short routes, and whether or not to route supply nets.

To set the Packaging Auto Router options:

1. In the Layout Editor, choose **Packaging > Auto Route**. The **Auto Route** toolbar appears.



2. On the **Auto Route** toolbar, do one of the following:
  - If you saved the **Packaging** options that you set previously, choose the **Option Set** you want to use. See [Managing Option Sets](#). To modify the **Packaging** settings, proceed to step 3. To start routing, see [Starting the Packaging Auto Router](#).
  - To set new **Packaging** options, click **Eject**  to open the **Auto Route** dialog box.



- Specify the **Nets** to be routed:
  - All:** All nets in the design.
  - Selected:** Specified nets in the design. Type the names of the nets in the **Nets** box, using space delimiters. Alternatively, select the nets in the Design Navigator.
- If you want to route neighboring microbumps (pins) on specified layers to reduce the number of escape vias, select **Short Route** and set the following:

| Option | Description                                                                                      |
|--------|--------------------------------------------------------------------------------------------------|
| Layers | Layers to be used for short routes.<br>Tcl preferences: <code>rtPackagingShortRouteLayers</code> |

Using Short Route can save routing resources. The Packaging Router only connects the microbumps using single any-angle straight paths.

Short routing requires the following:

- The net has neighboring microbumps in the DIEBOX region.
- The net is not a large-area supply net. (Large-area supply nets usually require every microbump to be escaped individually to optimize electrical results).

5. To route supply nets, select **Route Supply** and set the following:

| Option                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Route Supply                                  | Routes specified supply nets.<br>Tcl preference: <code>rtPackagingUseSupplyRoute</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Auto Create Plane Boundary                    | Automatically creates PG plane boundaries before routing starts, and assigns routes to the plane boundaries during the routing operation. The auto mode applies only to PG nets that do not have net-layer specifications.<br>For example, suppose the design contains the following PG nets: VSS, VDD, VDD1, and VDD2. You specify the following in Plane Boundary Nets and Layers: { VSS layer1 } { VDD layer2 }. The router will honor your specifications for VSS and VDD, and automatically assign the layer for VDD1 and VDD2.<br>See also <a href="#">Creating PG Plane Boundaries Manually</a> and <a href="#">Creating PG Plane Boundaries Using the Command-Line Interface</a> for other methods of creating PG plane boundaries. |
| Plane Boundary Nets and Layers                | Routes the specified supply nets using the corresponding PG plane boundaries. The syntax is:<br><code>{net1 layer1 layer2 ... layer n} {net2 layer1 layer2 ...} ...</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Nets Extend Plane Boundary to Design Boundary | Extends the PG plane boundaries to the design boundary and routes the supply nets.<br>This option is used only when you select <b>Auto Create Plane Boundary</b> , and you do not specify <b>Plane Boundary Nets and Layers</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

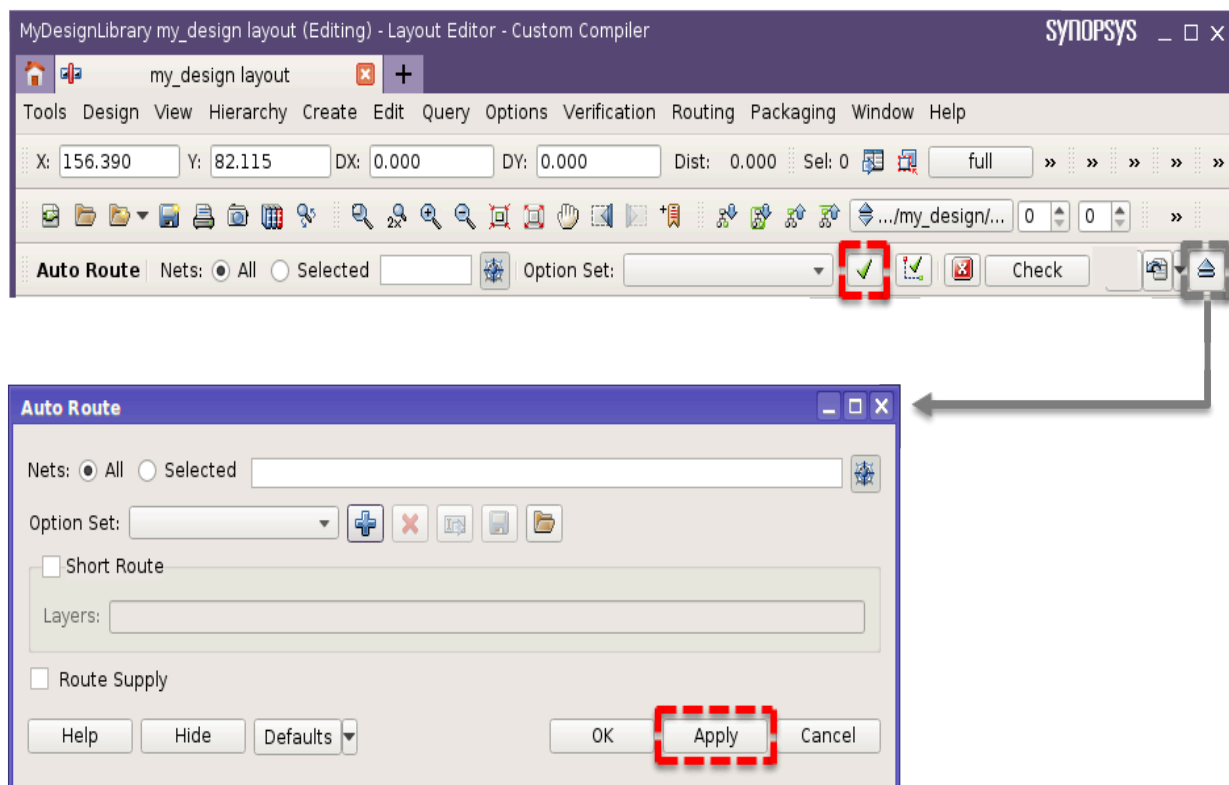
6. If you want to save the options and reuse them later, see Managing Option Sets.

## Starting the Packaging Auto Router

After you set all the routing options, you can start to route the packaging design.

To start routing the packaging design:

1. Use one of the following methods:
  - If you are in the **Auto Route** dialog box, click **Apply**.
  - On the **Auto Route** toolbar, click **Route** (  ).

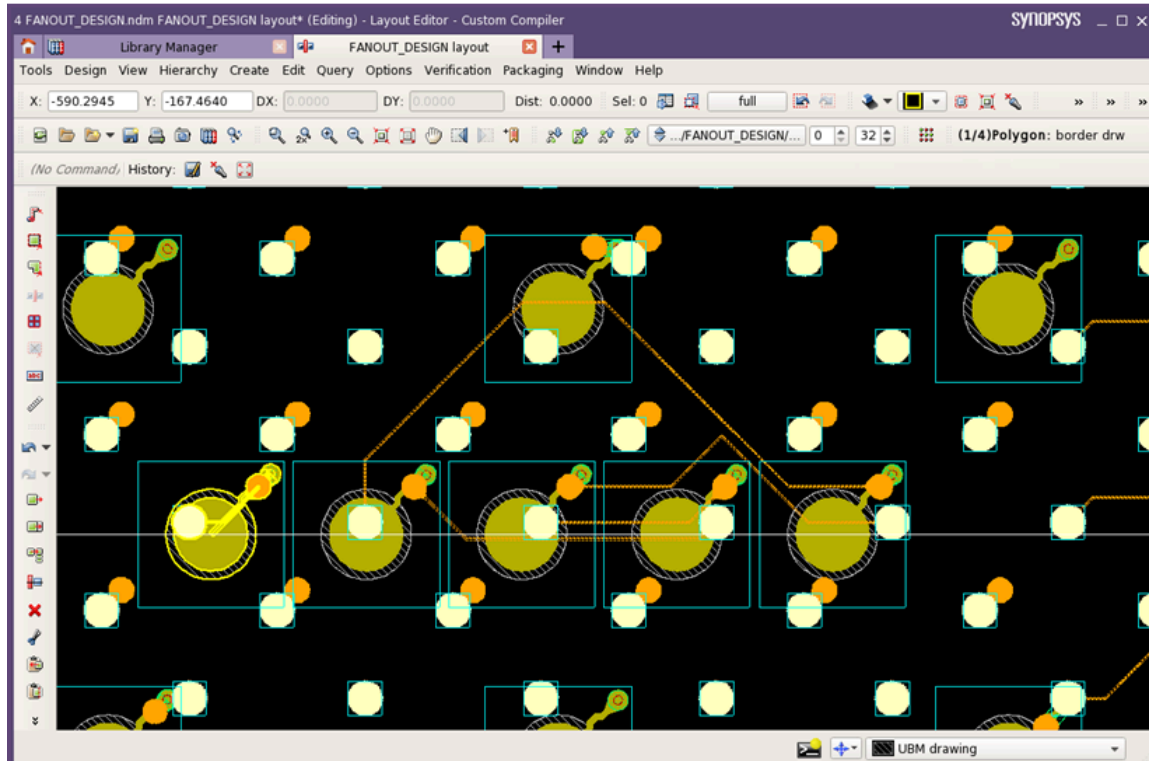


The router initializes, loads the settings, and starts the routing operation.

The Packaging Auto Router displays a progress bar. A message in the Console lets you know that the Auto Router is busy.

2. (Optional) To interrupt the routing operation and save the current routing state, press Ctrl+C or click **Stop** in the progress dialog box.

When the routing is finished, the routed design displays on the layout canvas. A [routing report](#) displays in the Console.



## Viewing the Packaging Routing Report

The packaging routing report provides a summary that lets you know the number of nets that were successfully routed and the nets that were not routed. The report also lets you know of any tear-length adjustments made, and reasons why teardrop shapes are not created.

When the Packaging Auto Router finishes routing, the Packaging Routing Report appears in the Console. The report consists of the following:

- [Routing Summary Sections](#)
- [Tear Length Adjustment Section](#)
- [Tear Drop Skipped Section](#)

```

Information: Packaging Routing Report
Information: *****
Information: * Selected Routed Unrouted Skipped *
Information: *      166      160          1          5 *
Information: *****
Information: * Unrouted Net                Pin
Information: * NetA                      NetA-PAD
Information: *****
Information: * Skipped Net                Reason
Information: * NetB                      Invalid pin count. [1 pin]
Information: * NetC                      Fully routed.
Information: * NetD                      Has Pre-routed shapes.
Information: * NetE                      Misaligned pins in LDI.
Information: * NetF                      Same channel net NetD has pre-routed shapes.
Information: *****
Information: Tear Length Adjustments
Information: *****
Information: * Bump Instance | Layer | Min Length | Adjusted Length *
Information: *      BUMP_1 | CU_PPI_1 |      10.00 |          31.50 *
Information: *****
Information: Tear Drop Skipped (Tear length <= via enclosure or wire width >= diameter.)
Information: *****
Information: * Instance | Layer | Diameter | Wire Width *
Information: *      BUMP_2 | CU_PPI_1 |      50.00 |          78.00 *
Information: *****
Information: *****
Information: * Die-to-Die Region Length Matching Results *
Information: *****
Information: * Tolerance 0.1%
Information: * G(length: 99.99) is 0.01% mismatch with I(length: 100.0) on layer CU_PPI_2 (INFO)
Information: * H(length: 99) is 1% mismatch with I(length: 100.0) on layer CU_PPI_2 (WARNING)
Information: * Total Mismatch Net Count: 2 (1 out of all exceed 1%)
Information: *****

```

---

## Routing Summary Sections

The first three sections of the report summarizes the routing results with general statistics:

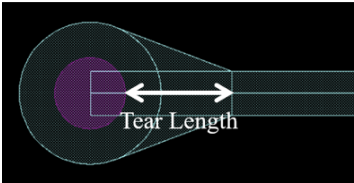
- The number of nets that were selected for routing
- The number of nets that were routed
- The number of unrouted nets
- For unrouted nets, the pins that were not connected during routing
- The number of nets that were skipped
- For skipped nets, the reasons the specific nets were skipped

See [Skipped Nets](#).

---

## Tear Length Adjustment Section

During routing, tear lengths are automatically extended if they do not meet the minimum tear length value in the packaging technology.



The Tear Length Adjustment section lets you know which tear lengths were extended by listing the corresponding microbumps:

| Entry           | Description                                           |
|-----------------|-------------------------------------------------------|
| Bump Instance   | Microbump instance name.                              |
| Layer           | Layer on which the teardrop shape is created.         |
| Min Length      | Minimum tear length specified in the technology file. |
| Adjusted Length | Tear length after the extension.                      |

## Tear Drop Skipped Section

Teardrop shapes are not created during routing for the following reasons:

- Tear length  $\leq$  via enclosure
- Wire width  $\geq$  teardrop diameter

The Tear Drop Skipped section lets you know which teardrop shapes were not created by listing the corresponding microbumps:

| Entry      | Description                                            |
|------------|--------------------------------------------------------|
| Instance   | Microbump instance name.                               |
| Layer      | Layer on which the teardrop shape is created.          |
| Diameter   | Diameter of the teardrop shape.                        |
| Wire width | Width of the wire to be covered by the teardrop shape. |

## Reporting PG Plane Boundaries That Have Non-Connecting Vias

After routing the packaging design, you can check for non-connecting vias in PG plane boundaries.

To check the number of PG plane boundaries that have at least one non-connecting via:

1. Type the following command in the Custom Compiler Console:

```
> rt::pkg::reportSupplyPlaneIntegrity
```

The Packaging Router Supply Plane Integrity Report appears in the Console.

```
Information:
*****
Information: Packaging Router Supply Plane Integrity Report
Information:
*****
Information: * 1 supply plane is broken
Information:
*****
Information: * Supply      Layer      Location
Crossing signal
Information: * VDD2        FA_SPP_1   (524.364000, 619.698000)    B
Information:
*****
```

2. Review the report.

| Column Title    | Description                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------|
| Supply          | Supply net name.                                                                                                        |
| Layer           | PG plane boundary layer name.                                                                                           |
| Location        | Location of the PG plane boundary.                                                                                      |
| Crossing signal | Signal routes that cross the PG plane boundary. To minimize the plane-crossings, see <a href="#">Before You Start</a> . |

3. Open the Error Viewer Assistant to view additional information. In the Error Viewer, click the plane-crossing error. The viewport zooms in to the region containing the error.

---

## Fixing Routing Violations

After routing the packaging design, you can use the **Push Route** and **Auto Taper** commands to automate route adjustment operations and fix DRC violations. Although the operations are automated, the commands allow you to control the precision of the fixes in your designs.

This topic describes the following:

- [Before You Start](#)
- [Fixing Spacing Violations](#)
- [Fixing Tapering Violations](#)

---

### Before You Start

Before using the **Push Route** and **Auto Taper** commands, make sure you remove realized PG plane boundaries.

The **Push Route** command cannot fix violations when PG plane boundaries are realized. In addition, PG plane boundaries must be re-realized after route modifications are made.

To remove realized PG plane boundaries, see [Deleting Objects](#).

### See Also

- [Creating PG Plane Boundaries](#)
- [Realizing PG Plane Boundaries](#)

---

## Fixing Spacing Violations

The **Push Route** command lets you move (or push) route shapes away from selected shapes or nets to comply with spacing rules in the packaging technology file. You use **Push Route** when making postroute adjustments.

This section describes the following:

- [About the Push Route Command](#)
- [Pushing Routes](#)
- [Viewing Push Route Reports](#)

### About the Push Route Command

The **Push Route** command moves (or pushes) violating shapes away from the selected shape or net to resolve spacing violations. Because you select the shapes or nets within an area that contains the violations, you control the precision of the fixes in the design.

This topic describes the following:

- [How Push Route Works](#)
- [When Vias Are Pushed](#)
- [Unmovable Objects](#)
- [Shapes and Nets That Cannot Be Pushed](#)

### How Push Route Works

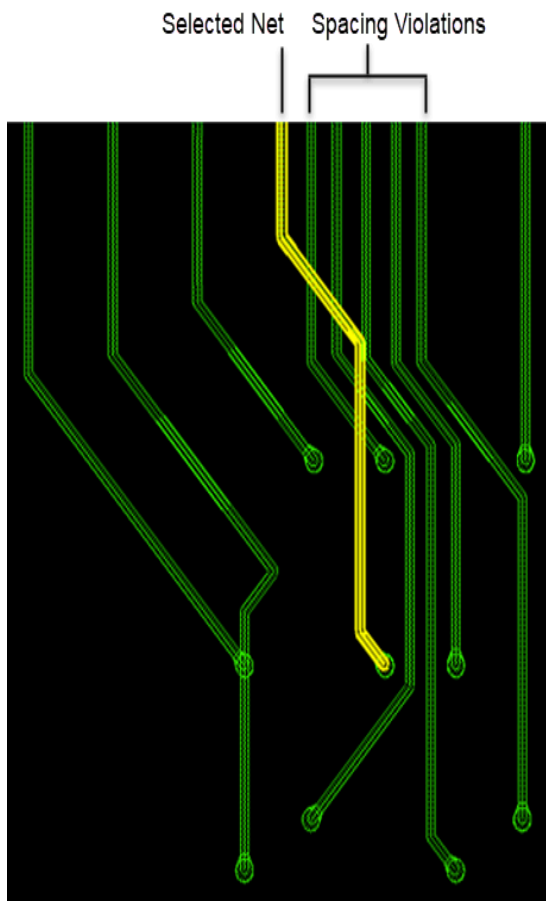
When using **Push Route**, you first select one or more route shapes or nets. During the operation, the command:

- Keeps the selected shapes or nets static.
- Uses the rules in the packaging technology file and non-default rules in the Constraint Editor to check for violations.
- Moves (or pushes) only the following neighboring shapes that are causing spacing violations.
  - Paths
  - Vias or bumps
  - Tapered shapes (see also [Fixing Tapering Violations](#))

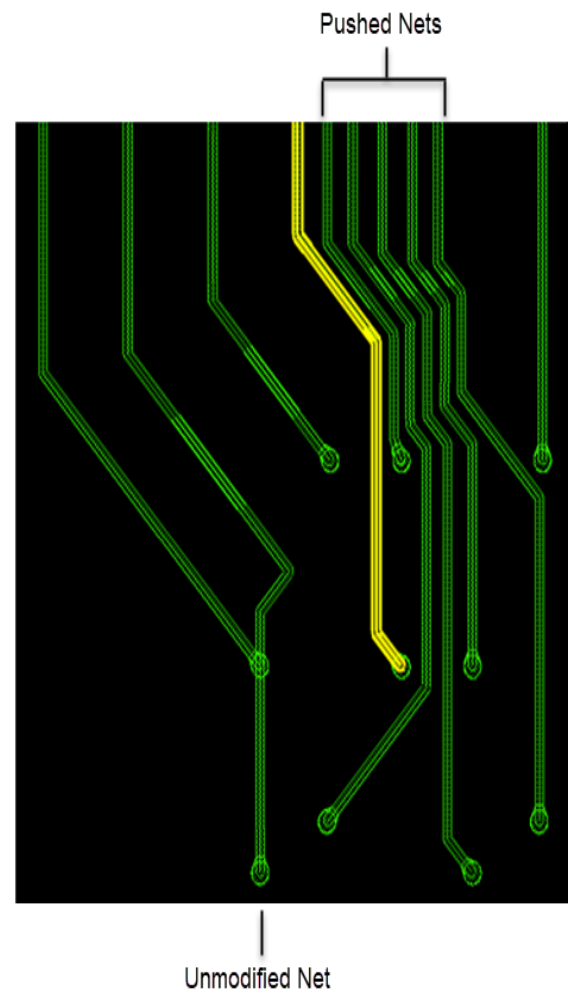
Neighboring shapes and nets that do not cause violations are not modified.

- Retains connectivity of the selected and pushed shapes or nets.

Before



After



The shapes or nets you select can impact the results of the operation. The following example shows two possible net selections and the results of their selections.

Before



After: Net A Selection Result



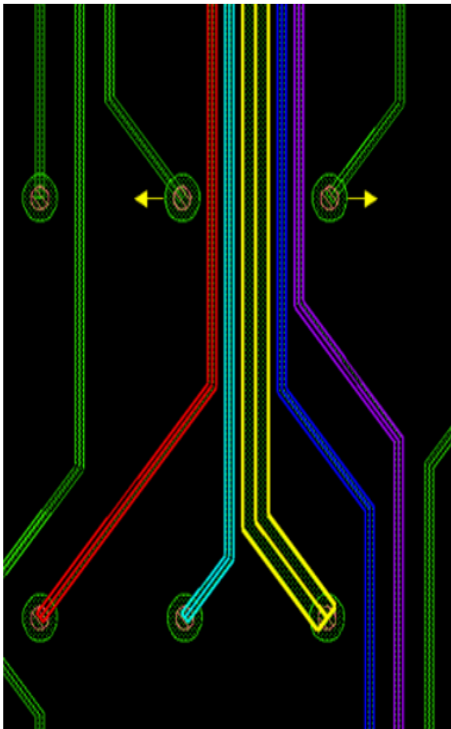
After: Net B Selection Result



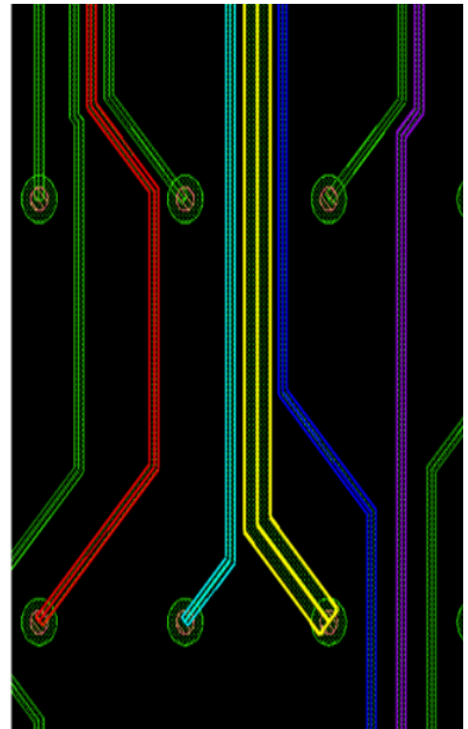
### When Vias Are Pushed

**Push Route** has a **Push Via** option that lets you specify whether vias are fixed or can be pushed during the operation. Allowing the vias to be pushed can result in fewer route detours in congested parts of the design.

☒ Push Via



☐ Push Via



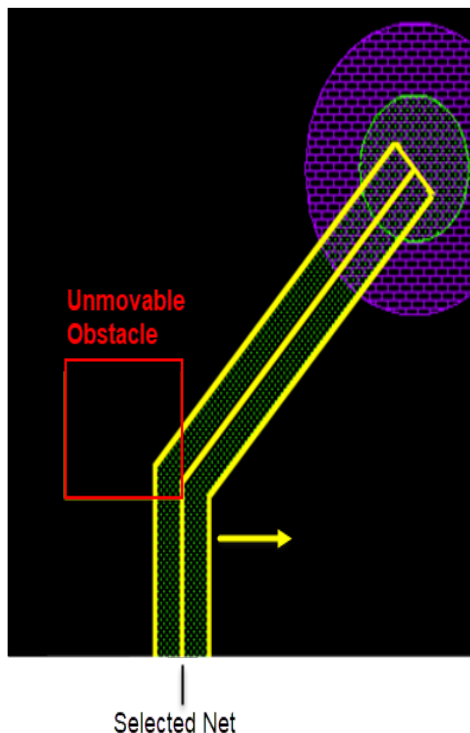
**Push Route** can also push stacked vias as follows:

- All vias in the stack are pushable, except:
  - If one via in the stack is fixed, the entire via stack is not pushable.
  - If the stacked vias are connected to bumps, which are considered unmovable, the via stack is not pushable.
- All vias are pushed together to the same target location. However, if one of the vias is unable to move to the target location, the entire via stack is not pushed.

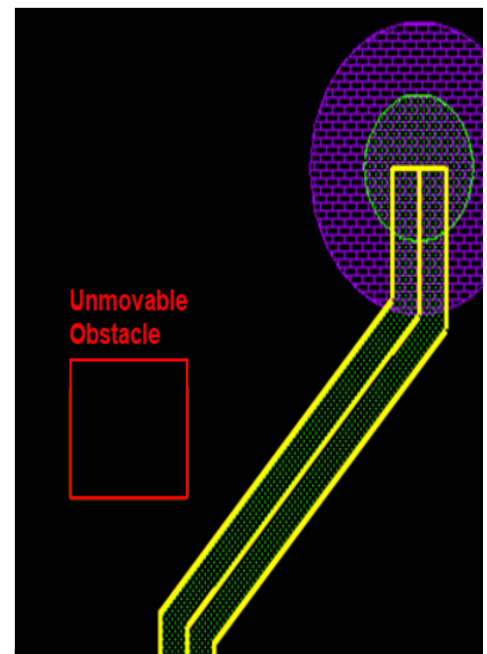
### Unmovable Objects

If the selected shapes or nets are adjacent to unmovable objects that are causing violations, **Push Route** will push the selected shapes or nets instead.

Before: Selected Net With Unmovable Obstacle



After: Selected Net is Pushed



Unmovable objects include:

- Bumps
- Routing blockages
- Rectangles or polygon shapes

- Path segments
- Paths that have the **Route Status: fixed** or **Route Status: locked** property  
See Viewing and Editing Route Shape Properties.
- Vias that are connected to bumps
- Vias, when the **Push Via** option is not selected
- Vias and tapered shapes, when the PCell instances that represent the vias and tapered shapes have the **Placement Status: fixed** or **Placement Status: locked** property  
See Viewing Design Object Properties.

### Shapes and Nets That Cannot Be Pushed

**Push Route** works only with shapes that are created by the Packaging Auto Router, or shapes that have the **Custom Route Type: Taper** or **Custom Route Type: Via** property. See Viewing Design Object Properties.

Because multi-branch routes that contain Steiner points are created manually or are in imported designs created by third-party tools, they cannot be pushed. However, you can select multi-branch routes and push neighboring violating shapes or nets.

## Pushing Routes

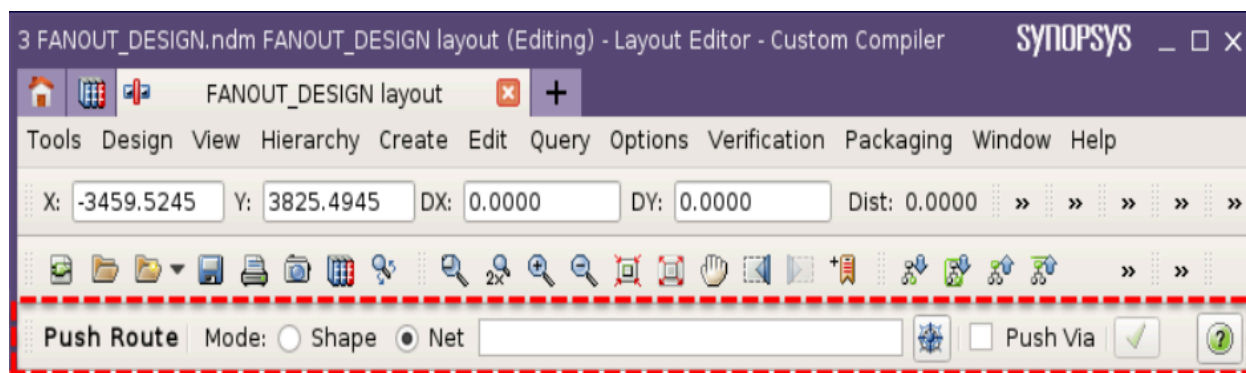
You use the **Push Route** command to move violating shapes away from selected shapes or nets to resolve spacing violations. Available options allow you to run the operation on one or more shapes or nets.

Before you start, review the [prerequisite information](#).

To use **Push Route** to fix spacing violations:

1. In the Layout Editor, choose **Packaging > Push Route**.

The **Push Route** toolbar appears.



2. On the **Push Route** toolbar, select the **Mode**:

| Argument | Description                               |
|----------|-------------------------------------------|
| Shape    | Lets you select shapes for the operation. |
| Net      | Lets you select nets for the operation.   |

3. (Optional) Select **Push Via** if you want to allow vias to be moved during the operation.

Allowing the vias to move can result in fewer route detours in congested parts of the design. See [About the Push Route Command](#).

4. (**Net** mode only) Specify the net names or net group name using one of the following methods:
- In the Design Navigator, select the nets or net group.
  - On the **Push Route** toolbar, type the net names or net group name in the box.

5. (**Shape** mode only) On the layout canvas, select the shapes.

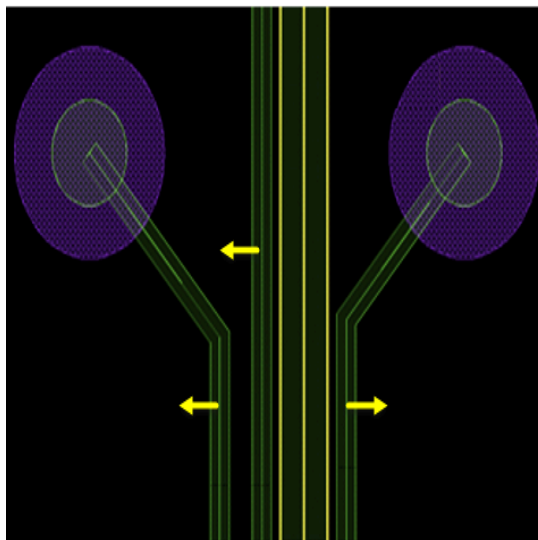
The shapes you can select are:

- Paths
- Rectangles
- Polygons
- Vias or bumps
- Layer blockages

6. On the **Push Route** toolbar, click **Apply** .

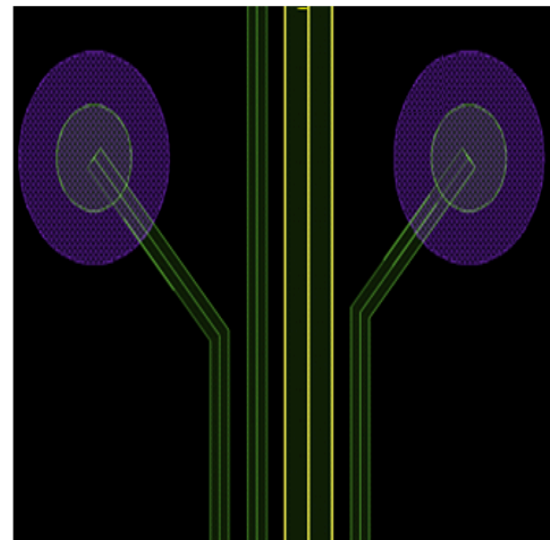
The violating shapes around the selected shapes or nets are moved.

Shape or Net Selection



Selected Shape or Net

Push Route Result



The operation can take longer than expected if the command encounters a lot of DRC errors in especially congested areas of the design.

- You can press Ctrl+C to interrupt the operation.
- The operation stops if it exceeds the timeout limit specified by the `rtPackagingPushRouteTimeout` Tcl preference.

### See Also

- [Creating PG Plane Boundaries](#)
- [Realizing PG Plane Boundaries](#)
- [Viewing Push Route Reports](#)
- [Packaging Auto Router Error Messages](#)

## Viewing Push Route Reports

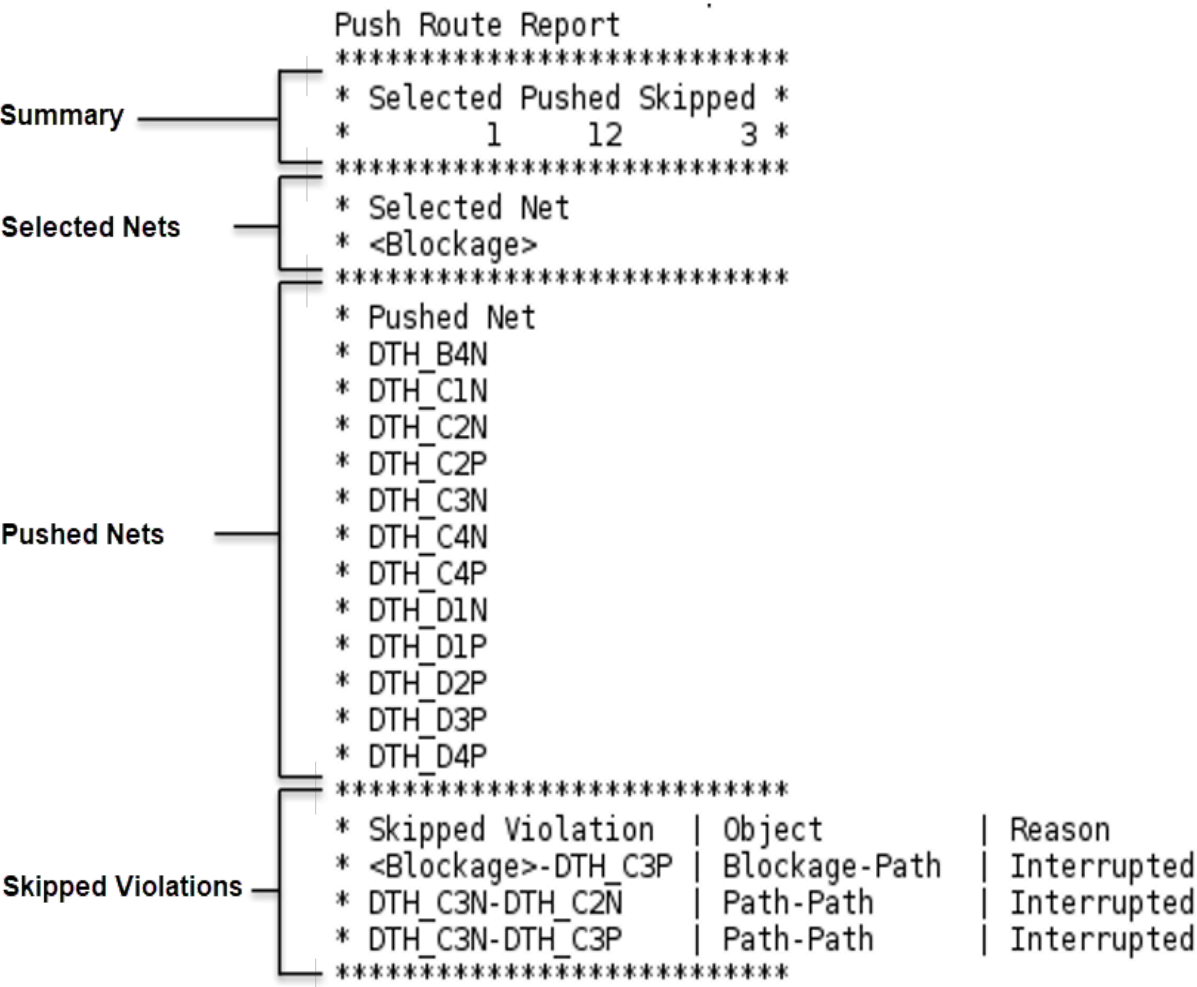
When the Push Route operation is finished, or when the operation is interrupted, the Push Route Report appears in the Console. You can review the information to determine whether the operation completed successfully or what you need to do to resolve errors.

To view the Push Route Report in the Console:

- 1. In the Layout Editor, display the Console tab or the Console assistant.
- 2. Scroll down to locate and review the [Push Route Report](#).
- 3. If shapes or nets were not pushed:
  - a. Review the [reasons](#) why they were not pushed.
  - b. Resolve the issues if needed.

**Push Route Report Components**

The Push Route report consists of the following sections:



| Report Section                     | Description                                                                                                                                                                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary                            | Displays the following statistics: <ul style="list-style-type: none"> <li>• <b>Selected:</b> The number of selected nets.</li> <li>• <b>Pushed:</b> The number of nets that were pushed to resolve spacing violations.</li> <li>• <b>Skipped:</b> The number of nets that were not pushed.</li> </ul> |
| Selected nets                      | Lists the names of the selected nets.                                                                                                                                                                                                                                                                 |
| Pushed nets                        | Lists the names of the pushed nets.                                                                                                                                                                                                                                                                   |
| <a href="#">Skipped violations</a> | Lists the following for the nets that were not pushed, and therefore still have violations: <ul style="list-style-type: none"> <li>• Names of the nets that were not pushed.</li> <li>• Types of objects of the skipped nets.</li> <li>• Reason for not pushing the net.</li> </ul>                   |

### Skipped Violation Section

The **Skipped Violation** section lets you know the reasons why the listed nets were not pushed:

| Object 1         | Object 2            | Not Pushed Because...                                                                                                                                                                |
|------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bump or blockage | Bump or blockage    | Bumps and blockages cannot be pushed. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                                                                                    |
| Bump or blockage | Fixed routing shape | Bumps and blockages cannot be pushed. Fixed routing shapes cannot be pushed. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                                             |
| Bump or blockage | Tapered shape       | Bumps and blockages cannot be pushed. Tapered shapes cannot be pushed to a different location. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                           |
| Bump or blockage | Via                 | Bumps and blockages cannot be pushed. The <b>Push Via</b> option is not selected. Try to select the <b>Push Via</b> option and try again. See <a href="#">When Vias Are Pushed</a> . |

| Object 1            | Object 2            | Not Pushed Because...                                                                                                                                                                 |
|---------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bump or blockage    | Via on bump         | Bumps and blockages cannot be pushed. The via is connected to a bump, which cannot be pushed. See <a href="#">When Vias Are Pushed</a> .                                              |
| Fixed routing shape | Fixed routing shape | Fixed routing shapes cannot be pushed. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                                                                                    |
| Fixed routing shape | Taper               | Fixed routing shapes cannot be pushed. Tapered shapes cannot be pushed to a different location. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                           |
| Fixed routing shape | Via                 | Fixed routing shapes cannot be pushed. The <b>Push Via</b> option is not selected. Try to select the <b>Push Via</b> option and try again. See <a href="#">When Vias Are Pushed</a> . |
| Fixed routing shape | Via on bump         | Fixed routing shapes cannot be pushed. The via is connected to a bump, which cannot be pushed. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                            |
| Taper               | Taper               | Tapered shapes cannot be pushed to a different location. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                                                                  |
| Taper               | Via                 | Tapered shapes cannot be pushed to a different location. However, you can try to select the <b>Push Via</b> option and try again. See <a href="#">When Vias Are Pushed</a> .          |
| Taper               | Via on bump         | Tapered shapes cannot be pushed to a different location. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                                                                  |
| Via                 | Via                 | The <b>Push Via</b> option is not selected. See <a href="#">When Vias Are Pushed</a> .                                                                                                |

| Object 1        | Object 2        | Not Pushed Because...                                                                                                                                          |
|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Via             | Via on bump     | The via connected to the bump cannot be pushed. For the standalone via, the <b>Push Via</b> option is not selected. See <a href="#">When Vias Are Pushed</a> . |
| Via on bump     | Via on bump     | Vias connected to bumps cannot be pushed. See <a href="#">Shapes and Nets That Cannot Be Pushed</a> .                                                          |
| Via             | Any object type | The via cannot be pushed to the target location without causing spacing violations.                                                                            |
| Any object type | Any object type | The operation was interrupted.                                                                                                                                 |

## Fixing Tapering Violations

The **Auto Taper** command lets you create or fix tapered shapes to comply with tapered shape rules in the packaging technology file. You use **Auto Taper** when making postroute adjustments.

This section describes the following:

- [About the Auto Taper Command](#)
- [Auto Tapering](#)

### About the Auto Taper Command

The **Auto Taper** command automatically creates or corrects tapered shapes based on tapered shape rules in the packaging technology file. In addition to the entire design, you can fix tapered shape violations by selecting shapes or nets, thus letting you control the precision of the fixes in the design.

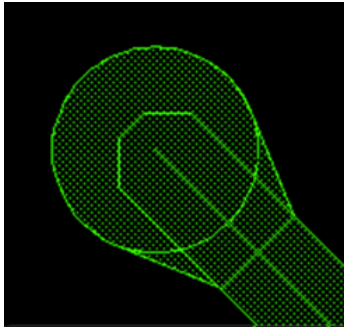
This topic describes the following:

- [Tapered Shapes](#)
- [Tapered Shape Violations](#)
- [How Auto Taper Works](#)
- [When to Use Auto Taper](#)

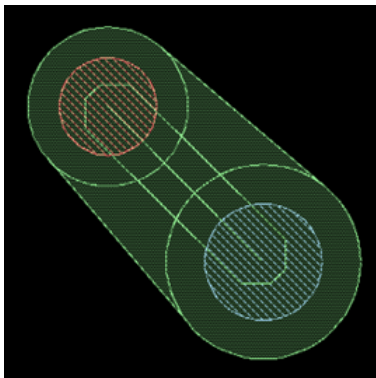
## Tapered Shapes

Tapered shapes connect two routing objects. Different types of tapered shapes are used in the connections:

- Teardrop tapered shapes connect paths to vias or bumps.



- Convex hull tapered shapes connect two vias or bumps.



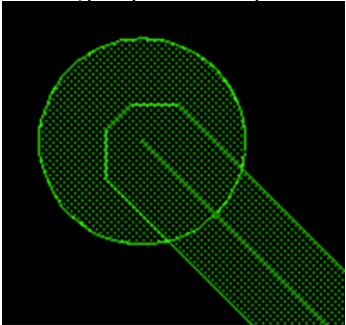
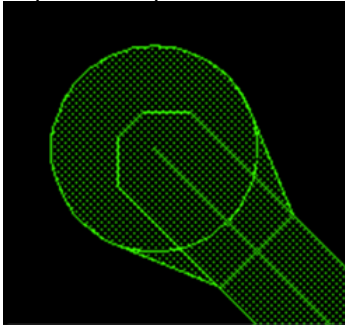
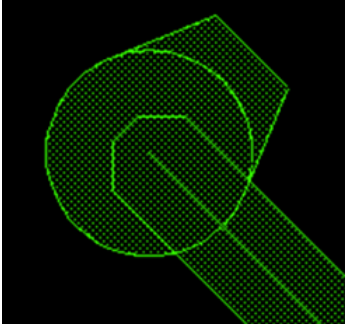
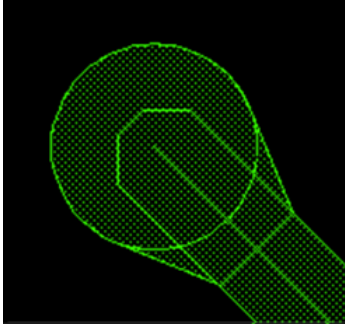
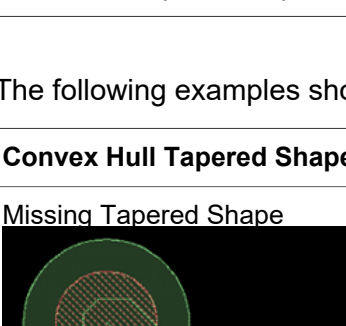
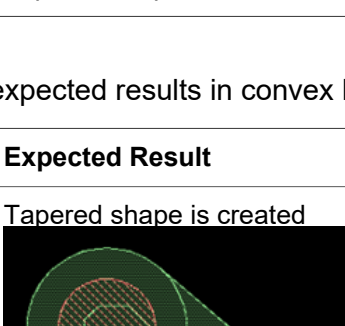
## Tapered Shape Violations

Tapered shape rules are defined in the packaging technology file. During the **Auto Taper** operation, the command checks for tapered shape rule violations.

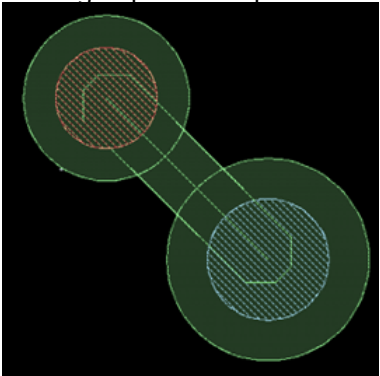
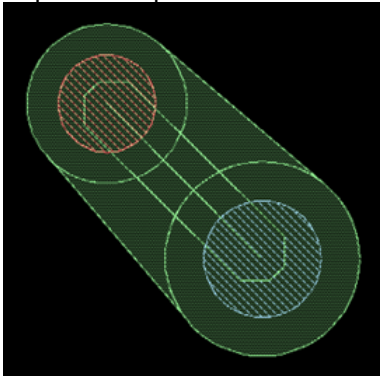
Three types of violations can occur in the design:

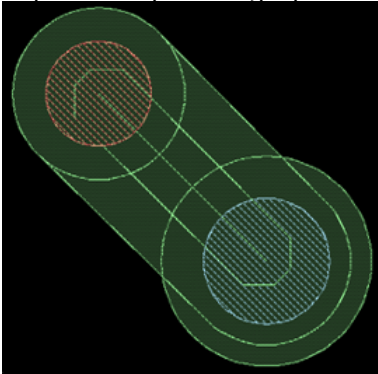
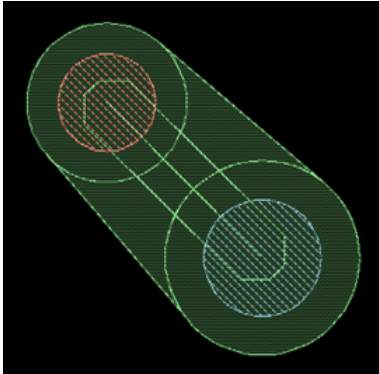
- Missing tapered shapes
- Incorrect tapered shape placement
- Redundant tapered shapes

The following examples show violations and expected results in teardrop connections.

| Teardrop Tapered Shape Violation                                                                                                     | Expected Result                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Missing Tapered Shape<br>                           | Tapered shape is created<br>         |
| Incorrect (or illegal) tapered shape placement<br> | Placement is correct<br>            |
| Redundant tapered shape<br>                       | Tapered shape is not redundant<br> |

The following examples show violations and expected results in convex hull connections.

| Convex Hull Tapered Shape Violation                                                                          | Expected Result                                                                                                  |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Missing Tapered Shape<br> | Tapered shape is created<br> |

| Convex Hull Tapered Shape Violation                                                                                        | Expected Result                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <p>Tapered shape in illegal position</p>  | <p>Placement is correct</p>  |
| <p>Redundant tapered shape</p>                                                                                             | <p>Tapered shape is not redundant</p>                                                                          |

### How Auto Taper Works

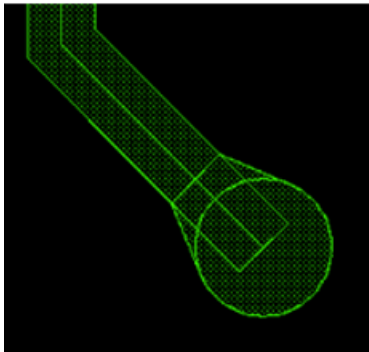
When using **Auto Taper**, you first select the entire design; or you can select specific paths, vias, or nets. During the operation, the command:

- Uses the tapered shape rules in the packaging technology file to check for violations.
- Fixes all tapered shape violations in the design, or fixes only tapered shapes associated with the selected paths, vias, or nets as follows:
  - Missing tapered shapes: Adds tapered shapes
  - Incorrect (or illegal) tapered shape placements: Corrects tapered shape placements
  - Redundant tapered shapes: Deletes redundant tapered shapes
- Does not check design rules, therefore existing paths and vias are not modified.

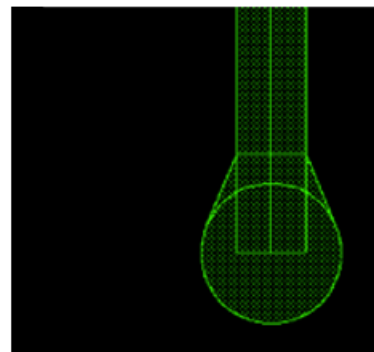
**Auto Taper** can also be triggered automatically during a **Push Route** operation if connected paths or vias are pushed. See [About the Push Route Command](#).

In the following example, when the path is pushed, the tapered shape is also modified (rotated) to retain the connection with the path.

Before Push



After Push



### When to Use Auto Taper

When you edit route paths and vias, the Layout Editor does not automatically update the associated tapered shapes. Because tapered shape instances are PCells defined in the packaging library, updating the tapered shapes manually involves iterative modifications of the PCell parameters.

Instead of manually fine-tuning the PCell parameters, you can use **Auto Taper** to automatically update the tapered shapes. Because **Auto Taper** does not modify existing paths and vias, you can use the command to fix tapered shapes in isolation without disturbing other shapes.

### Auto Tapering

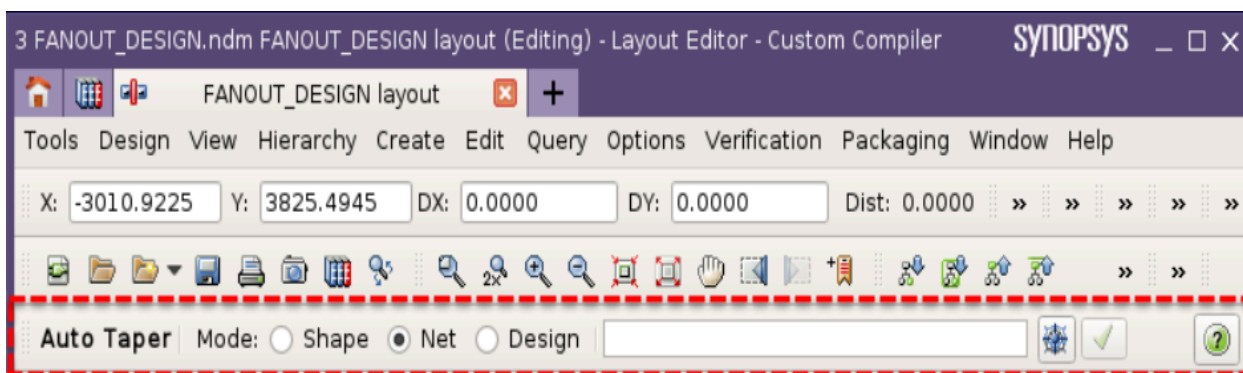
Instead of manually fine-tuning PCell parameters after editing route paths and vias, you can use the **Auto Taper** command to automatically update tapered shapes to resolve violations.

Before you start, review the [prerequisite information](#).

To use **Auto Taper** to fix tapering violations:

1. In the Layout Editor, choose **Packaging > Auto Taper**.

The **Auto Taper** toolbar appears.



2. On the **Auto Taper** toolbar, select the **Mode**:

| Argument | Description                                          |
|----------|------------------------------------------------------|
| Shape    | Lets you select shapes for the operation.            |
| Net      | Lets you select nets for the operation.              |
| Design   | Fixes tapered shape violations in the entire design. |

3. (**Net** mode only) Specify the net names or net group name using one of the following methods:
  - In the Design Navigator, select the nets or net group.
  - On the **Push Route** toolbar, type the net names or net group name in the box.
4. (**Shape** mode only) On the layout canvas, select the shapes.

The shapes you can select are:

- Paths
- Tapered shapes (polygons)
- Via shapes or bumps

5. On the **Auto Taper** toolbar, click **Apply** .

When the Auto Taper operation is finished, information appears in the Console to let you know of errors encountered during the operation. See [Packaging Auto Router Error Messages](#).

## Troubleshooting Package Design Routing Errors

This topic describes how to troubleshoot the following:

- [Skipped Nets](#)
- [Packaging Auto Router Error Messages](#)
- [Adjusting Routes](#)

### Skipped Nets

The [Packaging Routing Report](#) shows the number of skipped nets and the reasons for skipped nets. The following table lists the reasons for the skipped nets, the potential causes, and possible resolutions.

| Reason                                   | Causes                                                                                                          | Solutions                                                                                                    |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Invalid pin count                        | The net has fewer than two pins or an uneven number of pins.                                                    | Make sure the net has at least two pins, or an even number of pins.                                          |
| Fully routed                             | The net is already routed.                                                                                      | No need to take action.                                                                                      |
| Has pre-routed shapes                    | The net contains pre-routed shapes.                                                                             | To route the net, remove the pre-routed shapes and route again.                                              |
| Misaligned pins in LDI                   | The net has microbumps that do not align with the LDI frontside pads on LDI layers, so vias could not be added. | Check the design and make sure the microbumps align with the LDI frontside pads, and then try routing again. |
| Same channel net % has pre-routed shapes | Pre-routed nets exist in the same channel.                                                                      | To route all nets in the die-to-die channel, remove the pre-routed shapes on the reported nets.              |

### Packaging Auto Router Error Messages

The following table lists the error and warning messages you might encounter when using the Packaging Auto Router. Use the list to determine solutions you can try when troubleshooting the errors.

See [Types of Error Messages](#) for the description of the types of messages.

If you are still experiencing problems after trying the solutions, or if an error not on the list is displayed, contact [Synopsys Support](#).

**Note:**

You must be logged in to SolvNetPlus for the link to connect directly to the Synopsys Support page. If you are prompted to log in to SolvNetPlus upon clicking the link, log in, then click the link again to reach the support page.

See also Troubleshooting Error Messages for the Custom Router.

| ID                          | Type    | Message                                                                                                                                                      | Causes                                                                                                                                   | Solutions                                                                                                                                   |
|-----------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| PACKAGIN<br>G_ROUTE-<br>001 | Warning | Push Route is stopped because the timeout limit of %d seconds is exceeded. The timeout limit is specified by the rtPackagingPushRouteTimeout Tcl preference. | Push Route is unable to fix DRC errors within the specified time. The operation is interrupted to prevent hanging and undesired results. | Make sure the selected objects do not create shorts on nets. Try again by selecting different objects. See <a href="#">Pushing Routes</a> . |
| PACKAGIN<br>G_ROUTE-<br>003 | Warning | Supply planes of non-existing net(s) "%s" are skipped.                                                                                                       | The supply plane guide belongs to a net that does not exist.                                                                             | Delete all the supply plane guides belonging to the net that does not exist.                                                                |
| PACKAGIN<br>G_ROUTE-<br>004 | Error   | Packaging router requires layers defined in packaging tech file to be consecutive and either in same or reverse order in OA tech.                            | Layers defined in the packaging techfile are not consecutively defined in the same or reverse order in the OA technology file.           | Make sure the order of the layers in the packaging technology file or OA technology file are in the same or reverse order.                  |
| PACKAGIN<br>G_ROUTE-<br>005 | Error   | Packaging library does not exist.                                                                                                                            | Packaging library is not defined correctly.                                                                                              | Define the packaging library. To define the library correctly, refer to the packaging library specification.                                |
| PACKAGIN<br>G_ROUTE-<br>006 | Error   | LDI region is enabled without specifying LDI layers.                                                                                                         | If you have LDI regions in the design, you must specify LDI layers.                                                                      | Specify the LDI layers. See <a href="#">Setting Global Routing Options for Packaging Designs</a> .                                          |
| PACKAGIN<br>G_ROUTE-<br>007 | Error   | Packaging technology file is not specified.                                                                                                                  | The Packaging Auto Router needs the packaging technology file to consider the routing rules.                                             | Specify the packaging technology file. See <a href="#">Defining Packaging Libraries and Technology Files</a> .                              |

| ID                          | Type  | Message                                                                                                                                          | Causes                                                                                                                          | Solutions                                                                           |
|-----------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| PACKAGIN<br>G_ROUTE-<br>008 | Error | %s region is enabled, but there are %d objects on its specified LPP. Expected number of objects is %s.                                           | The LDI/DIEBOX/D2D/FOX region is enabled but there are no objects or unexpected number of objects defined on its specified LPP. | Create regions with the expected number of objects on the LPP.                      |
| PACKAGIN<br>G_ROUTE-<br>009 | Error | Channel region %s is skipped, because there are pre-routed shapes of net %s. Please delete the pre-routed shapes within the region and re-route. | Incremental routing is not supported in D2D regions, because the routing takes all pins in a channel into consideration.        | Delete the pre-routed shapes in the reported region and re-run routing.             |
| PACKAGIN<br>G_ROUTE-<br>010 | Error | %s rules format error. Should be %s. Input string %s.                                                                                            | The rule has incorrect format.                                                                                                  | Revise the rule to fit the format.                                                  |
| PACKAGIN<br>G_ROUTE-<br>011 | Error | In Packaging tech file, layer %s is missing.                                                                                                     | The reported layer is not defined in the packaging technology file.                                                             | Add the missing layer in the packaging technology file.                             |
| PACKAGIN<br>G_ROUTE-<br>012 | Error | In packaging tech file, %s on layer %s contradicts with %s on layer %s.                                                                          | The packaging technology file has conflicting XML tag settings.                                                                 | Review and update the packaging technology file to resolve the conflicts.           |
| PACKAGIN<br>G_ROUTE-<br>013 | Error | %s rules %s value %s is invalid. Should be %s.                                                                                                   | The value for the specified rule is invalid.                                                                                    | Revise the value defined in packaging technology file to a valid value.             |
| PACKAGIN<br>G_ROUTE-<br>015 | Error | In packaging tech file, layer %s is set to %s, but there is no %s.                                                                               | Required parameter is missing in the packaging technology file.                                                                 | Add the required parameter for the reported layer in the packaging technology file. |
| PACKAGIN<br>G_ROUTE-<br>016 | Error | In packaging tech file, %s is missing on layer %s.                                                                                               | Required parameter is missing for a layer in the packaging technology file.                                                     | Add the required parameter for the reported layer in the packaging technology file. |
| PACKAGIN<br>G_ROUTE-<br>018 | Error | Layer %s defined in packaging tech file is not a physical layer defined in oa tech.                                                              | The layer defined in the packaging technology file does not exist in the OA (Custom Compiler) technology file.                  | Update the Custom Compiler technology file to add the layer.                        |

| ID                          | Type    | Message                                                                        | Causes                                                                     | Solutions                                                                                                           |
|-----------------------------|---------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| PACKAGIN<br>G_ROUTE-<br>024 | Warning | Skip to auto create plane boundary for net %s on layer %s (No shape is found). | There is no pre-routed or routed shape for the net on the specified layer. | Remove the specified layer from the auto-created plane boundary net layers or route the net on the specified layer. |

## Adjusting Routes

During troubleshooting, you can use the available Packaging menu commands to adjust routes. You can also use the Layout Editor commands to fine-tune the routes.

When resolving DRC errors, you can use the following methods:

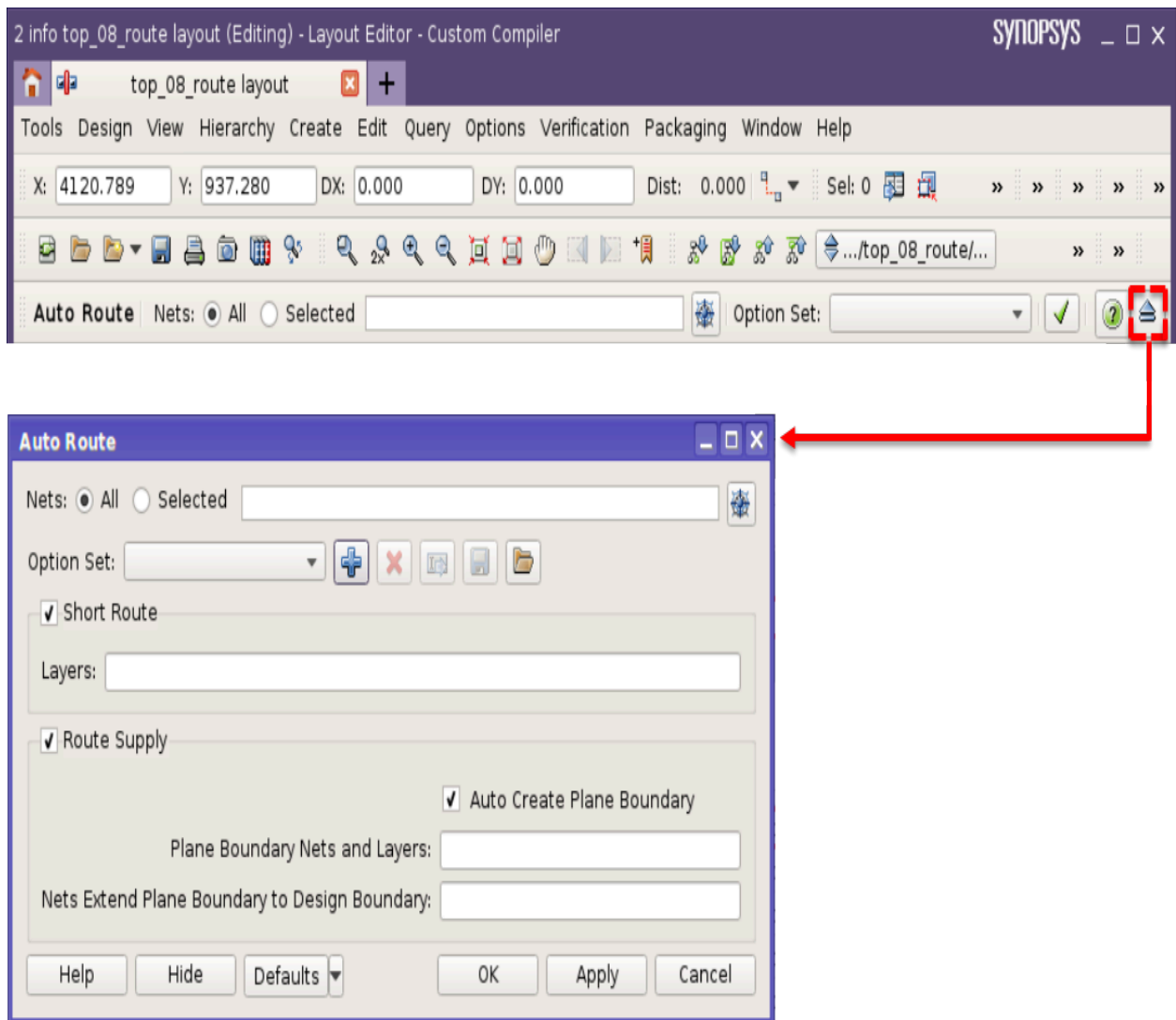
- The **Push Route** and **Auto Taper** commands. See [Fixing Routing Violations](#).
- Layout Editor editing commands to adjust the PG planes and routing shapes. For example, you can move, chop, trim, stretch, or reshape paths or path segments. See [Creating and Editing Interconnects and Editing Objects - Basic Operations](#) for commands that are applicable to PG plane and route editing.

## Packaging Auto Route Toolbar and Dialog Box Reference

This section provides a quick reference of the Packaging **Auto Route** toolbar and dialog box. For information about how to add routes in the packaging design, see [Workflow for Routing Packaging Designs](#).

The **Packaging > Auto Route** dialog box contains the following sections:

- [Basic Options](#)
- [Short Route](#)
- [Route Supply](#)



## Basic Options

This section describes the options that appear in both the toolbar and dialog box. In the dialog box, the common options appear above and below the tabs.

| Option      | Description                                                                                                                                                                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nets</b> | Specifies the nets you want to route: <ul style="list-style-type: none"><li>• <b>All:</b> Routes all nets.</li><li>• <b>Selected:</b> Routes only the selected nets. Selected net names appear in the box. You can select nets in the Design Navigator, or you can type the net names in the Nets box.</li></ul> |

| Option                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Displays or hides the Design Navigator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Option Set                                                                          | Displays the chosen set of options. See Managing Option Sets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|    | Starts the auto routing based on the options and specifications you provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (toolbar)<br>OK/Apply (dialog box)                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|    | Displays the Custom Compiler help.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| (toolbar)<br>Help (dialog box)                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Defaults (dialog box)                                                               | <p>Creates new default options and values, or resets to the selected default options and values. Click the button, and then select one of the following:</p> <ul style="list-style-type: none"><li>• <b>Make Defaults for:</b> Uses the current information to create a new set of defaults for cellview, cell, library, or user.</li><li>• <b>Save:</b> Saves your current preferences as an XML file.</li><li>• <b>Load from:</b> Loads from the default cellview, cell, library, or user.</li><li>• <b>Load:</b> Lets you load a previously saved preference file.</li><li>• <b>Purge from:</b> Lets you delete the default options and values from the cellview, cell, library, or user.</li></ul> |
|  | Opens or hides the Packaging <b>Auto Route</b> dialog box.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| (toolbar)<br>Hide (dialog box)                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Short Route

Connects neighboring microbumps (pins) on the specified layers to reduce the number of escape vias, thus saving routing resources. The Packaging Router only connects the microbumps using single any-angle straight paths.

Short routing requires the following:

- The net has neighboring microbumps in the DIEBOX region.
- The net is not a large-area supply net. (Large-area supply nets usually require every microbump to be escaped individually to optimize electrical results).

Tcl preferences: `rtPackagingUseShortRoute`

| Option | Description                                                                                      |
|--------|--------------------------------------------------------------------------------------------------|
| Layers | Layers to be used for short routes.<br>Tcl preferences: <code>rtPackagingShortRouteLayers</code> |

## Route Supply

| Option                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Route Supply                                  | Routes specified supply nets.<br>Tcl preference: <code>rtPackagingUseSupplyRoute</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Auto Create Plane Boundary                    | Automatically creates PG plane boundaries before routing starts, and assigns routes to the plane boundaries during the routing operation. The auto mode applies only to PG nets that do not have net-layer specifications.<br>For example, suppose the design contains the following PG nets: VSS, VDD, VDD1, and VDD2. You specify the following in Plane Boundary Nets and Layers: { VSS layer1 } { VDD layer2 }. The router will honor your specifications for VSS and VDD, and automatically assign the layer for VDD1 and VDD2.<br>See also <a href="#">Creating PG Plane Boundaries Manually</a> and <a href="#">Creating PG Plane Boundaries Using the Command-Line Interface</a> for other methods of creating PG plane boundaries. |
| Plane Boundary Nets and Layers                | Routes the specified supply nets using the corresponding PG plane boundaries. The syntax is:<br><code>{net1 layer1 layer2 ... layer n} {net2 layer1 layer2 ...} ...</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Nets Extend Plane Boundary to Design Boundary | Extends the PG plane boundaries to the design boundary and routes the supply nets.<br>This option is used only when you select <b>Auto Create Plane Boundary</b> , and you do not specify <b>Plane Boundary Nets and Layers</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

# 7

## Adjusting Post-Route Designs

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*This chapter describes how to adjust the routed packaging designs.*

This chapter contains the following topics:

- [Adjusting the Routes](#)
- [Adjusting the PG Plane Boundaries](#)

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### Adjusting the Routes

If the routing results show that some nets are not routed, or if you need to make route adjustments to ensure they comply with design rules, you can create new routes or edit the routes manually. In some cases, you might need to create new PCells, and add new or modify existing terminals so that the routes can escape to the target bumps. This section describes how to adjust routes manually to escape to target bumps.

To adjust routes manually to escape to target bumps:

1. Open the routed packaging design and look for bumps that are not connected to routes.
2. Turn off the `leNetlistMode` Tcl preference so that you can make modifications to the connections in the layout. In the Custom Compiler Console, type:

```
db::setPrefValue leNetlistMode -value off -scope
[de::getActiveContext]
```

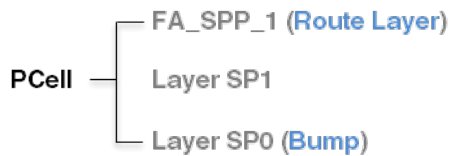
3. Create a PCell at the bump location. See [Creating Packaging Instances](#).

At the step where you set the options in the **Parameters** tab, specify the following:

- **Number of Layers:** 3
- **Layer 1:** Bump layer, with a **Diameter** value that matches the diameter of the bump.
- **Layer 2:** Connecting layer. Set the **Diameter** to the width that is specified in the packaging technology file.

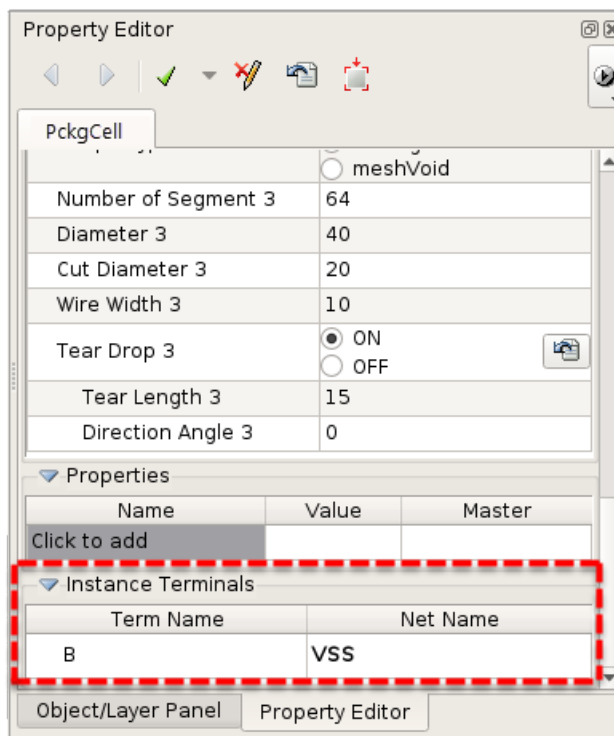
- **Layer 3:** Route shape layer. This layer is for the new route you are creating. Set the **Diameter** value based on the cut width and enclosure. Turn on **Tear Drop**. Set the **Tear Length** to the length specified in the packaging technology file, and **Direction Angle** to 45.

In the following example, Layer 1 is SP0, Layer 2 is SP1, and Layer 3 is FA\_SPP\_1.



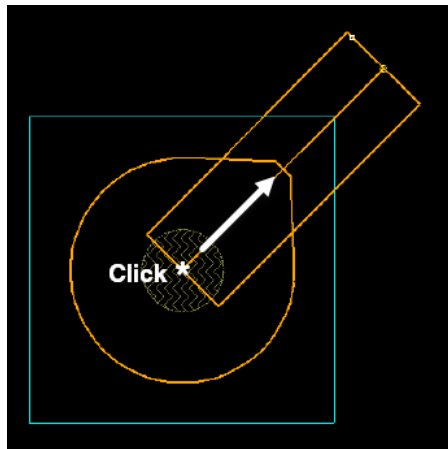
Place the instance at the location of the bump.

4. On the layout canvas, the PCell that you created. In the Property Editor, and set the **Net Name** property for the instance terminals. The net name should correspond to the net of the route and bump.



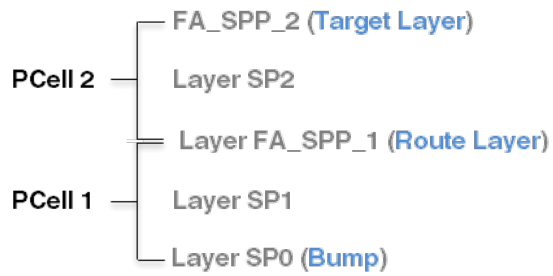
5. Add a path to align with the teardrop. The width should match the teardrop, the length is dictated by the technology requirement.

- a. In the Object/Layer Panel, select the route layer (the Layer 3 name that you specified in step 3).
- b. In the Layout Editor, choose **Create > Interconnect**.
- c. On the **Interconnect** toolbar, set the **Width** to match the teardrop. In the **Net** box, type the name of the net (the same net as the bump). Choose the **Angle: Diagonal**.
- d. On the layout canvas, click the center of the teardrop and move the pointer in the direction of the teardrop. Stop when you reach the target length, and double-click to commit the path.

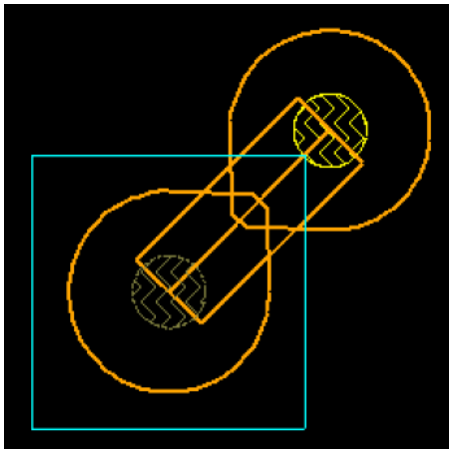


6. To escape to the target layer, repeat step 3 to create another PCell.
  - **Number of Layers:** 3
  - **Layer 1:** Route shape layer. Turn on **Tear Drop**. Make sure the **Diameter** and **Tear Length** values match the corresponding values you set for Layer 3 in PCell 1 (step 3). Set the teardrop **Direction Angle** to -135.
  - **Layer 2:** Connecting layer. Set the **Diameter** to the width that is specified in the packaging technology file.
  - **Layer 3:** Target layer. Set the **Diameter** to the cut layer width.

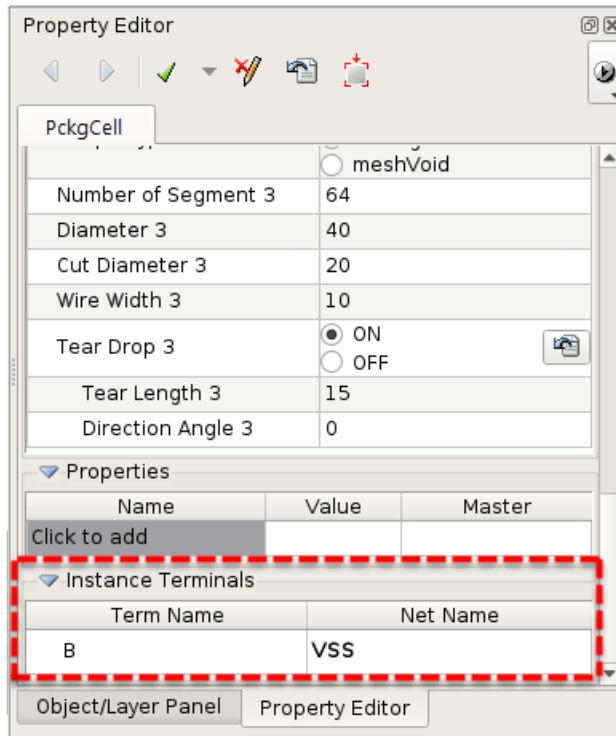
In the following example, PCell 1 was created previously to connect the bump to FA\_SPP\_1. PCell 2 is created to continue the connection from the FA\_SPP\_1 to the target layer (FA\_SPP\_2).



7. Place the second PCell instance so that the center of the via aligns with the end of the path centerline.



8. On the layout canvas, select the newly created PCell. In the Property Editor, and set the **Net Name** property for the instance terminals. The net name should correspond to the net of the route and bump.



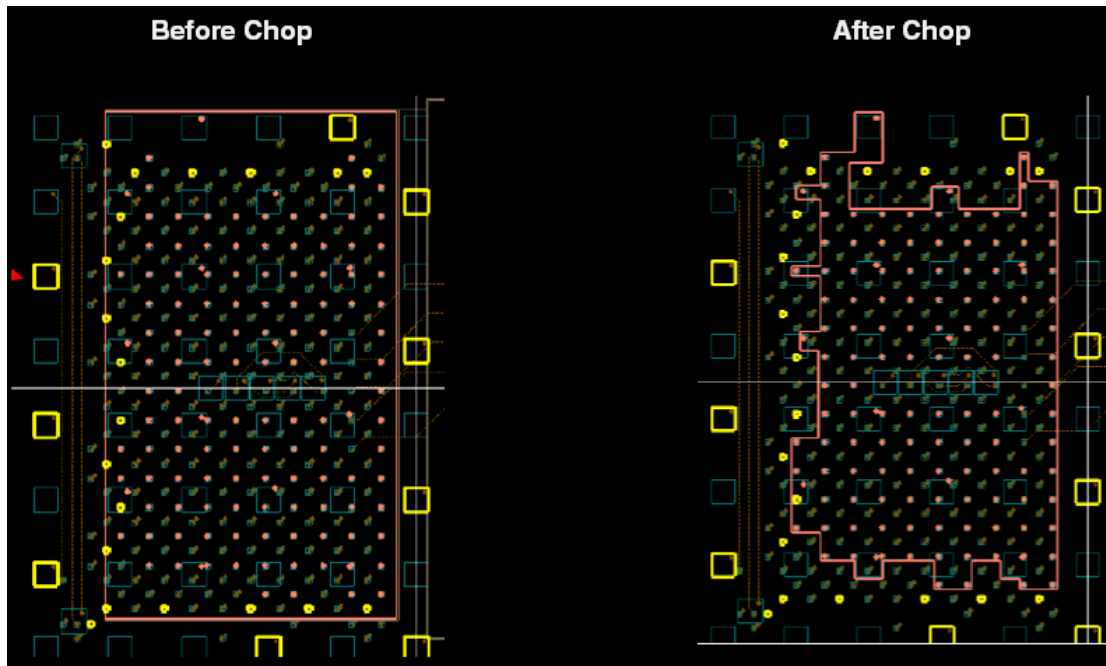
- Set the `leNetlistMode` Tcl preference to the read-only mode to prevent further connection modifications. In the Custom Compiler Console, type:

```
db::setPrefValue leNetlistMode -value read -scope
[de::getActiveContext]
```

## Adjusting the PG Plane Boundaries

To prevent shorts and other potential issues, PG plane boundaries should only cover bumps and vias on the same net. For example, the plane boundary for VDD1 should only cover the VDD1 bumps and vias.

After routing, review the design, and adjust the PG plane boundaries as needed. In the following example, an existing VDD1 PG plane boundary is edited to exclude bumps and vias on other nets.



To adjust the PG plane boundaries:

1. Open the routed packaging design.
2. Check the design. Make sure each PG plane boundary encloses only the bumps and vias on the same net.
3. Use one or more of the following methods to adjust the PG plane boundaries:
  - Use the available Layout Editor editing commands to edit the existing plane boundaries. For example, you can use the Chop command to exclude coverage of bumps and vias. See [Editing Objects - Basic Operations](#).
  - Delete and create new plane boundary.

To delete a PG plane boundary, select the boundary shape, and press Delete. To create a new PG plane boundary, see [Creating PG Plane Boundaries Manually](#).

# 8

## Finishing the Packaging Design

---

*This chapter describes how to realize (or place) power and ground (PG) planes in the design, change long plane boundary lines to shorter zigzag lines, and create degassing holes.*

- [Realizing PG Plane Boundaries](#)
- [Replacing PG Plane Boundaries](#)
- [Creating and Reporting Redundant Vias](#)
- [Generating Zigzags](#)
- [Creating Dummy Metals and Degassing Holes](#)
- [Deleting Dummy Metals and Degassing Holes](#)

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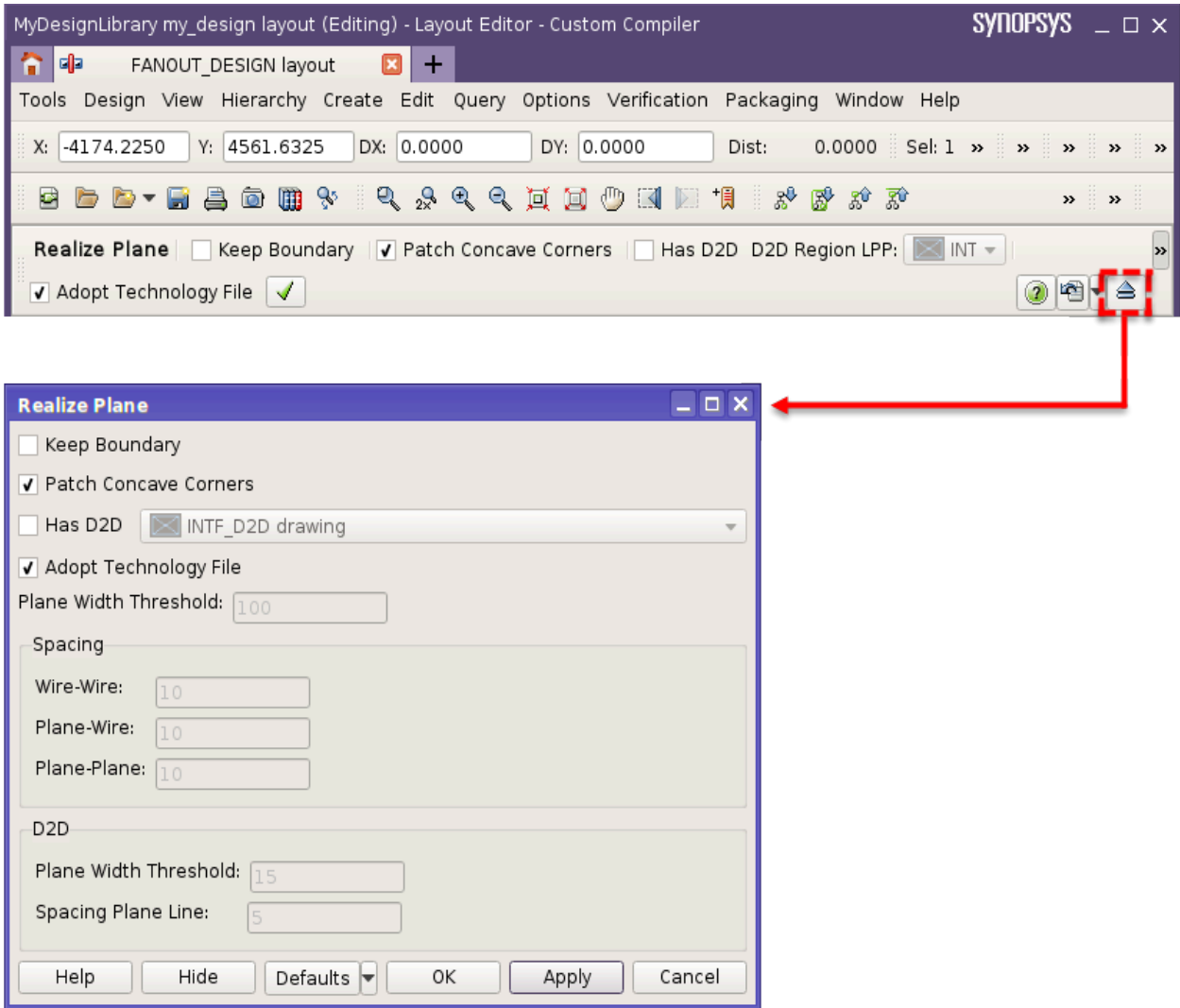
### Realizing PG Plane Boundaries

After routing, you can realize (or place) the power and ground (PG) plane that you created earlier using the **Create Plane Boundary** command. See [Creating PG Plane Boundaries Manually](#). To create zigzag lines on long edges of the realized PG plane boundaries, see [Generating Zigzags](#).

To realize the PG plane boundary:

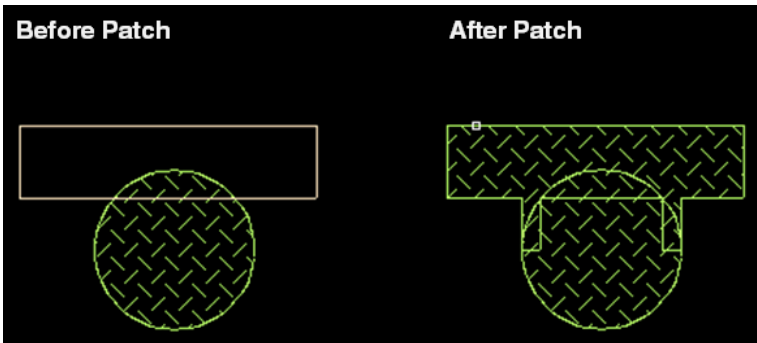
1. In the Layout Editor, choose **Packaging > Realize Plane**.

The **Realize Plane** toolbar appears. To view additional options, click **Eject**  to open the dialog box.



2. Set the options on the **Realize Plane** toolbar.

| Field         | Description                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keep Boundary | Keeps the plane boundary after realization. Clear the check box to delete the plane boundary after realization.<br>Tcl preference: lePackagingKeepBoundary |

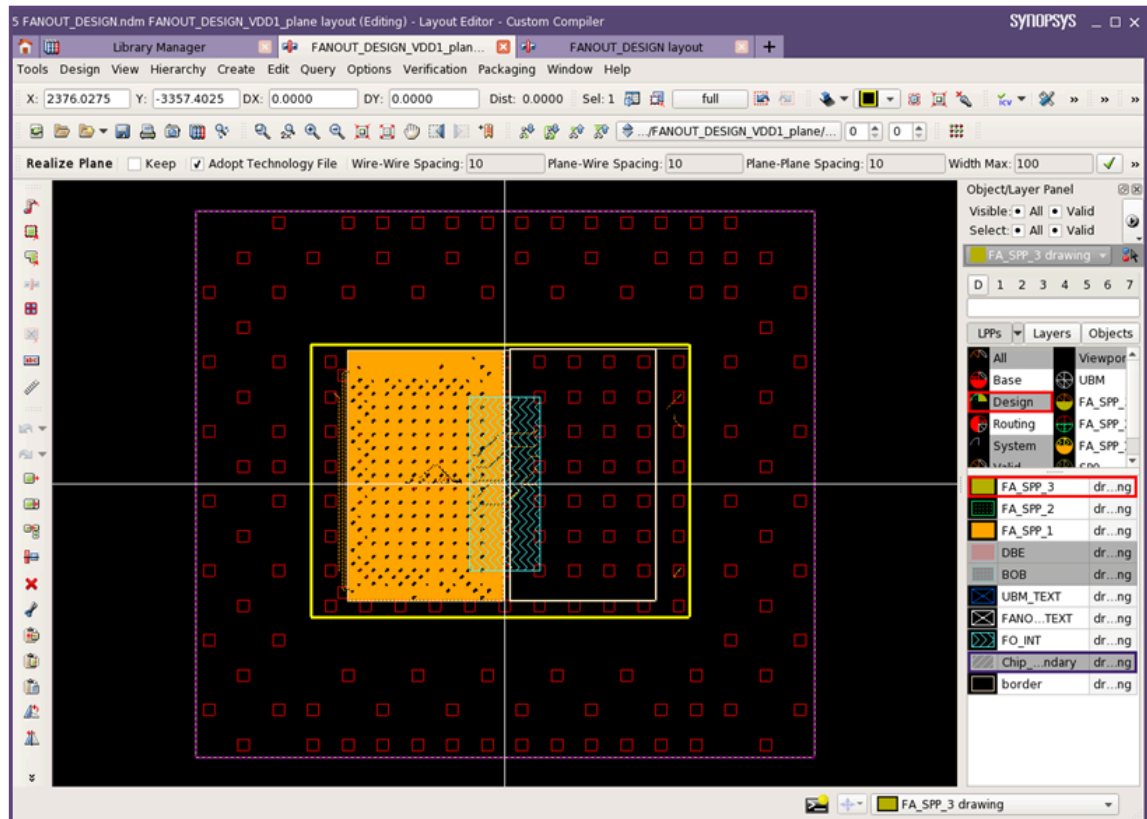
| Field                                      | Description                                                                                                                                                                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patch Concave Corners<br>(Dialog box only) | <p>Adds patches to rectangular PG plane boundaries, thus eliminating concave angles produced by the plane boundary creation.</p>  <p>Tcl preference: <code>lePackagingPatchConcaveAcute</code></p> |
| D2D                                        | <p>Turns on and off the die-to-die (D2D) options.</p> <p>Tcl preference: <code>rtPackagingUseD2dRegionLpp</code></p>                                                                                                                                                                 |
| D2D Region LPP                             | <p>Layer-purpose pair (LPP) that is used to define the die-to-die (D2D) connection region.</p> <p>Tcl preference: <code>rtPackagingD2dRegionLpp</code></p>                                                                                                                           |
| Adopt Technology File                      | <p>Uses only the packaging technology file and ignores all rules in the OA technology file. When this option is selected, all other options on the toolbar become unavailable.</p> <p>Tcl preference: <code>lePackagingAdoptTechnologyFile</code></p>                                |
| Plane Width Threshold                      | <p>Maximum width threshold. Any shape that has a width wider than this value is treated as a boundary plane. Otherwise, the shape is treated as a wire.</p> <p>Tcl preference: <code>lePackagingWidthMax</code></p>                                                                  |
| Spacing:                                   |                                                                                                                                                                                                                                                                                      |
| Wire-Wire Spacing                          | <p>Distance between two wires.</p> <p>Tcl preference: <code>lePackagingWireWireSpacing</code></p>                                                                                                                                                                                    |
| Plane-Wire Spacing                         | <p>Distance between the PG plane boundary and wire shapes.</p> <p>Tcl preference: <code>lePackagingPlaneWireSpacing</code></p>                                                                                                                                                       |
| Plane-Plane Spacing                        | <p>Distance between adjacent PG plane boundaries.</p> <p>Tcl preference: <code>lePackagingPlanePlaneSpacing</code></p>                                                                                                                                                               |
| D2D:                                       |                                                                                                                                                                                                                                                                                      |

| Field                 | Description                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plane Width Threshold | Maximum width threshold within the die-to-die region. Any shape that has a width wider than this value is treated as a boundary plane. Otherwise, the shape is treated as a wire. |
| Spacing Plane Line    | Minimum spacing between the PG plane boundary and the routing object (wire, via, or teardrop).                                                                                    |

- On the layout canvas, select the PG plane boundaries that you want to realize.

If the `deDirectApply Tcl` preference is `true`, the command automatically realizes the selected plane boundaries as soon as you select them. Otherwise, proceed to step 4.

- On the **Realize Plane** toolbar, click **Apply** . The layout canvas updates to show the realized plane.



---

## Replacing PG Plane Boundaries

Before creating redundant vias, you must make sure PG plane boundaries that are on adjacent layers are on the same net. Instead of manually adjusting the PG plane boundaries, you can use the **Replace Plane Boundary** command.

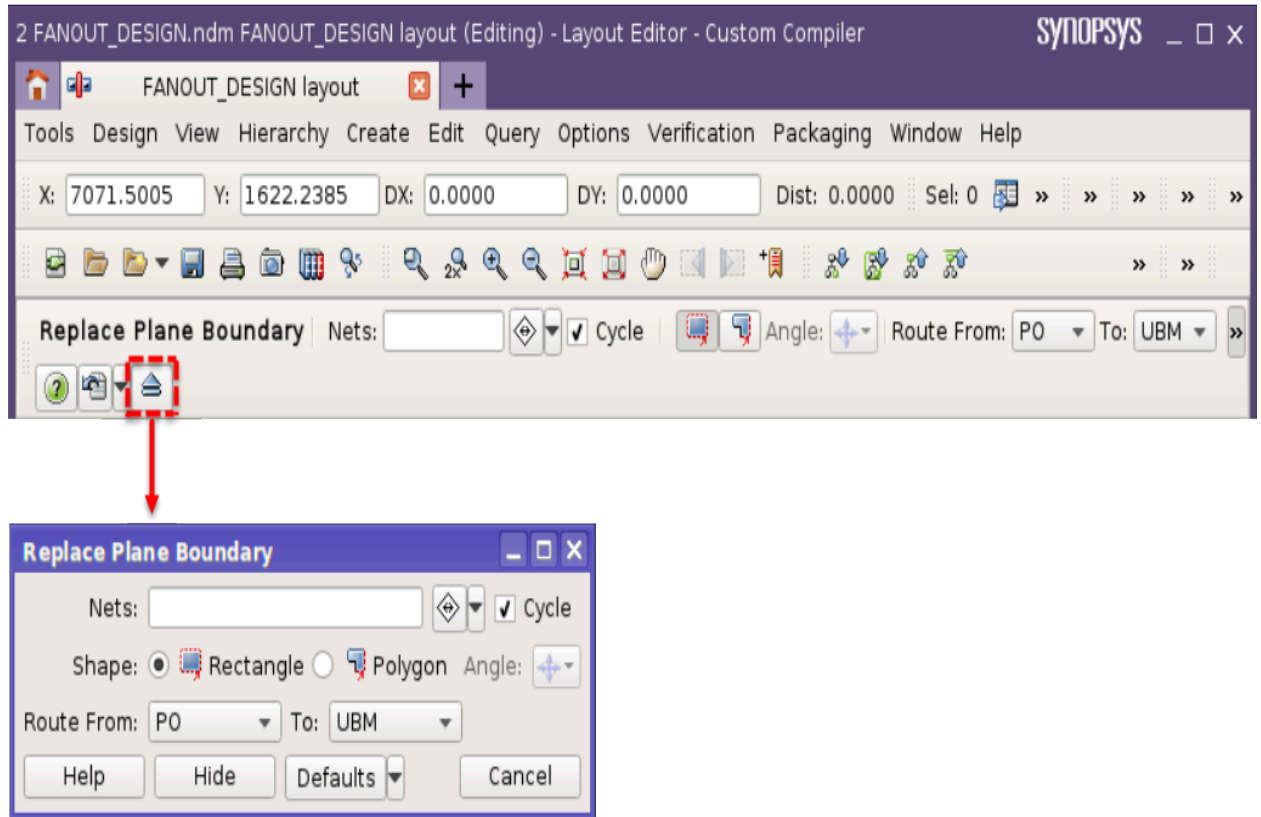
The **Replace Plane Boundary** command:

- Creates new PG plane boundaries within the specified region, nets, and range of layers.
- Merges existing unnetted PG plane boundaries with the newly created PG plane boundaries if they overlap.
- Updates existing PG plane boundaries that are on different nets, and ensures the correct spacing is applied.
- Realizes modified and newly created PG plane boundaries.

To replace PG plane boundaries:

1. In the Layout Editor, choose **Packaging > Replace Plane Boundary**.

The **Replace Plane Boundary** toolbar appears. If you prefer to work in the dialog box version of the command, click **Eject** .



2. Specify the **Nets**. When specifying multiple nets, add a character space between the net names.
3. Select the **Shape** of the region you plan to draw on the canvas:
  - **Rectangle** 
  - **Polygon** 
4. (Polygon only) Choose the **Angle** of the polygon:
  - **Orthogonal** : The polygon is 90° from the starting angle.
  - **Diagonal**: The polygon is 45° from the starting angle.
  - **Any Angle**: The polygon can be at any angle from the starting angle.
  - **X First**: Two orthogonal shapes are created. The first shape is horizontal.
  - **Y First**: Two orthogonal shapes are created. The first shape is vertical.

5. In the **Route From** and **To** lists, choose the starting and ending layers, respectively, to specify the range of layers.
6. On the layout canvas, draw the rectangle or polygon.

The PG plane boundary is created based on your specifications.

---

## Creating and Reporting Redundant Vias

To reduce voltage (IR) drops and to improve reliability, you can use the **Generate Redundant Vias** command to insert additional vias after routing and realizing PG plane boundaries. The command adds vias between two or more 0x planes that are on adjacent layers and on the same net.

This topic describes the following:

- [Creating Redundant Vias](#)
- [Reporting Redundant Vias](#)
- [Deleting Redundant Vias](#)

---

### Creating Redundant Vias

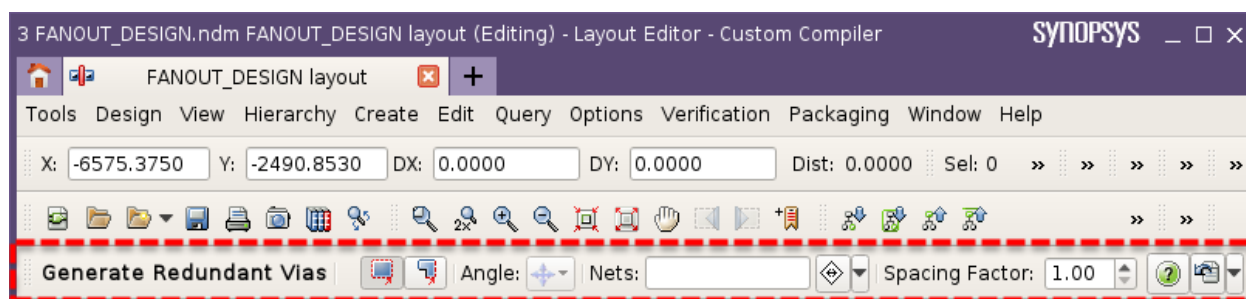
To create redundant vias:

1. In the Object Layer Panel, make sure all LPPs associated with the net of interest are visible. See Working With LPPs. The PG plane boundaries on these layers must overlap.
2. In the Design Navigator, click the **Nets** tab, and select one or more nets.

To facilitate viewing the selected nets, on the Design Navigator toolbar, turn on the

**Cross Shadow** .

3. In the Layout Editor, choose **Packaging > Generate Redundant Vias**. The **Generate Redundant Vias** toolbar appears.



4. On the **Generate Redundant Vias** toolbar, set the following:

| Option                                                                                           | Description                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>(Rectangle) | Region within which the vias are created.<br>Tcl preferences: <code>leCreateShapeType</code>                                                                                                                                                                                                                                                          |
| <br>(Polygon)   |                                                                                                                                                                                                                                                                                                                                                       |
| Angle                                                                                            | Orientation of the polygonal region. See Angle Mode for Edges and Centerlines.<br>Tcl preferences: <code>leCreateAngleMode</code>                                                                                                                                                                                                                     |
| Nets                                                                                             | Nets on which you want to generate the vias.<br>Click  . The nets you selected in the Design Navigator appear in the <b>Nets</b> box.<br>Tcl preferences: <code>leNetName</code>                                                                                     |
| Spacing Factor                                                                                   | Multiplier of the minimum spacing between redundant vias. Use this option to control the density of the vias. Specify a value $\geq 1$ . Default: 1<br>For example, if the minimum spacing is 0.5 $\mu\text{m}$ , and Spacing Factor = 2, then minimum spacing used is 1 $\mu\text{m}$ .<br>Tcl preferences: <code>leRedundantViaSpacingFactor</code> |

5. On the layout canvas, draw a rectangular or polygon shape.

Vias are automatically generated within the region. In the Console, a report shows the number of vias generated for the specified nets.

```
Information: Redundant vias generated on net: VDDPST
Information: +-----+-----+
Information: | Layer | Number |
Information: +-----+-----+
Information: | SP2   | 30     |
Information: +-----+-----+
```

## Reporting Redundant Vias

Although a report is displayed after redundant vias are created, you can use the following Tcl command to generate a report at any time, independently of running the **Generate Redundant Vias** command:

```
le::pkg::reportRedundantVias -design <oaDesign> [-net <oaNet | string>]
```

If you do not specify nets, the command reports redundant vias for all nets in the design.

The report appears in the Console, and is identical to the report that appears when you run the **Generate Redundant Vias** command.

In the following example, the command generates a redundant via report for the VDDPST net in the design that is currently opened for editing.

```
le::reportRedundantVias -design [ed] -net VDDPST
```

```
Information: Redundant vias generated on net: VDDPST
Information: +-----+-----+
Information: | Layer | Number |
Information: +-----+-----+
Information: | SP2   | 30     |
Information: +-----+-----+
```

---

## Deleting Redundant Vias

Inserting redundant vias can increase the component density of your design. If you need to remove some of the redundant vias, use the following Tcl command:

```
(le::pkg::deleteRedundantVias-box <box> | -points [<point>]) -design  
<oaDesign>
```

The command deletes all redundant vias within a specified area in the design, regardless of net or layer. You can either draw the area on the layout canvas, or you can specify coordinates to indicate the area.

In the following example, the command deletes all redundant vias within the area specified by the coordinates. The command is operating in the design that is currently opened for editing.

```
le::deleteRedundantVias -design [ed] -points {{-92.4235 -916.3485}  
{94.5595 -734.8445}}
```

---

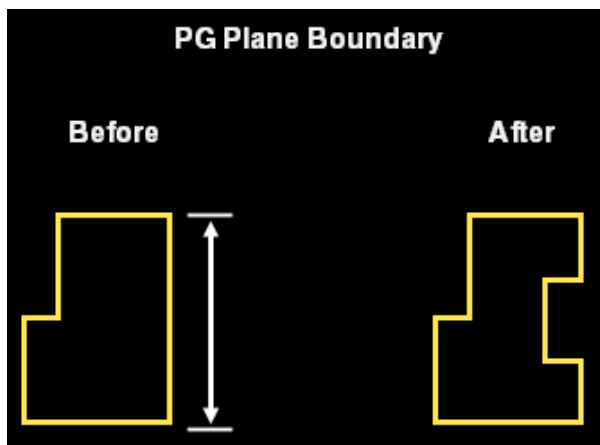
## Generating Zigzags

This topic describes the following:

- [About Zigzags](#)
- [Before You Start](#)
- [Creating Zigzags in PG Plane Boundary Shapes](#)

## About Zigzags

You use the **Packaging > Generate Zigzag** command to change long lines into several shorter lines. The command applies only to long edges of **realized (placed) PG plane boundaries** and ignores all other shapes.



The **Packaging > Generate Zigzag** command applies rules from the packaging technology file to determine the length, width, spacing, and other requirements for the operation. Contact the foundry for details about the technology file.

## Before You Start

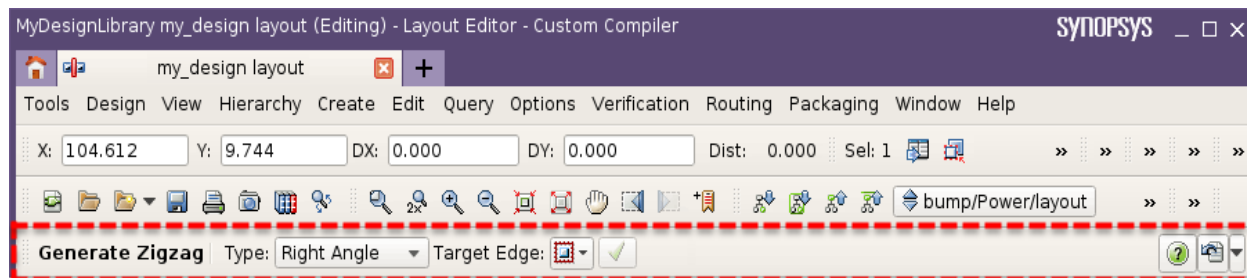
Make sure the PG plane boundary is realized (placed). See [Realizing PG Plane Boundaries](#).

## Creating Zigzags in PG Plane Boundary Shapes

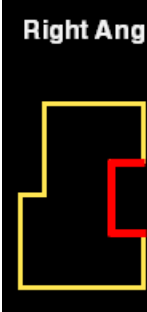
To generate a zigzags:

1. In the Layout Editor, choose **Packaging > Generate Zigzag**.

The **Generate Zigzag** toolbar appears.



- 2. On the layout canvas, select the realized (placed) PG boundary plane.
- 3. On the **Generate Zigzag** toolbar, set the options:

| Field | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type  | <div><p>The type of zigzag line:</p><ul style="list-style-type: none"><li>◦ <b>Right Angle</b>: Creates lines that connect at 90°.</li><li>◦ <b>Obtuse Angle</b>: Creates lines that connect at 135°.</li></ul></div> <div><div><div>Right Angle</div></div><div><div>Obtuse Angle</div></div></div> |

Tcl preference: `lePackagingZigzagType`

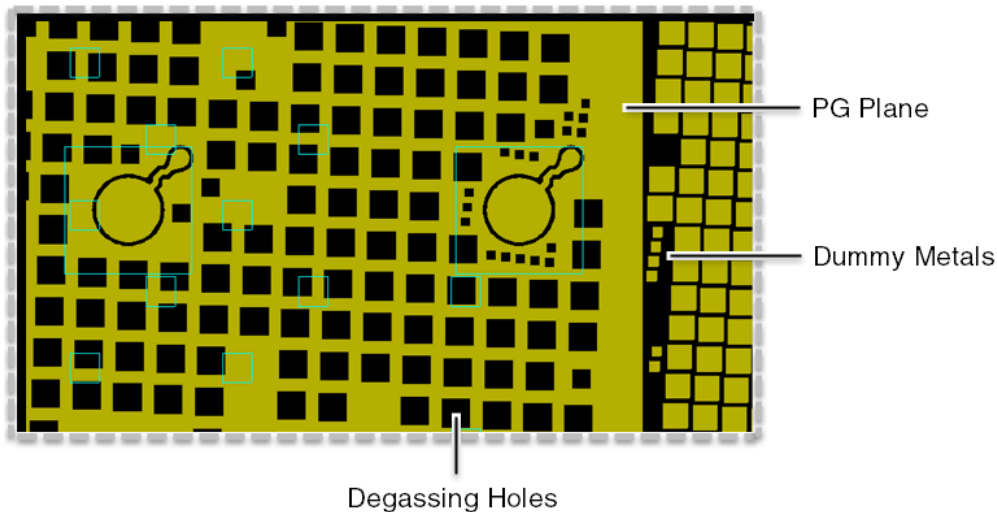
| Field       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target Edge | <p>Target of the zigzag operation:</p> <ul style="list-style-type: none"><li>◦ <b>Boundary</b>: All boundary edges.</li><li>◦ <b>Top</b>: Top boundary edge only.</li><li>◦ <b>Bottom</b>: Bottom boundary edge only.</li><li>◦ <b>Left</b>: Left boundary edge only.</li><li>◦ <b>Right</b>: Right boundary edge only.</li><li>◦ <b>Center</b>: Cuts the selected plane boundary from the center.</li></ul> <div><p>Target Edge: Center</p><div><p>Before</p></div><div><p>After</p></div></div> <p><b>Note:</b><br/>The <b>Top</b>, <b>Bottom</b>, <b>Left</b>, and <b>Right</b> selections can be combined to form multiple target edges.</p> <p>Tcl preference: <code>lePackagingZigzagTarget</code></p> |

4. Click **Apply** .

## Creating Dummy Metals and Degassing Holes

After routing, you can add dummy metals, or metal fills, to ensure uniform metal density across the design. You must also create degassing holes in the power and ground (PG) planes. Degassing holes in packaging designs let gas escape to prevent bubbles from forming during the manufacturing process.

### PG Plane With Degassing Holes and Dummy Metals



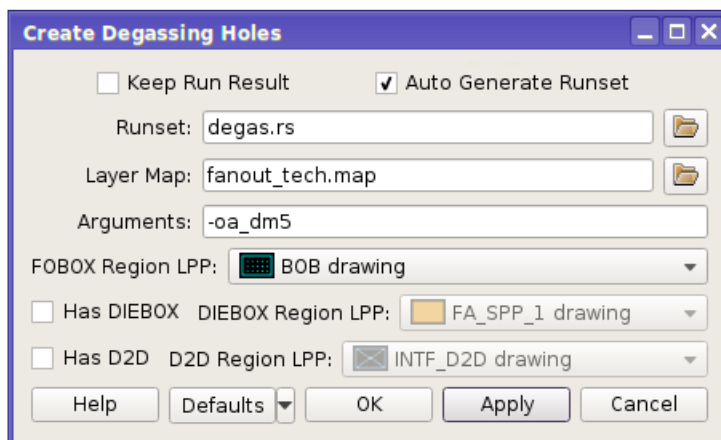
You use the **Packaging > Create Degassing Hole** command to:

- Generate the GDSII Stream format for the design.
- Run IC Validator to create degassing holes and dummy metals.
- Import the degassing holes into the packaging design, replacing the PG planes.

To create degassing holes:

1. In the Layout Editor, choose **Packaging > Create Degassing Holes**.

The **Create Degassing Holes** dialog box opens.



2. Set the options in the **Create Degassing Holes** dialog box.

| Option               | Description                                                                                                                                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keep Run Result      | Creates a synopsys_custom/degassing_hole_<name> directory and stores the output of the run in the directory.<br>Tcl preference: lePackagingKeepRunResult                                                                                                                                                                                           |
| Auto Generate Runset | Automatically generates the required runset file.                                                                                                                                                                                                                                                                                                  |
| Runset               | Name and path of the IC Validator degassing runset file.<br>Tcl preference: lePackagingRunset                                                                                                                                                                                                                                                      |
| Layer Map            | Name and path of the layer mapping file.<br>Tcl preference: lePackagingLayerMapFile                                                                                                                                                                                                                                                                |
| Arguments            | IC Validator arguments you want to run. For example, because the degassing operation can be lengthy, you can reduce the processing time by specifying -host_init to increase the number of cores or hosts.<br>See the <a href="#">IC Validator User Guide</a> on SolvNetPlus.<br>Default argument: -oa_dm5<br>Tcl preference: lePackagingArguments |
| FOBOX Region LPP     | Fanout box (FOBOX) region layer-purpose pair (LPP).                                                                                                                                                                                                                                                                                                |
| Has DIEBOX           | Turns on and off the die box (DIEBOX) region LPP option.                                                                                                                                                                                                                                                                                           |
| DIEBOX Region        | Die box (DIEBOX) region layer-purpose pair (LPP) .<br>Tcl preferences: rtPackagingUseDieRegionLpp, rtPackagingDieRegionLpp                                                                                                                                                                                                                         |
| Has D2D              | Turns on and off the die-to-die (D2D) options.<br>Tcl preference: rtPackagingUseD2dRegionLpp                                                                                                                                                                                                                                                       |
| D2D Region LPP       | Die-to-die region layer-purpose pair (LPP).<br>Tcl preference: rtPackagingD2dRegionLpp                                                                                                                                                                                                                                                             |

3. When you are finished, click **OK**.

### See Also

- [Deleting Dummy Metals and Degassing Holes](#)

## Deleting Dummy Metals and Degassing Holes

The **Delete Degassing Holes** command returns the design to the state of the realized power and ground (PG) planes before you added dummy metals (or metal fills) and

created degassing holes. The command eliminates the need to create backup copies of the design.

**Caution:**

You cannot undo the **Delete Degassing Holes** operation. The dummy metals, degassing holes, and all modifications made to the design after you created the dummy metals and degassing holes will be deleted permanently.

The dummy metals and degassing holes data are stored in the layout.oa.degas file. The **Delete Degassing Holes** command deletes the layout.oa.degas file.

To delete the dummy metals and degassing holes:

- In the Layout Editor, choose **Packaging > Delete Degassing Holes**.

**See Also**

- [Creating Dummy Metals and Degassing Holes](#)

# 9

## Reviewing Custom Compiler Edits in the 3DIC Compiler Tool

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*After modifying designs using the Custom Compiler tool, you can continue editing the design using the 3DIC Compiler tool. Before running the 3DIC Compiler tool, you should confirm that the changes you made to the design in the Custom Compiler environment will appear correctly in the 3DIC Compiler tool.*

The Layout Editor **3DIC Compiler > Open in 3DIC Compiler** command lets you open and view the design using the 3DIC Compiler tool while you are still in the Custom Compiler environment.

1. Make sure the **3DIC Compiler** menu is displayed. See [Enabling the Co-design Assistant](#).
2. In the Layout Editor, choose **3DIC Compiler > Open in 3DIC Compiler**.

When you are finished, close the design in the Custom Compiler environment and open it in the 3DIC Compiler tool to make any necessary revisions. For instructions, see the [3DIC Compiler user documentation](#) available on SolvNetPlus.

**Note:**

You must be logged in to SolvNetPlus for the link to connect directly to the 3DIC Compiler user documentation. If you are prompted to log in to SolvNetPlus, log in, then click the 3DIC Compiler user documentation link again to reach the help topics.